OECD

OECD ECONOMIC SURVEYS

1999

SPECIAL FEATURES

Enhancing tax revenue

Strengthening the banking system

MEXICO

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1998-1999

MEXICO

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF MEXICO

THE LAND

Area (sq. km)	1 967 183
Agricultural area (sq. km) (1990)	394 600

Inhabitants in major cities (thousands, 1990):	
Mexico City	15 048
Guadalajara	3 044
Monterrey	2 651

THE PEOPLE

Population (thousands, 1997)	94 184
Inhabitants per sq. km (1997)	47.9
Annual population growth (1980-1997)	2.0

Employment (thousands, 1997)	35 904

PRODUCTION

Structure of production, 1997 (per cent of total, 1993 prices):	
Agriculture	5.2
Industry	25.8
of which: Manufacturing	19.3
Services	60.9

GDP in 1997 (US$ billion)	402.1
GDP per capita in 1997 (US$, using PPPs)	7 697
Gross fixed capital formation in 1997 (per cent of GDP, 1993 prices)	17.2

THE GOVERNMENT

Public sector indicators (per cent of GDP, 1997)	
Public sector total expenditure	23.7
Federal government current expenditure	8.7
Federal government capital expenditure	1.9
Federal government revenue	16.0
Public sector debt (December 1997)	19.4

Composition of Parliament (1998):	Senate	Chamber of Deputies
PRI	77	238
PAN	33	121
PRD	16	126
Other	2	15

FOREIGN TRADE

Exports of merchandise (per cent of GDP, 1997)	27.5
Main exports (per cent of total, 1997):	
Manufactures	85.8
Petroleum products	10.3
Agriculture	3.5

Imports of merchandise (per cent of GDP, 1997)	27.4
Main imports (per cent of total, 1997):	
Intermediate goods	77.7
Capital goods	13.8
Consumer goods	8.5

THE CURRENCY

Monetary unit: Peso

Currency units per US dollar, average of daily figures:	
Year 1998	9.15
February 1998	10.00

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Mexico by the Economic and Development Review Committee on 18 February 1999.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 10 March 1999.

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The previous Survey of Mexico was issued in February 1998.

Assessment and recommendations

Following a strong recovery from the 1995 recession...

Mexico's economic performance has been positive in the past three years or so. Real GDP growth has averaged more than 5 per cent a year, at first export driven and then supported also by gross fixed investment and private consumption. With a lag, the recovery has resulted in important job creation in the formal sector. Substantial progress has also been recorded in reducing inflation, with the rise in the CPI coming down to 16 per cent on average in 1998, from 34 per cent in 1995. But the disinflation process reversed and economic activity started losing some buoyancy in the second half of 1998. The international financial crisis and the fall in oil prices caused the peso to depreciate and required successive tightenings of macroeconomic policies to preserve confidence and curb the deterioration of the external balance. Nonetheless, as domestic demand continued to buoy imports and declining oil prices dampened export earnings, the current-account deficit is estimated to have widened by some 2 percentage points in 1998, to nearly 4 per cent of GDP.

... activity is expected to slow down in 1999, before picking up in 2000, but the degree of uncertainty is considerable

Repercussions from global financial turbulence, increases in short-term interest rates and successive budget cuts throughout 1998 are bound to dampen economic activity in the period ahead. In the OECD Secretariat projections prepared in November 1998, real GDP growth was seen as decelerating substantially in 1999 to 3½ per cent, reflecting restrained domestic demand and weaker manufacturing exports as a result of slower growth in foreign markets. Since then, there have been further adverse developments. The Brazilian crisis unsettled financial markets again in January 1999. Although the impact on Mexico was much less severe than after the Russian crisis and was quickly

reversed, the external environment remains uncertain. Hence, the authorities may have to maintain a tight policy stance for longer than previously expected, or even take further restrictive measures, to preserve confidence, reduce inflation and keep the current-account deficit in line with available financing. Inward foreign direct investment remained resilient in 1998, but portfolio investment turned negative in the third quarter. Although the size and direction of capital flows is notoriously difficult to predict, net inflows may be somewhat restrained in the foreseeable future, as risk perception is unlikely to return quickly to pre-crisis levels. Another negative development since the finalisation of the OECD Secretariat projections in November 1998 has been a further decline in oil prices. On the other hand, economic growth in the United States has remained surprisingly strong. On balance, the November projections now seem somewhat optimistic, and growth is likely to be lower. For instance, the Mexican authorities expect real GDP growth to be 3 per cent in 1999. Barring new major shocks, growth could pick up again in 2000. For the time being, however, the external environment is still not fully stable, and the degree of uncertainty surrounding the outlook remains considerable.

Macroeconomic policy was tightened in 1998, but the peso depreciated and inflation rose

The authorities responded to the external shocks of 1998 – *i.e.* the international financial crisis and the fall in oil prices – with a tightening of monetary policy and fiscal policy which, in the framework of a floating exchange-rate regime, limited the depreciation of the peso. This approach proved broadly effective in preserving confidence and protecting the real economy, although it was less successful in containing the inflationary impact of the depreciation of the peso – which was compounded by administered price hikes and increases in fresh food prices. In the midst of a highly volatile external environment, when the Bank of Mexico estimated that the pressure against the peso was inconsistent with the fundamentals of the economy and that the deterioration in inflation expectations was becoming worrying, it moved forcefully to engineer a rise in short-term interest rates. Following record increases in September, Mexico's interest rates eased slightly in the latter part of the year, against a backdrop of abating international turbu-

lence. Although the tightening of monetary conditions in several stages combined with successive budget cuts prevented the exchange rate from getting out of control, inflation did rise significantly in the second half of 1998, reaching 18½ per cent in the twelve months to December – well above the 12 per cent official target.

Bringing inflation under control should be at the top of the policy agenda in 1999

In the context of an uncertain external environment, the authorities decided to keep a restrictive stance of macroeconomic policy in 1999, as evidenced by the approved budget and the monetary programme for the same year. In response to the Brazilian crisis in January 1999, monetary policy was tightened further. This timely response coupled with the fiscal stance kept the contagion effect to a minimum: the increase in Mexican short-term interest rates and the weakening of the peso were short-lived. Looking to the future, while a degree of exchange-rate flexibility remains appropriate as the immediate response to an external shock, if the peso depreciates excessively, inflation will start rising again from an already high level, with the risk of seriously undermining the credibility of monetary policy and confidence in the currency. Unless international financial markets settle down durably and the peso strengthens, monetary policy will have to remain tight to put the country firmly back on a disinflationary path. At the same time, fiscal policy should be sufficiently restrictive and the pace of structural reform sufficiently rapid – especially in the financial sector – to signal the determination of the authorities to regain control of inflation, take some of the burden off monetary policy, and make high interest rates more credible – hence, more effective in underpinning the peso. This stabilisation strategy is likely to entail some costs, especially in the short term, with lower output growth. It is also uncertain to what extent high interest rates will affect the cost of the various financial rescue programmes and the position of banks, which remains fragile even after the recent approval of parts of the financial reform package. Nonetheless, unlike alternative policy courses, inflation control holds the promise of a reasonably rapid return to economic growth and rising living standards. As the experience of many countries has shown, half-hearted attempts at stabilisation carry the risk of a protracted period of high

interest rates and tight fiscal policy, with little prospect for a sustained recovery. In sum, a floating exchange rate regime can provide a useful degree of flexibility, but only if it is accompanied by rigorous economic policies. Floating – like any other exchange-rate regime – is no substitute for necessary corrective measures.

Fiscal policy needs to remain tight

In response to lower revenue from the oil sector, public expenditure was cut repeatedly in 1998, so that the outturn of the budget deficit was on target. Weak oil prices require continuing fiscal restraint in the near future. Beyond short-run considerations and regardless of oil price movements, a tight fiscal policy stance is needed because of the key role it can play in preserving macroeconomic stability. By reducing the public sector deficit, it can increase total domestic saving and limit the deterioration of the current account. This, in turn, would contribute to underpinning the exchange rate. In addition, sound public finances could enhance investors' confidence in the run up to the presidential elections in 2000 – a period typically marked by increased risk perception and instability. The 1999 budget, adopted at the end of 1998, aims at maintaining the public sector financial deficit at 1¼ per cent of GDP, through reduced primary spending and some increase in non-oil government revenue. Barring new external shocks, this deficit target seems broadly appropriate.

A medium-term policy framework could help foster confidence...

With the budget proposal for 1999 the government presented an outline of its strategy for 1999-2000, restating the objectives announced in the National Programme to Finance Development (PRONAFIDE) of June 1997, including a macroeconomic scenario. One of the stated objectives of the policy strategy is to create the conditions required for an orderly transition to the next administration. The budget projections that are part of this strategy imply maintaining a tight fiscal stance but, as noted above, further tightening may be necessary if the peso comes under renewed pressure. Such a course, if fully implemented, would constitute a key element of the discipline required to foster macroeconomic stability. Recent experience has shown that under adverse circumstances, such as the fall in oil revenue in 1998, the government is ready to take corrective measures

as needed to meet budget targets. This episode has certainly enhanced credibility in the government's resolve, and by providing information on the policy orientation over the coming two years, the government may have helped to further reassure markets on the continuity of its strategy. Nonetheless, a truly medium-term framework would require a somewhat longer time horizon. For this purpose it would be useful to present annually an updated set of multi-year objectives, such as those contained in the PRONAFIDE of June 1997 which covered the four-year period 1997-2000.

... and reduce the need for budget adjustments under pressure

In the case of Mexico, actions on the revenue side should be emphasised as a means of obtaining the desired fiscal stance: spending cuts under the pressure of events are not without a cost: curtailing spending implies postponing or reducing the scale of programmes which may have high economic and social benefits. Indeed, while government primary spending, at 10 per cent of GDP, is one of the lowest among OECD countries, there is a clear lack of physical infrastructure and human capital development, including basic education and health, as well as a need to alleviate extreme poverty. Social spending was largely spared in the 1998 adjustments and the 1999 budget. Nevertheless, there is still considerable scope for strengthening public expenditure programmes that will be beneficial for the overall development of the Mexican economy. However, there are many other pressures on public finances in Mexico which also need to be taken into account in setting fiscal policy. Contingent liabilities, reflecting the debt incurred to support banks and debtors and the cost of the social security reform (pension and health insurance systems), will continue to weigh heavily on public finances for years to come. In addition, lessons from past crises in Mexico point to the need to maintain solid public finances, so as to allow a build up of public saving which can help to increase domestic saving. Recourse to foreign saving can usefully contribute to finance investment, but only in as far as this is done through relatively stable capital flows. An effective management of public finances along these lines, as recommended in the 1998 *Economic Survey*, thus requires a strengthening of the government's tax revenue to allow the funding of key spending programmes without relying too

heavily on oil revenues, which are highly vulnerable to exogenous shocks. Indeed, as oil prices recover, the government should aim to develop a stabilisation fund outside the government accounts to absorb the direct budgetary effects from short-term fluctuations in oil prices.

The tax system needs to be strengthened to support justifiable public spending increases

The Mexican tax system has been substantially reformed and improved over the past two decades. However, the system still contains serious flaws that limit its capacity to raise revenue – as reflected in the fact that the overall level of taxation in Mexico is the lowest in the OECD area relative to GDP. The low capacity for raising revenue necessitates correspondingly low levels of public spending in order to secure satisfactory outcomes for the budget balance, and thus limits the scope for more public expenditure even where, as noted above, this would have a potentially high social return and allow for improved equity through public spending. Strengthening the revenue generating potential of the tax system is a key policy challenge for the coming years.

The general statutory features of the tax system are relatively neutral and non-distortive...

The general statutory features of the system are relatively neutral; and disincentives to work, save and invest appear to be less severe than in many other OECD countries. Average effective tax rates on consumption, labour and capital are low compared with other OECD countries and statutory marginal tax wedges on labour income are also relatively modest for most wage levels and family types. The adverse effects from high and volatile inflation are largely neutralised through indexation and the personal and corporate income taxes are integrated so as to avoid double taxation of dividends and distributed profits. This helps to make the tax system reasonably neutral in terms of its impact on different types of investment and financing instruments.

... but special regimes, exemptions and weak enforcement contribute to low revenues and insufficient horizontal equity

However, the tax system still contains several major deficiencies, which make the system inequitable in terms of incidence across different types of income and activities and contribute to keeping tax revenues low. Even though the package of tax measures in the 1999 budget addresses some of the shortcomings (see below), problems remain in a number of areas. The main priority as regards reform of the tax system is to broaden the tax base by reducing to a minimum the major preferential regimes, *i.e.* the large number of zero-rated and exempted goods and services in the VAT regime, the favourable treatment of specific sectors in the taxation of corporate income, as well as the exemption of fringe benefits in the taxation of individual income. These preferential regimes, besides diluting revenue and complicating tax administration *per se*, create significant loopholes and facilitate evasion, which in turn impinge on the effectiveness of the administration and creates major problems of horizontal inequity (some pay taxes, others do not). Another important challenge is to continue to improve the tax administration itself by strengthening surveillance and enforcement.

Although some progress has been made, more needs to be done...

The Government's agenda, as laid out in the National Development Plan 1995-2000 and reiterated in the PRONAFIDE medium-term programme in 1997, stressed the need for raising the effectiveness of tax collection and broadening the tax base, as well as granting local governments more taxation and expenditure responsibilities. Although some progress has been recorded, in particular in modernising tax administration, changes made to the tax system in recent years have generally not succeeded in increasing the tax-to-GDP ratio. The budget for 1999 contains a package of tax measures aiming at increasing revenues by around 1 percentage point of GDP relative to the outturn in 1998. Top marginal tax rates are raised to 40 per cent on individual income, whereas the corporate income tax rate on retained earnings is lowered to 30 per cent. Some broadening of the corporate tax base is achieved through the elimination of the special scheme for immediate deduction of some categories of investment and the tightening of corporate consolidation rules, which are thereby brought closer to practices in most other OECD

countries. Furthermore, the package includes some initiatives to reduce evasion and informal sector activity. Whereas the move towards a broader corporate tax base is warranted, the lowering of corporate statutory rates would not seem strictly necessary in light on Mexico 's low taxation of capital and the need to raise revenues. Moreover a non-negligible wedge is created between retained earnings and distributed profits. However, the package as a whole is a step in the right direction – although not sufficient to accommodate longer-term spending requirements. More needs to be done and the focus should now shift from changing statutory rates to broadening the tax base, through the reduction of preferential regimes, and further improving administration.

... and various options should be considered to broaden the tax base and improve tax collection

A number of specific options to strengthen tax revenue should be examined. With respect to the VAT regime, zero-rated goods should be limited to at most a few basic staples; the zero rating of agriculture and fisheries as well as the exemption of land transportation should be eliminated and the preferential border rate at 10 per cent should be brought into line with the standard rate. Exempting selected goods and services from consumption tax is generally not an efficient way to achieve equity objectives. Low-income groups could be compensated for the base broadening through targeted transfers at a much lower cost. In the corporate tax system, the favourable regimes for certain sectors, including agriculture, fisheries, land-transportation and publishing, should be abolished and a further tightening of consolidation rules could be considered. With respect to income tax on individuals, it would be appropriate to tax fringe benefits and eliminate the fiscal subsidy. Furthermore, increased taxation of property, in particular real estate, would be warranted from both revenue raising and distributional points of view. Finally, administration could be further improved through different measures, including training of personnel, registration, auditing and enhancing legal certainty. Although each of these suggestions in isolation would improve the system, they would achieve a much greater effect if several were implemented simultaneously. If implemented altogether, these initiatives could raise tax revenue by some 3 percentage points of GDP, which seem

to provide a reasonable benchmark for Mexico in the medium term. Nevertheless, some measures aimed at compensating vulnerable groups from the effects of such initiatives would have to be considered to make them feasible. The changes, besides increasing the revenue generating capacity of the tax system *per se*, would also reduce the sensitivity of the budget to world oil prices.

Action taken since 1995 has improved the banks' situation, but the financial sector remains fragile...

The financial sector is another area of major importance for structural reform. Addressing problems in this sector has been a high priority of the authorities, and vigorous action has been taken since 1995 to clean up and strengthen the banks, and certain categories of debtors. This has required the introduction of several programmes, which have a sizeable fiscal cost – the associated debt was estimated at over 14 per cent of GDP at the start of 1998, although the final cost will need to be revised in the light of recent developments. While the short-term costs will most likely be higher than anticipated a year ago, the final cost remains uncertain. The wide-ranging support operations have salvaged the banking system from collapse. The financial situation of the banks has improved, and regulation and supervision have been strengthened. But progress has been slow: normal debtor/creditor relationships have not resumed, and banks remain fragile. Just when banking credit was beginning to grow again in mid-1998, renewed financial market instability and the hike in Mexican interest rates aggravated the situation. Although in 1998, economic activity does not seem to have been seriously constrained by the virtual absence of banking credit, the financial system will need to be significantly strengthened over the longer run to provide the financial intermediation consistent with strong and sustained growth.

... financial volatility has worsened the situation, and unfinished business will have to be addressed rapidly

Proposals for a financial reform were discussed in Congress during most of 1998 before being partially approved in December. The controversy surrounding some elements of the reform slowed the needed process of bank restructuring and recapitalisation. The most important items approved included the removal of the remaining restrictions regarding foreign ownership of Mexican banks, and creating the basis for transforming the debt incurred for supporting banks into

tradable government-guaranteed instruments. The clarification of the coverage of the newly-created deposit guarantee fund – where a cap on deposit insurance will be introduced gradually – is also of crucial importance, since it can bolster market discipline which has been weakened by the succession of rescue packages over the past three years. At the same time, a new relief programme for small debtors was announced. A swift decision on the items of the financial reform still under discussion and an effective implementation of measures already approved would greatly help in tackling remaining difficulties, although this may not be sufficient. The non-performing loan problem will be satisfactorily solved only through lower interest rates and strong sustainable growth. Moreover, to put bank lending on a better footing and give the authorities early warning of trouble, supervisory and regulatory mechanisms should continue to be strengthened. Clarification of creditors' rights and the application of effective collection, foreclosure and bankruptcy procedures are also necessary to underpin soundly-based credit expansion.

While the social security reform is well under way, the modernisation of labour-market institutions has been delayed

Progress has been more pronounced in the area of social security, where the 1997 reform is being carefully implemented. A fully-funded capitalisation pension system is now operating for private sector workers, with contributions being deposited on individual accounts which are managed by private administrators. As part of the reform of the private sector health fund, payroll taxes to finance compulsory health insurance are being progressively reduced – with a fixed contribution replacing the proportional employer-employee contribution; and the cost of voluntary affiliation has been reduced. One of the possible beneficial effects of the social security reform is that – by developing the interaction between social and labour market policies – it can enhance the attractiveness of employment in the formal sector. Substantial job creation in the formal sector over the recent past warrants a certain optimism. However, it will take some time before it is possible to assess whether the reforms are delivering their full benefits in this respect, and to measure the extent to which these favourable developments reflect cyclical influences or more lasting structural factors. In the areas of education and training, progress has

continued, but tight budget constraints have implied delaying some programmes in basic education, health and poverty alleviation. As the political debate during most of 1998 was focused on the financial reform and budget proposals, little progress was made on the modernisation of the labour market institutional structure which has been under discussion for some time. Although in practice the labour market has demonstrated its flexibility in coping with adverse shocks, the reform agenda should include a review of institutional arrangements and regulations to bring them closer to actual practices. Initiatives in this area could contribute to lowering barriers that keep many activities in the informal sector.

Further regulatory reform should yield important benefits in the electricity sector…

Product market competition has been sharpened. The privatisation programme of the previous administration, although impressive in scope, had left public services almost untouched. A number of initiatives have been taken in this area over recent years, to open key sectors – railroads, airports, satellite services, gas and electricity – to private participation. By mid-1997, the revision of the legal framework governing these sectors and the introduction of relevant regulations were well advanced. The auctioning of concessions (or outright selling off of public assets) has been proceeding since. In the electricity sector in particular, at first, generation was opened to private investment. Then, in February 1999, a proposal to reform the sector in depth was sent to Congress. It aims at restructuring the state-owned companies, splitting them into separate segments specialised in generation and distribution, then opening these activities to private investment. At the same time, a new regulatory and legal framework is to be established. After the year 2000, privatisation of state-owned companies would be allowed. Full implementation of this proposal is highly desirable.

… as well as in other areas

Structural reform should be pursued in several other areas. Experience shows that for private participation to yield consequent benefits, it is important to establish and carefully implement a regulatory framework that is oriented at enhancing competition in practice. For instance, in the telecommunication sector, where privatisation was completed

by 1991 and the basic regulations were established in 1995, under the terms of the concession the private company, Telmex, enjoyed a virtual monopoly until 1997 for the long-distance market and until 1998 for the local market. In comparison with all other OECD countries, Mexico's performance in this sector is still lagging considerably, with the lowest service penetration and the highest price levels. Regarding regulatory reform in the private sector, the current administration has made significant progress in streamlining regulations by systematic scrutiny and simplification to alleviate the administrative burden on enterprises. But here too, it is necessary to pursue this process by broadening its scope and carefully overseeing the implementation stage. Even when regulations appear to be appropriate, greater transparency, predictability and accountability of the administrations involved may usefully contribute to improving the regulatory environment.

Summing up

Mexico's economy has recovered strongly from the 1995 recession but, mainly as a result of external shocks, 1998 was again a difficult year. The disinflation process reversed and output growth started slowing, as lower oil prices and the international financial crisis prompted the authorities to tighten monetary and fiscal policy in the framework of a floating exchange-rate regime. This combination of currency flexibility and macroeconomic adjustment was, on the whole, well received by the market: it allowed the authorities to preserve confidence and limit the slowdown in output growth. But inflation rose, exceeding its target. Early in 1999, the Brazilian crisis had only a relatively minor and short-lived impact; however, with international financial markets remaining somewhat unsettled, the outlook is uncertain. Mexico faces the challenge of, first, putting the economy back on a course of steady disinflation; and then, securing strong sustainable output and employment growth to absorb the rapid increase in the labour force and further progress toward social goals in the areas of education, health and poverty alleviation. But until the external constraint loosens and inflation is brought firmly under control, Mexico will have to pursue a tight monetary and fiscal policy, in a coherent and well spelled-out medium-term framework. It would also seem urgent to strengthen tax

revenue through a broadening of the tax base – by reducing preferential regimes to a minimum – and increasing the effectiveness of the tax collection system. This would enhance public saving, and hence make it possible to limit recourse to foreign saving to the amount that can be financed through relatively stable capital flows. It would also allow the authorities to preserve, if not increase, public spending in areas such as physical infrastructure and human capital development, where the social and economic returns are potentially high. Moreover, to allow the economy to better cope in this difficult environment, structural reform should continue, especially in the financial sector, where a decision on a number of items of the reform long submitted to Congress has still not been taken.

I. Recent trends and short-term prospects

The current economic cycle in perspective

The economic recovery from the 1994 peso crisis continued at a rapid pace in the first half of 1998, stimulated by buoyant private spending (Table 1). However, signs of a slowdown emerged in the second half of the year, as both fiscal policy and monetary policy were tightened in response to the global financial crisis and the sharp fall in oil prices. The depreciation of the peso during 1998 helped to limit the deterioration of the external balance, but led to a reversal in the disinflation process (Figure 1). When put in a medium-term perspective, these developments suggest a mixed picture of Mexico's performance: at the end of 1998, after three years of recovery, output per capita was just 3 per cent above its pre-crisis level; employment in the formal sector was some 12 per cent above levels in 1994, while real wages were still about 20 per cent below those prevailing in 1994. The three-year old expansion is likely to have buoyed income of the self-employed and corporate profits.[1]

A number of features differentiate the recovery since 1995 from the one following the 1982 debt crisis. In particular, in the current cycle exports have not been the only engine of growth: investment in export-oriented firms has also expanded at very rapid rates, followed more recently by private consumption. Moreover, the longevity of the export boom has been different. While in the initial stages of both cycles Mexican producers were able to re-orient their activities towards the export sector (Figure 2), in 1982-85 the export boom fizzled out as soon as domestic demand started to recover. This was due to the fact that productive capacity had not been increased sufficiently and the manufacturing export base was narrow.[2] In the current upswing, on the other hand, high levels of capacity created by several years of buoyant investment in export-oriented firms have allowed the rapid expansion of exports to continue even as domestic demand picked up. This may be a harbinger of the sustainability of the current cycle and a sign that the Mexican economy has become more responsive. Another positive indication is that the investment boom and the recovery of the economy

Table 1. **Demand and output**

Percentage changes, volume, 1993 prices

	1994[1] current prices	1989-94[2]	1995	1996	1997	1998[3]	1998 January-September[4]
Demand							
Private consumption	1 016.1	4.9	–9.5	2.2	6.3	6.3	7.9
Government consumption	164.2	3.0	–1.3	–0.7	1.8	–2.0	–1.9
Gross fixed capital formation	274.9	7.8	–29.0	16.4	20.9	11.4	14.8
Public sector	53.3	3.2	–19.7	20.2	4.3	–2.0	–13.2
Private sector	221.5	9.1	–31.2	15.3	25.7	14.6	20.2
Final domestic demand	**1 455.2**	**5.1**	**–12.3**	**4.0**	**8.3**	**6.4**	**8.2**
Change in stockbuilding[5]	36.7	0.2	–2.2	1.6	0.9	–0.3	–1.3
Total domestic demand	**1 491.9**	**5.2**	**–14.1**	**5.6**	**9.0**	**5.9**	**6.5**
Exports of goods and services	239.0	7.8	30.2	18.2	13.0	11.2	10.9
Imports of goods and services	307.5	15.9	–15.0	22.8	22.0	15.5	16.2
Change in foreign balance[5]	–68.5	–1.3	8.5	–0.2	–1.7	–1.0	–1.0
GDP at market prices	**1 423.4**	**3.9**	**–6.2**	**5.2**	**7.0**	**4.8**	**5.3**
Output							
Agriculture, forestry, fishing	78.2	1.8	0.9	3.6	0.2	0.5	–1.1
Mining (including petroleum)	17.4	1.6	–2.7	8.1	4.5	3.4	4.7
Manufacturing	245.0	4.3	–4.9	10.8	10.0	7.4	8.6
Construction	69.1	5.7	–23.5	9.8	9.3	4.6	5.7
Electricity	19.2	3.0	2.1	4.6	5.2	4.7	5.1
Commerce	275.7	4.8	–15.5	4.8	10.6	4.2	6.4
Transport and communication	124.8	5.0	–4.9	8.0	9.9	10.1	10.6
Financial services	211.5	4.7	–0.3	0.6	3.7	3.8	4.2
Community services	311.0	2.8	–2.3	1.0	3.3	2.6	3.1

1. Billion pesos.
2. Annual average growth rate.
3. Demand components are OECD Secretariat estimates. GDP by sector is actual data.
4. Changes from same period of previous year.
5. As a percentage of GDP in the previous period.

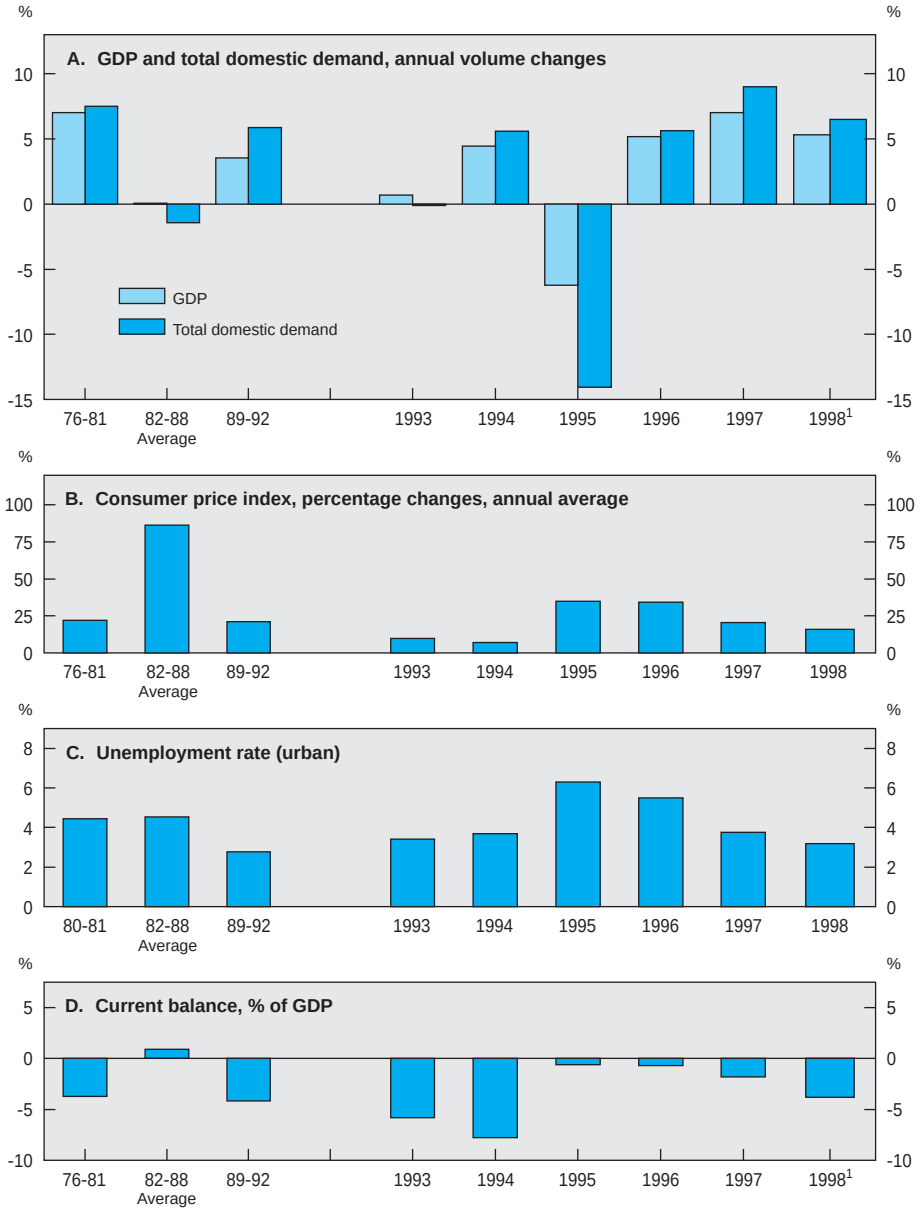
Source: OECD Secretariat.

have proceeded despite severe problems in the banking sector and the near drying-up of bank credit (Chapters II and IV). Some of the financing has come from external sources, but non-bank domestic sources seem to have predominated in 1998: according to business surveys in manufacturing firms, suppliers' credit and reinvested profits have been the most important of these sources.[3]

The economy is slowing down

Boosted by buoyant domestic demand, real output growth reached 7 per cent in 1997 (Figure 3), but it decelerated to 4.8 per cent in 1998. This deceleration took place despite a household consumption boom and the continued

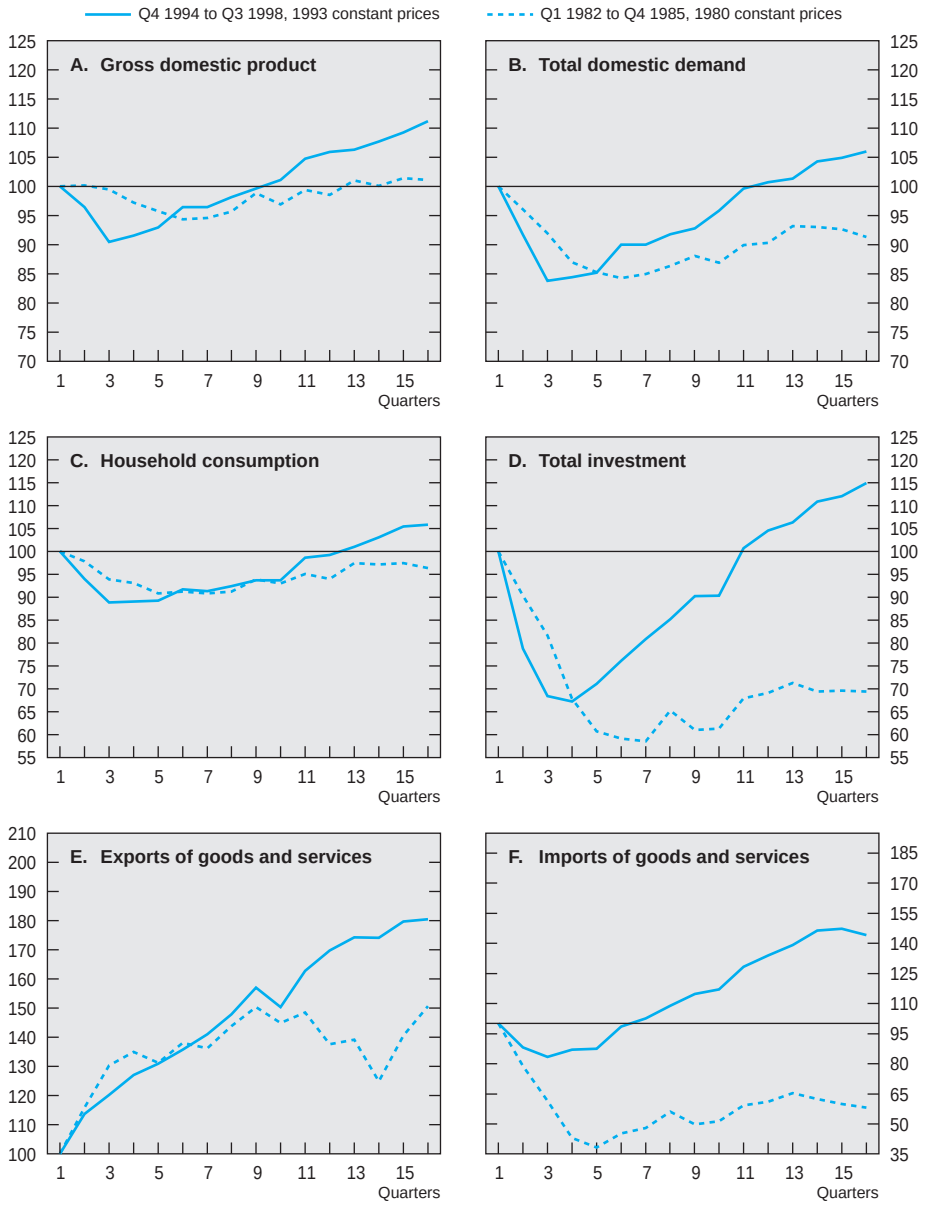
Figure 1. **Key aspects of economic activity**



1. January-September 1998 over January-September 1997.
Source: OECD Secretariat and INEGI.

Figure 2. **Comparison of two cycles**

Volume indices of seasonally-adjusted data



Source: OECD Secretariat.

strength of business investment, which resulted in private domestic demand growing by 10 per cent. But real exports lost buoyancy, in line with slower market growth, and the negative contribution of the foreign balance to growth remained substantial. Public demand has been sluggish since the middle of 1997, as a result of cutbacks in budget spending in response to falling oil revenues. Some signs of a slowdown in private domestic demand emerged in the latter part of 1998, as a result of policy tightening and the negative impact on confidence of global financial turbulence.[4]

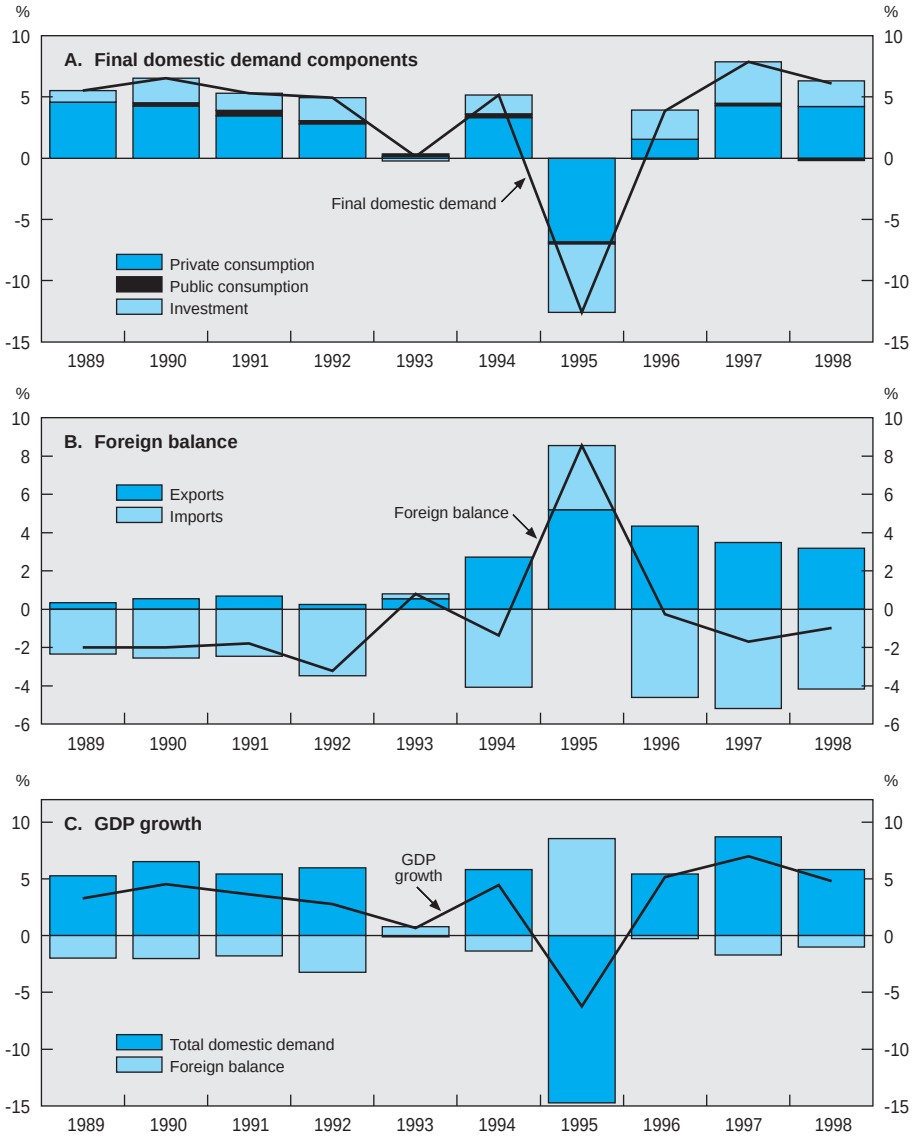
Gross fixed investment has expanded by a cumulative 50 per cent in real terms from its trough in late 1995. Since the middle of 1997, the investment boom has been entirely driven by private investment, while public investment has been curtailed. The three fiscal packages announced between January and July 1998 in response to the fall in oil-related revenues led to a decline of public investment of some 13 per cent in real terms over the first three quarters of the year. PEMEX, the state oil company, and CFE, the federal electricity commission, accounted for the bulk of the budget cuts.[5] Private investment on the other hand has been buoyed by improved profits, strong demand factors and the incipient recovery of bank credit: it has thus remained the most dynamic demand component, with a growth rate of some 20 per cent in the first three quarters of 1998. Within private investment, demand for machinery and equipment increased again at above-average rates, particularly in firms oriented to the domestic market, which had been lagging in the recovery. Both domestic activity and imports of equipment goods have benefited from this investment boom (Table 2). Finally, it should be noted that, since 1997, some investment programmes have been undertaken under the initiative of the public sector with financing by the private sector.[6] These projects have aimed at building capacity for public enterprises – such as the oil company, PEMEX, and the electricity company, CFE – as well as gas pipelines and highway sections: they have helped support construction activity otherwise hit by budget cuts.

Total investment (including stockbuilding) was close to 25 per cent of GDP in 1997 and 1998, 3 percentage points higher than in 1994 (Table 3). After rising during the past three years, domestic saving declined somewhat in the first half of 1998. Nonetheless, at 21 per cent of GDP, it was about 6 percentage points higher than its pre-crisis level – entirely due to stronger private saving, since public saving was virtually unchanged. Foreign saving (reflecting the external deficit), at a little more than 3 per cent of GDP, was significantly lower than in 1994. The strong rebound of private domestic saving has undoubtedly contributed to sustaining the investment boom without creating undue pressures on the external account.

After a rapid increase in 1997, private consumption picked up further in 1998, growing by 8 per cent (in the first three quarters over a year earlier). While

Figure 3. **Contributions to real GDP growth**[1]

Percentage points



1. Changes as a percentage of real GDP in the previous period. OECD Secretariat estimates for 1998.

Source: OECD Secretariat.

Table 2. **Indicators of investment activity**

Percentage changes from same period of previous year[1]

	1995	1996	1997	1997		1998		
	Average	Average	Average	Q3	Q4	Q1	Q2	Q3
Gross fixed capital formation	–29.0	16.4	20.9	22.8	18.3	22.9	11.4	10.2
Machinery and equipment	–36.3	23.0	34.4	38.2	29.6	36.8	20.5	16.2
Residential building and construction	–22.7	11.6	10.2	11.4	8.0	10.2	2.6	4.6
Domestic production of machinery and equipment	–36.1	20.5	26.7	33.0	22.5	36.9	33.1	23.7
Imports of capital goods	–36.6	25.2	40.5	38.2	31.7	36.9	12.6	10.8

1. Volume at 1993 constant price basis.
Source: INEGI.

Table 3. **Aggregate saving and investment**

Percentage of GDP

	1993	1994	1995	1996	1997	1998 S1[1]
Gross capital formation	21.0	21.7	19.8	23.3	26.4	24.5
Gross domestic saving	15.1	14.8	19.3	22.6	24.6	21.3
Private	10.0	10.8	14.5	18.0	21.1	18.0
Public[2]	5.1	4.0	4.8	4.6	3.5	3.3
Foreign saving	5.9	7.0	0.6	0.7	1.9	3.2

1. Preliminary.
2. The public sector comprises federal government and enterprises under budgetary control. The estimate is based on Ministry of Finance accounts.
Source: Banco de México; Ministry of Finance; OECD Secretariat.

durable goods were by far the most dynamic component of total consumption, non-durables and services also grew significantly.[7] The strength of consumption has its roots in the growth of total earnings. Employment has increased strongly – especially until mid-1998 – and real wages have risen continuously since mid-1997. Hence, the real wage bill has grown rapidly, with its manufacturing component moving up by 6.5 per cent in January-September 1998 from a year earlier.[8] This development, although not sufficient to restore overall wage income to pre-crisis levels, has buoyed consumer confidence. In addition, although data are not available, the income of the self-employed is likely to have benefited significantly from the three-year-old expansion, contributing to the growth of household disposable income. The "feel-good" factor and pent-up demand for consumer goods seem to have boosted household spending between early 1997 and mid-1998, depressing the private saving ratio. However, private consumption lost some strength in the second half of the year. Public consumption has been

rather sluggish. After a slight increase in 1997, it fell by 2 per cent in the first three quarters of 1998. Budget adjustments during 1998 included a hiring and salary freeze in the public sector and cuts in operating expenses across Ministries.

The labour market has improved

The labour market has improved continuously since 1996: by mid-1998, the number of insured employees – an indicator of activity in the formal sector –

Table 4. **Labour market indicators**

	1993	1994	1995	1996	1997	1998	1998 S1	1998 S2
	Thousands	Percentage changes from same period of previous year						
Insured employment (private sector)	11 318	1.1	–4.2	3.7	8.0	8.0	7.4	8.7
Permanent	10 033	1.4	–1.6	4.8	8.9	5.6	6.9	4.4
of which:								
Manufacturing	2 967	–0.8	–2.4	7.0	12.9	12.2	13.9	10.4
Trade	1 704	1.4	–3.1	1.4	7.2	8.8	9.1	8.6
Personal services	1 671	1.9	–1.9	2.7	11.6	10.4	11.4	9.4
Temporary	1 286	–1.4	–24.4	–5.8	–2.1	44.7	13.4	76.0
Employment in manufacturing								
Non-maquiladoras		–3.1	–9.1	2.7	4.9	3.0	3.5	2.4
Maquiladoras		7.6	11.2	16.4	19.1	12.2	13.8	10.6
	Per cent							
Urban areas[1]								
Participation rate[2]	55.2	54.7	55.4	55.4	56.2	56.6	56.5	56.6
Unemployment rate	3.4	3.7	6.2	5.5	3.7	3.2	3.4	3.0
Unemployment rate (wider definition)[3]	5.6	6.1	8.6	6.4	4.6	4.2	4.3	4.0
Employment in small enterprises/total employment[4]	42.3	42.2	44.2	44.6	43.7	42.9	43.4	42.3
Uninsured wage earners/total wage earners	21.3	22.5	24.9	29.6	24.7	23.5	23.4	23.7
Part-time workers/total employment[5]	24.9	23.9	25.4	25.5	25.0	23.6	25.4	21.8

1. The *Encuesta Nacional de Empleo Urbano* covers a limited number of urban areas (initially 16, raised to 32 in 1992, and gradually thereafter to 43, as of 1997).
2. Labour force ("Economically active" population) as a percentage of population aged 12 year and over.
3. Open unemployment *plus* people who gave up seeking employment and are no longer included in the labour force but are available for work.
4. Small enterprises are defined as those with 1 to 5 employees.
5. Part-time workers are defined as those working less than 35 hours weekly.

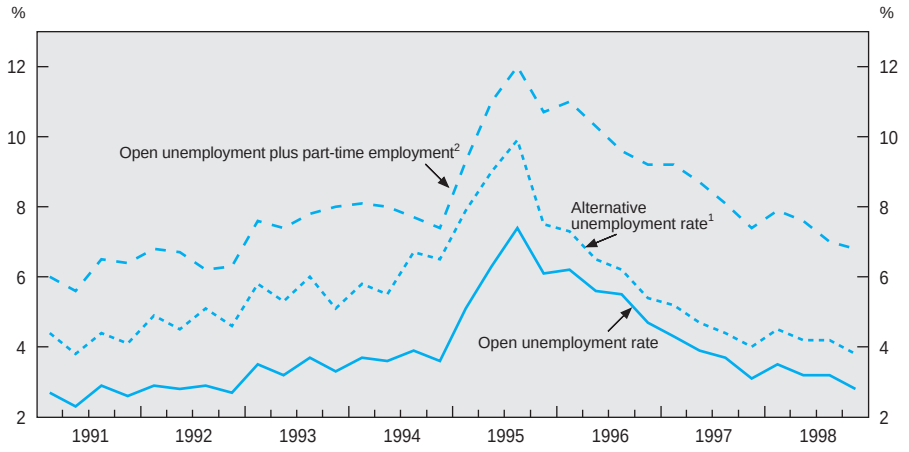
Source: INEGI, *Encuesta Industrial Mensual, Estadistica de la Industria Maquiladora de Exportación, Encuesta Nacional de Empleo Urbano (ENEU)*; IMSS.

was 14 per cent above pre-crisis levels. After mid-year, however, affiliation of permanent workers at the social security institute, IMSS – which had been increasing at rates above 7 per cent – started to decelerate, pointing to a slowdown in the economy. The manufacturing sector remained by far the most important source of job creation in 1998. Employment growth in the *Maquiladora* sector (in-bond industries) decelerated somewhat, after mid-1998, but at more than 10 per cent year-on-year, it continued to outpace the rest of the economy and, with 1 million workers, the sector now accounts for about a third of total manufacturing employment (up from 20 per cent in 1993). Other manufacturing enterprises and the service sector have also recorded strong – though decelerating – employment growth.[9]

In spite of a fast growing labour force – boosted by a rapid expansion of the working-age population and rising female participation rates – open unemployment has continued to decline.[10] The urban unemployment rate (the best available indicator of "open" unemployment) fell to 2.8 per cent in the last quarter of 1998, compared with 3.1 per cent a year earlier; at its peak in August 1995, it had reached 7.6 per cent.[11] Open unemployment fell in all of the three main urban areas surveyed – Mexico City, Guadalajara and Monterrey – with the last one recording both the lowest rate and the most significant drop.

Figure 4. **Unemployment in urban areas**

As a percentage of labour force (12 years and over)



1. Includes those who stopped searching for a job but are still available for work.
2. Part-time refers to employees working less than 15 hours a week.

Source: INEGI.

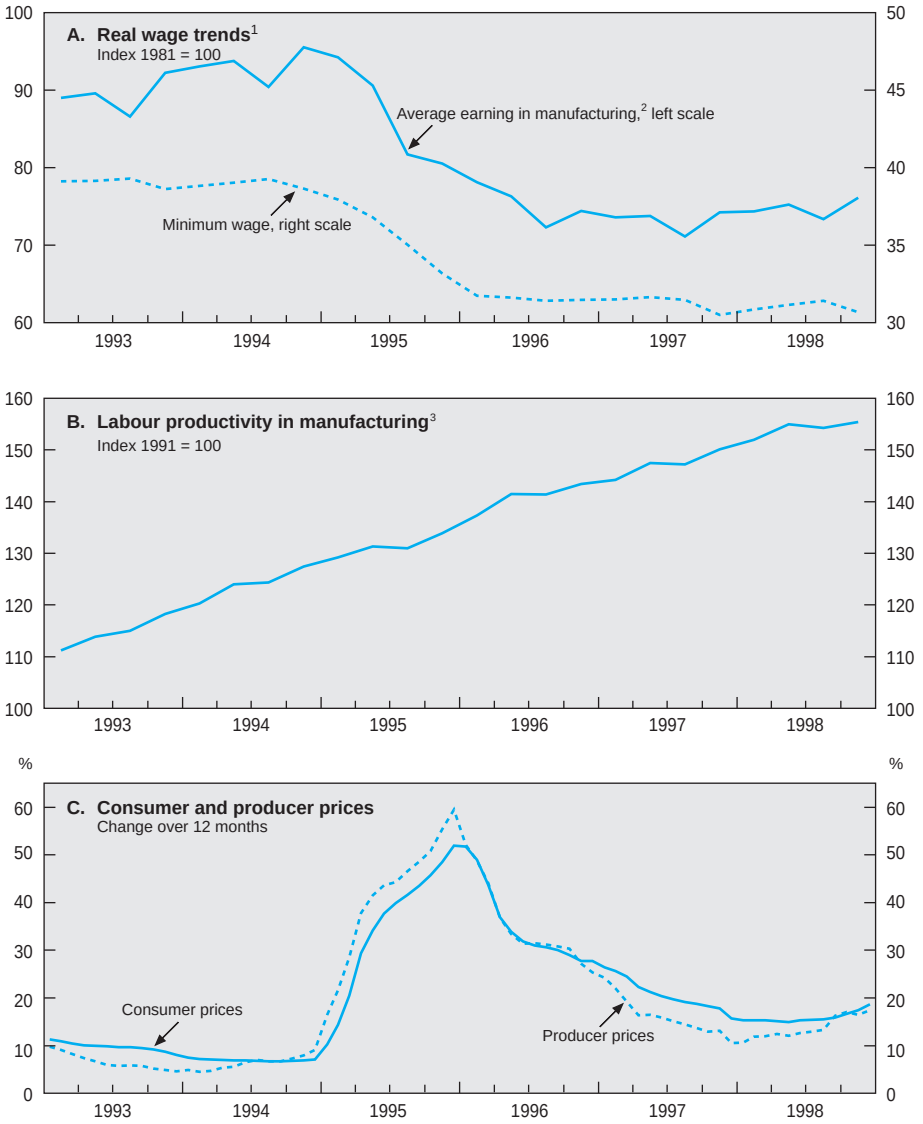
Alternative and broader measures of unemployment, which also count discouraged workers and part-time employees, have been following a parallel downward trend (Figure 4).

The disinflation process reversed in 1998

The process of disinflation, which had started in mid-1995, reversed in 1998, as the effects of currency depreciation began to be felt. Other factors which affected inflation during 1998 were: increases in publicly-controlled prices (including, *inter alia*, gasoline, electricity and tortillas), as subsidies were phased out; and weather-related disruptions in agricultural production, which led to a jump in fresh food prices. Increases in the price of tradables started to accelerate in April, following a drop of 8.4 per cent in the peso exchange rate *vis-à-vis* the US dollar (6.7 per cent in effective terms) in the six months to February 1998. On the other hand, wage moderation at the start of the year, helped to foster a deceleration of non-tradable prices, so that the national consumer price index (CPI) remained on a slightly downward trend until May, when it bottomed at 15 per cent (year-on-year)[12] (Figure 5, Panel C). A second bout of exchange rate weakness related to the Russian crisis saw the peso depreciating by a further 15.3 per cent *vis-à-vis* the dollar (16.1 per cent in effective terms) over the six months to October. In this episode, price rises in the non-tradable sector accelerated, fuelled by increased inflation expectations (as proxied by an upward revision of private-sector inflation forecasts[13]), which also contributed to higher nominal wage settlements. As a result, CPI-inflation increased to 18.6 per cent in the twelve months to December 1998, more than 6 percentage points above the Bank of Mexico's 12 per cent target. Upward adjustments in government-controlled prices in November contributed to the upturn in inflation, despite the dampening impact of fiscal and monetary policy tightening (Chapter II), which restrained domestic demand somewhat and curbed exchange rate depreciation and the rise in inflation expectations.

Real wages, which had turned up only after mid-1997, continued to grow in 1998. Over the January-September period, real average earnings in manufacturing increased by 3.5 per cent compared with the same period of 1997, although in September 1998 they were still about 20 per cent below 1994 levels. Some sectors, such as construction and retail trade, which had been lagging in this recovery, saw even higher real wage increases, in the 4 to 6 per cent range. Real average earnings in the in-bond exporting sector *(Maquiladora)*, where skills and salaries tend to be lower and turnover higher than in the rest of the economy, rose at relatively high rates – above 6 per cent in the second half of 1998. The traditional gap between earnings in this sector and domestic-oriented firms has been narrowing since the early stages of the cycle, when growth was led mainly by

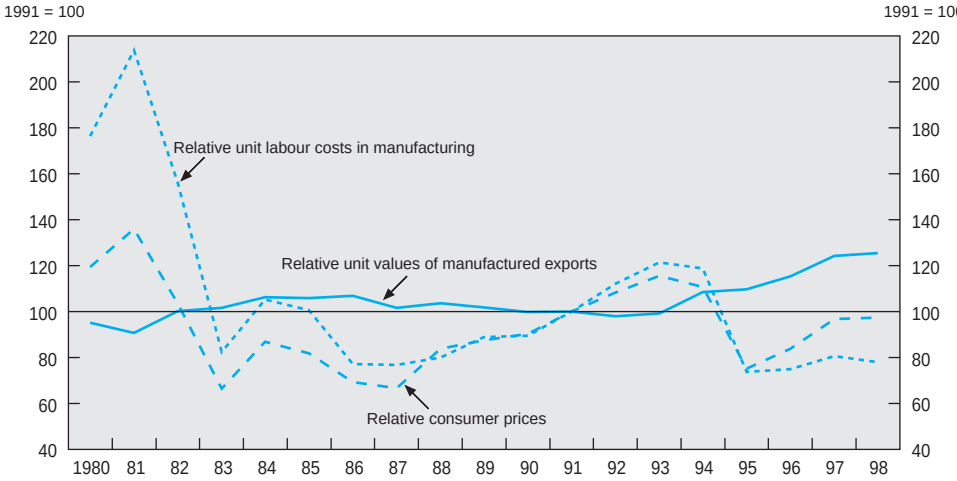
Figure 5. **Real wages, labour productivity and price trends**



1. Moving average over three quarters.
2. Wages, salaries and social benefits, based on INEGI monthly industrial survey.
3. Real output per man-hour, moving average over three quarters.

Source: INEGI and Banco de México.

Figure 6. **Mexico's relative competitive position**
Indices in US$ terms



Source: OECD Secretariat estimates.

external demand.[14] Real wage growth in manufacturing was more than offset by persistently high productivity growth, which reached 4.7 per cent in January-September 1998, so that real unit labour costs declined somewhat. Mexico's international competitive position was helped by the peso depreciation, although it seems to have changed little in 1998. Various indicators provide a mixed picture of Mexico's cost-price competitiveness compared with the early 1990s. While relative unit labour costs in manufacturing point to an improvement in competitiveness, relative consumer prices suggest a virtually unchanged position, and relative unit values of manufactured exports show a deterioration[15] (Figure 6).

The external balance has deteriorated

Reflecting buoyant domestic demand and the drop in oil prices, the trade balance deteriorated significantly and the current-account deficit widened in 1998. The capital account was severely affected by the turmoil in international financial markets, especially in the third quarter, when net capital inflows dwindled, putting pressure on the peso. The tightening of economic policy and the consequent slowing of the economy, combined with the growing response of

trade flows to the depreciation of the peso and a respite on international financial markets, may have stopped or reversed these trends in the final quarter of 1998. The new episode of financial turbulence originating in Brazil in January 1999 seems to have had only a relatively minor and short lived impact on capital flows.

Current-account developments

The adjustment in Mexico's current account was swift in the aftermath of the 1994 peso crisis, the deficit falling from 7 per cent of GDP in that year to less than 1 per cent in both 1995 and 1996. Starting in 1996 however, as domestic demand recovered, import growth picked up, leading to a steady deterioration in the trade balance, which moved into a deficit in the second half of 1997. Despite increased tourism revenues, the current-account deficit widened gradually (Table 5). Starting in October 1997, Mexico's trade balance has also been affected by falling oil prices. In 1998, this factor resulted in an estimated shortfall in export revenues equivalent to 1.2 per cent of GDP.[16] Its impact on the trade balance was, however, partly offset by a significant increase in non-oil exports and a reduction in import growth as a result of less buoyant domestic demand. The trade deficit reached about 2 per cent of GDP in 1998.

Table 5. **Current external account**

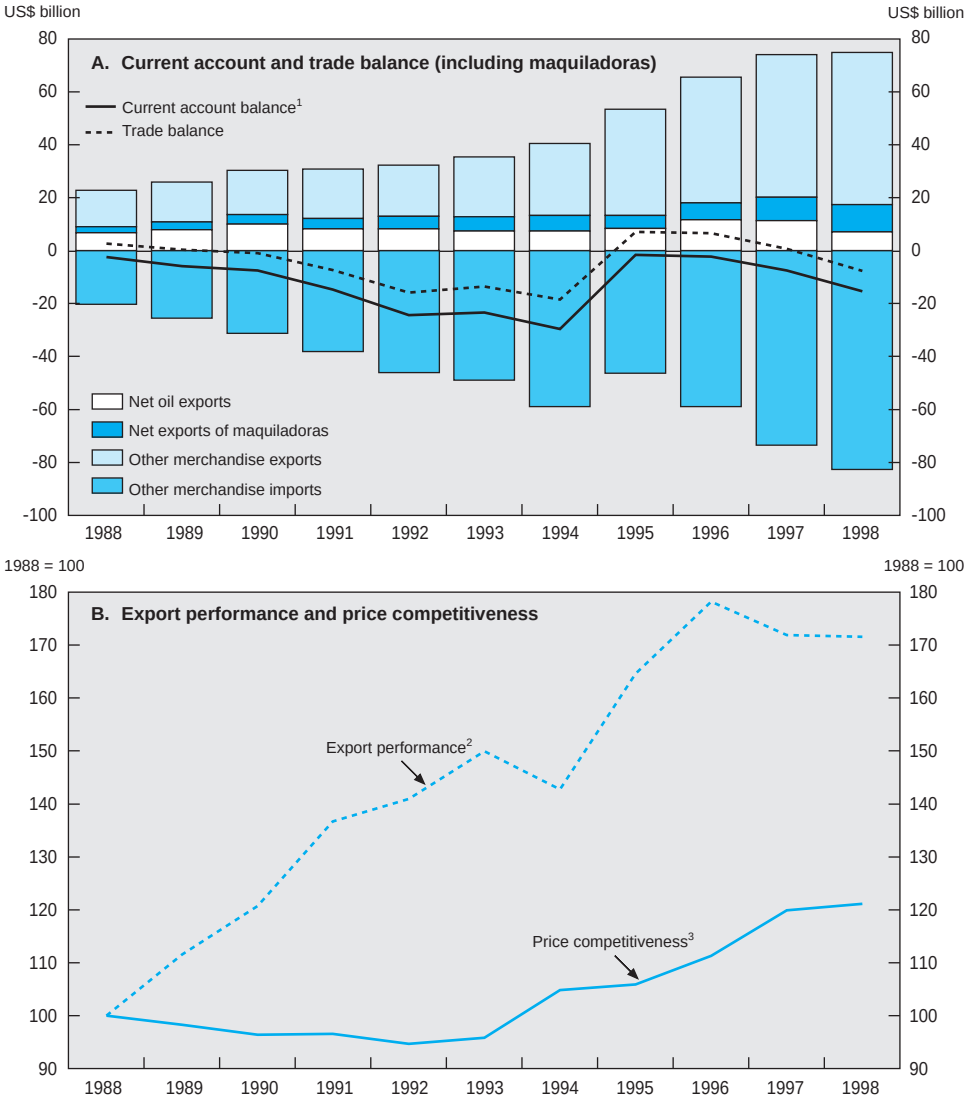
US$ billion

	1994	1995	1996	1997	January-September[2]	
					1997	1998
Exports, f.o.b.[1]	60.9	79.5	96.0	110.4	80.7	86.7
of which:						
Oil	7.4	8.4	11.7	11.3	8.5	5.6
Non oil	53.4	71.1	84.3	99.1	72.2	81.1
Imports, f.o.b.[1]	79.3	72.5	89.5	109.8	78.8	91.9
Trade balance	**–18.5**	**7.1**	**6.5**	**0.6**	**1.9**	**–5.2**
(% GDP)	(–4.4)	(2.5)	(2.0)	(0.2)	(0.6)	(–1.7)
of which: "Maquiladoras"	5.8	4.9	6.4	8.8	6.4	7.6
Non factor services, net	–2.0	0.7	0.5	–0.5	0.0	–0.5
of which: Tourism	1.0	3.0	3.5	3.7	2.9	2.8
Investment income, net	–13.0	–13.3	–13.9	–12.8	–9.7	–10.2
Transfers, net	3.8	4.0	4.5	5.2	4.0	4.4
Current balance	**–29.7**	**–1.6**	**–2.3**	**–7.4**	**–3.8**	**–11.5**
(% GDP)	(–7.0)	(–0.6)	(–0.7)	(–1.9)	(–1.3)	(–3.7)

1. Including trade by "Maquiladoras" (in-bond industries).
2. For 1998 as a whole, total exports reached US$117.5 billion, of which 7.1 billion on account of oil, while total imports totalled 125.2 billion. The trade deficit reached US$7.7 billion.

Source: Banco de México.

Figure 7. **Foreign trade, current account and export performance**



1. January-September 1998 annualised data, for the current account balance; 1998 full year for the trade balance and its components.
2. Ratio of Mexico's exports of manufactures to its main partners' imports of manufactures in volume terms.
3. Relative export unit value of manufactured goods. A rise indicates a deterioration.

Source: Banco de México; OECD Secretariat estimates.

Non-oil exports, which were still strong in early 1998, decelerated throughout the year, as Mexico's main export markets slowed[17] (Figure 7). Nonetheless, led by the machinery and equipment-producing sector of the *Maquiladora* industry, manufacturing exports continued to increase rapidly, their dollar value rising by 11.7 per cent in 1998 from a year earlier. This, as noted, helped offset the drop in oil-related exports.[18]

The cooling off of the economy during 1998 and the depreciation of the peso had a dampening impact on merchandise imports, which decelerated through the year from the exceptionally high rates reached in the previous two years. After recording a 21 per cent increase in the first half of 1998, imports grew by a more moderate 8 per cent in the second half (based on dollar values). Strong exports continued to require large purchases of foreign intermediate goods, the import content of exports being high in Mexico.[19] Fast-growing private investment and the consumption boom boosted imports of capital goods and consumer durables until July. Particularly buoyant were automobile imports, which rose by 84 per cent in dollar terms in the first eight months of 1998, but represent still a very small share of total imports. Given the drop in oil export prices and the depreciation of the exchange rate, Mexico suffered a significant terms-of-trade deterioration in 1998.

The deficit in the invisibles account (services and transfers, net) narrowed sharply after the 1994 peso crisis, falling from US$11.2 billion that year to US$8.7 billion in 1995, reflecting increased tourism receipts and workers remittances. Since then, the invisible deficit has remained broadly stable. In the first three quarters of 1998, the non-factor service balance deteriorated somewhat, reflecting an increase in insurance and freight costs, and there was a small widening of the deficit on net investment income from its post-crisis low in 1997 – as lower public interest payments (the fruit of external debt restructuring) were more than offset by higher private payments and increased dividend payments on foreign direct investment (FDI). On the other hand, net tourism revenues remained strong, despite an increase in expenditures by Mexican tourists abroad, and net transfers received (workers remittances for the most part) increased somewhat. All considered, the current-account deficit widened from 1.9 per cent of GDP in 1997 to around 4 per cent in 1998, according to preliminary estimates.

Capital account

Over the past few years, Mexico has recorded sizeable net capital inflows, mostly accounted for by foreign direct investment. The capital account remained in surplus between January and September 1998, despite a drying up of net inflows in the third quarter, when the global financial turbulence intensified (Table 6). Portfolio investment flows, in particular, turned negative after mid-1998, as the placement of Mexican foreign currency securities was curtailed by the

Table 6. **Capital account and balance of payments**

US$ billion

	1994	1995	1996	1997	Jan.-Sep. 1998	1998 SI	1998 Q3
Capital account	**14.6**	**15.4**	**4.1**	**15.4**	**7.5**	**6.8**	**0.7**
Liabilities	**20.3**	**22.8**	**10.4**	**8.7**	**7.4**	**8.5**	**–1.1**
Loans and deposits	1.1	23.0	–12.2	–8.8	0.8	1.0	–0.2
Public sector	–0.4	11.5	–8.9	–6.1	–1.7	–1.0	–0.7
Development banks	1.3	1.0	–1.2	–1.0	–0.4	–0.4	0.0
Non-financial public sector	–1.7	10.5	–7.7	–5.0	–1.3	–0.6	–0.7
Bank of Mexico	–1.2	13.3	–3.5	–3.5	–0.4	0.0	–0.4
Private sector	2.7	–1.8	0.2	0.7	2.9	2.0	0.9
Commercial banks	1.5	–5.0	–1.7	–2.0	1.2	0.8	0.4
Non-financial private sector	1.2	3.1	2.0	2.7	1.8	1.3	0.5
Total foreign investment	19.2	–0.2	22.6	17.5	6.5	7.5	–0.9
Direct investment	11.0	9.5	9.2	12.5	6.9	4.9	2.0
Portfolio investment	8.2	–9.7	13.4	5.0	–0.3	2.6	–2.9
of which:							
Stock market	4.1	0.5	2.8	3.2	–1.0	–0.8	–0.2
Money market	–2.2	–13.9	0.9	0.6	–0.1	0.4	–0.5
Foreign currency securities	6.3	3.6	9.7	1.2	0.7	3.0	–2.2
Assets	**–5.7**	**–7.4**	**–6.3**	**6.7**	**0.2**	**–1.7**	**1.8**
In banks abroad	–3.7	–3.2	–6.1	4.9	–0.8	–2.1	1.2
Credits to non-residents	0.0	–0.3	–0.6	–0.1	0.2	0.1	0.1
External debt garantees	–0.6	–0.7	0.5	–0.7	–0.6	–0.4	–0.2
Other	–1.3	–3.3	–0.2	2.7	1.4	0.6	0.7
Memorandum items:							
Current account	–29.7	–1.6	–2.3	–7.4	–11.5	–6.8	–4.7
Capital account	14.6	15.4	4.1	15.4	7.5	6.8	0.7
Errors and omissions	–3.3	–4.2	0.0	2.5	4.5	1.8	2.7
Changes in net international reserves (increase = –)	18.4	–9.6	–1.8	–10.5	–0.5	–1.8	1.3

Source: Banco de México.

worsening of risk perception regarding emerging markets. Purchases of Mexican stocks by foreign investors were affected already early in the year. Foreign direct investment, which typically is less affected than financial flows by short-term monetary considerations, continued to show a net inflow in the third quarter. Nonetheless, for the year as a whole, this inflow may have been lower than in 1997 – when it reached a record US$12.5 billion. Net inflows related to other private sector liabilities continued to grow, reaching US$3 billion in January-September. Furthermore, the surplus in "errors and omissions" – an item often regarded as largely representing capital movements – widened to US$4.5 billion, somewhat surprisingly in view of the downward pressure on the peso. Even though complete

data are not yet available, there are signs suggesting that in the last quarter of 1998, as financial tensions abated and international capital markets reopened to new borrowing by Mexico – including a large public sector (PEMEX) debt placement – portfolio investment inflows may have resumed. Throughout 1998, the exchange rate of the peso was allowed to bear some of the brunt of incipient capital outflows, and the Bank of Mexico used only a small proportion of its reserves to reduce foreign exchange volatility (Chapter II). Reserve accumulation reached US$1.8 billion in the first half of 1998; after turning negative in the third quarter, it was positive again in the fourth – resulting in a net accumulation of US$0.4 billion in the second half of the year. For 1998 as a whole, the sum of recorded net capital inflows and the surplus in the "errors and omissions" item may have been close to that of 1997 – when it amounted to US$18 billion – and it apparently exceeded the current-account deficit, since net international reserves increased by a little over US$2.1 billion.

The short-term outlook

Global financial turbulence, higher short-term interest rates and successive budget cuts throughout 1998 are likely to reduce real GDP growth in the period ahead. Signs of a slowdown of the economy became manifest in the second half of 1998, following a major policy tightening in response to falling oil prices and pressures on the exchange rate (Chapter II). In the OECD Secretariat projections prepared in November 1998 (and published in *Economic Outlook 64* of December 1998), real GDP growth was projected to slow to 3½ per cent in 1999, reflecting restrained domestic demand and weaker manufacturing exports. Since then oil prices have fallen further, but economic growth in the United States has been stronger than anticipated. In January 1999, international financial markets, after a short respite in the later part of 1998, were unsettled again by the Brazilian crisis. Although the impact on Mexico was much less severe than after the Russian crisis and was quickly reversed, the external environment remains uncertain. On balance the *Economic Outlook 64* projections now seem somewhat optimistic and a downward revision to output growth in 1999 would seem warranted.[20] The Mexican official projections, which put GDP growth at 3 per cent in 1999, point in the right direction. The slowdown will essentially reflect lower consumer spending in response to growing uncertainties about employment and real wages, cuts in public spending and less buoyant business fixed investment, as a result of high real interest rates, and the deterioration of export prospects. Under the assumption of a progressive return of confidence in international financial markets, real GDP growth could pick up in 2000, and disinflation could resume once the first-round effects of the recent depreciation have worked through.

Uncertainties surrounding the outlook are greater than usual. These relate mostly to developments abroad, notably in emerging markets, and Brazil in particular. Although direct effects on Mexico's trade from a recession in Brazil would be limited, a deepening of the Brazilian crisis and prolonged volatility in international financial markets could force the Mexican authorities to maintain a tight policy stance for longer than anticipated, or even to take further corrective measures to preserve confidence, reduce inflation and keep the current-account deficit in line with available financing. Although inward foreign direct investment showed considerable resilience in 1998, portfolio investment turned negative in the third quarter. The size and nature of capital flows are difficult to predict, but net inflows may be smaller than in the past couple of years, as risk perception is likely to subside only gradually, especially after the new flare up in early 1999. Nevertheless, compared with the situation in 1994, the exchange rate is now floating and the authorities are committed to taking action to stabilise the economy. The last year has thus shown that it is now possible to deal with external shocks in a more timely and, hence, more effective way than in the past.

II. Economic policies

Overview

In 1998, the Mexican authorities had to face a more adverse environment than expected when the policy guidelines for the year were prepared in November 1997. At that time, confidence was improving and despite some turbulence in international financial markets, contagion to Mexico was relatively limited: real GDP growth was strong and evenly spread across the economy, inflation was coming down steadily, and the current-account deficit, though widening, was still moderate. The government's National Programme for Financing Development, 1997-2000 (PRONAFIDE), presented in June 1997, sketched a cautious scenario for output growth and disinflation to the year 2000. In line with this medium-term strategy, the authorities presented a prudent budget for 1998 to secure a further decline in the debt/GDP ratio – which, at 25 per cent, was already low compared with most other OECD countries. The monetary programme for 1998 focused on bringing down inflation gradually and fostering stable financial markets. However, global financial turbulence and lack of confidence in emerging markets continued in the first half of 1998, and greatly increased after the Russian debt crisis in August; moreover, the fall in oil prices continued. In response, the Mexican authorities tightened policies forcefully. This policy reaction was broadly effective in preserving confidence and helping to stabilise the exchange rate of the peso – albeit at a significantly lower level. It did not succeed, however, in preventing a rise in inflation. After a lull towards the end of 1998, international financial markets were unsettled again by the Brazilian crisis in January 1999, but the impact was much smaller and more short-lived than after the Russian crisis. Nonetheless, the international environment remains uncertain and, given the need to bring inflation firmly under control, economic policies in Mexico will have to remain restrictive for some time, and the possibility of additional corrective measures cannot be ruled out.

The present chapter first reports on monetary policy and financial market developments. The following paragraphs review the broad lines of fiscal policy and debt management, highlighting the vulnerability of the budget to exogenous shocks and the challenges that fiscal policy will be facing in the longer run.

Chapter III reviews the main features of the tax system, and discusses related policy requirements, while progress in several areas of structural reform is considered in Chapter IV.

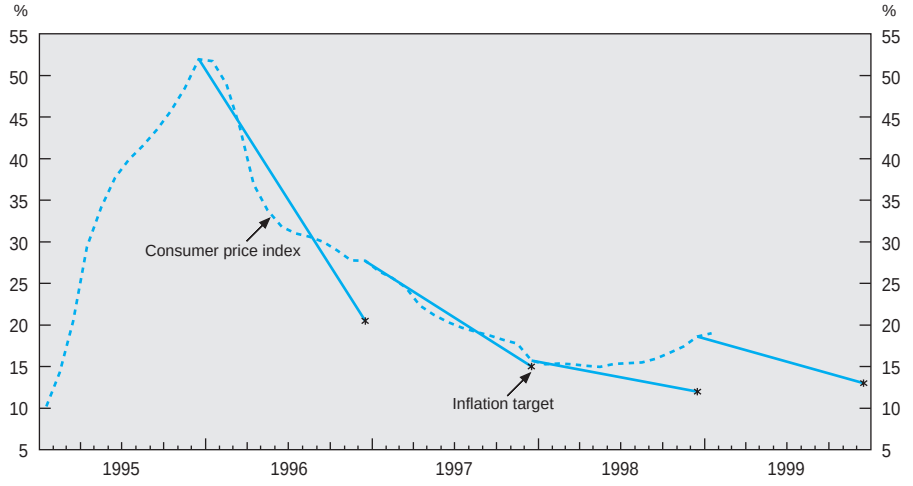
Monetary policy: coping with external shocks

The policy strategy

The Bank of Mexico's overriding objective is to reach brice stability in the long term.[21] Each year since 1995 the Bank has explained in its policy programme (issued in January), how it intends to pursue this objective. A yearly inflation target, determined jointly by the Bank and the Finance Ministry (in terms of the December-to-December growth rate of the CPI), is announced, establishing a path of gradual reduction towards the long-term objective. In this framework, the monetary base is used as an intermediate target; the Bank publishes its time path, set in accordance with the inflation target and consistent with the general macroeconomic projections for the year. As part of its policy commitments, the Bank also announces an annual ceiling for net domestic credit expansion and a target for the increase in net international assets.[22] Given the floating exchange

Figure 8. **Inflation targets**

Percentage changes over 12 months



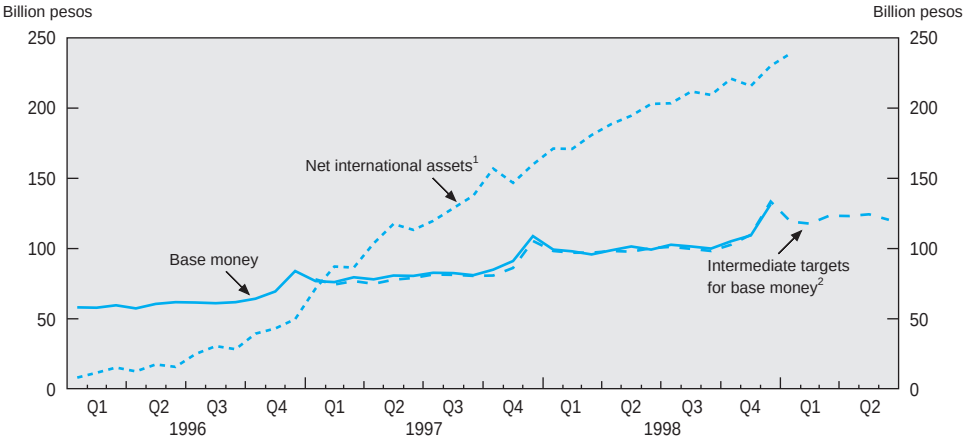
Source: Banco de México.

rate regime in force,[23] intermediate targets for monetary aggregates are also meant to provide the market with valuable signals and credible commitments, and hence play a more conspicuous role than in other countries pursuing explicit inflation targets.

The monetary programme for 1998 was designed in a context of global financial volatility related to the Asian crisis which, however, still had only a limited impact on Mexico. The programme stressed the monetary authorities' preference for a gradual approach to inflation reduction in order to ensure that progress would be sustainable. The inflation targets announced in the monetary programmes for 1996 and 1997 had established a clear downward path for inflation. Despite some slippage from the official targets (by 7 percentage points in 1996), the outcomes had broadly respected this downward path, with inflation dropping from 52 per cent at end-1995 to 15.7 per cent at end-1997 (Figure 8). For 1998, the inflation target was set at 12 per cent, within a macroeconomic framework that allowed for 5 per cent real GDP growth. The projection for base money growth was set at 22.5 per cent, which was deemed consistent with the projected nominal GDP growth (18.3 per cent) and a small degree of re-monetisation (Figure 9).

Figure 9. **Base money and net international assets**

End of period balances



1. Net international assets are defined as gross reserves, minus longer term liabilities, such as IMF loans.
2. The acceleration of base money at the end of each year is a stable seasonal pattern which is taken into account in the central bank's target definition.

Source: Banco de México.

In 1998, as in 1997, maintaining orderly conditions in foreign exchange and money markets were highlighted as an important objective guiding the Bank's operations. The Bank announced that in general it would seek to maintain a neutral monetary policy but, should unanticipated circumstances warrant it, it would adjust the policy stance symmetrically, either tightening it or relaxing it, to confront foreign or domestic shocks.[24] The normal way for the Bank to do this is by causing the banking system to be in "short" or "long" position, which pushes interest rates up and down, respectively. The Bank also reaffirmed the importance of maintaining the floating exchange rate regime, although a number of rules of the policy framework are aimed at reducing exchange rate volatility and uncertainties that may heighten perceptions of exchange rate risks. These rules include: the ceiling on domestic credit expansion; the target for net international asset accumulation; the fact that the central bank does not act as a price setter in money markets (a situation which increases the role of interest rates in absorbing liquidity shocks); and the put-options mechanism and the predetermined dollar-auction mechanism used by the Bank for its foreign exchange transactions (1998 Survey, Box 1).[25] Finally, the Bank emphasised the importance of policy

Box 1. New instruments of monetary policy

At end-August 1998, the Banco de Mexico started using instruments which, although already in existence, had been seldom employed, and introduced new ones.

i) Faced with a feeble response of interest rates to the expansion of the "short" position of the banking system in August 1998, the central bank decided to reduce the overdraft margin used in the intra-day settlement operations, thereby reducing liquidity in the money market on a daily basis.

ii) At the end of August 1998, during three days, the central bank fixed a floor to the interest rate applicable to daily operations, thus reducing market influence on rate determination.

iii) Starting in September 1998, the central bank established a remunerated mandatory deposit for commercial banks 8 set as a fixed amount (25 billion pesos) to be accumulated during 20 working days. The Bank is committed to replacing all the missing liquidity to the banking system through auctions on the open market; by becoming a net creditor to the system, it reinforces control over short-term interest rate behaviour. This instrument was used again in February 1999.

iv) The central bank, in August and September 1998, made swap offers of flows from fixed-rate securities (or long-term real instruments) for flows from variable-rate instruments. The assignation was done through auctions, with a ceiling per bank based on the required reserves accumulated at the central bank. This operation enabled banks to reduce exposure to the interest-rate variability risk.

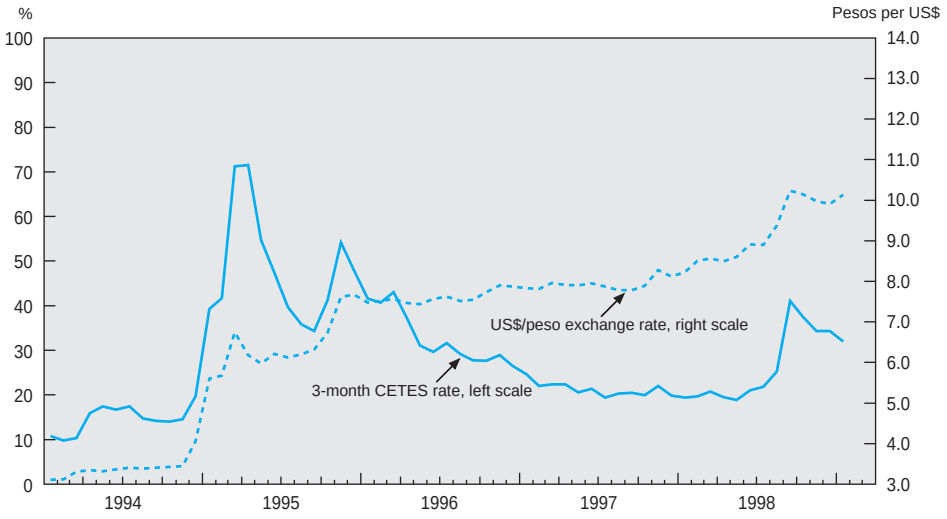
co-ordination and transparency: in particular, a tight fiscal stance was seen as a key element in the global strategy to fight inflation, monetary policy being unable to achieve the objective by itself; and an active communication policy on the part of the monetary authorities was stressed as an essential tool to provide market guidance.

Since monetary policy works slowly through the economy, the Bank monitors a number of leading and concomitant indicators to assess the appropriateness of the policy stance; these include the divergence of the daily monetary base from its projected path, the current-account deficit, the size and composition of capital inflows, exchange rate movements, wage settlements and inflation expectations (as proxied by private sector consensus forecasts for inflation).

The response to the international financial crisis

As the international environment in 1998 turned out to be much worse than expected, monetary policy had to be adjusted. The outbreak of the Asian crisis in the summer of 1997 at first, had a relatively small impact on Mexico, where nominal and real interest rates were on a downward trend and the nominal exchange rate was relatively stable in a climate of increasing confidence and declining inflation expectations (Figure 10). But in the fourth quarter of the year,

Figure 10. **The exchange rate and interest rates since 1994**



Source: OECD Secretariat.

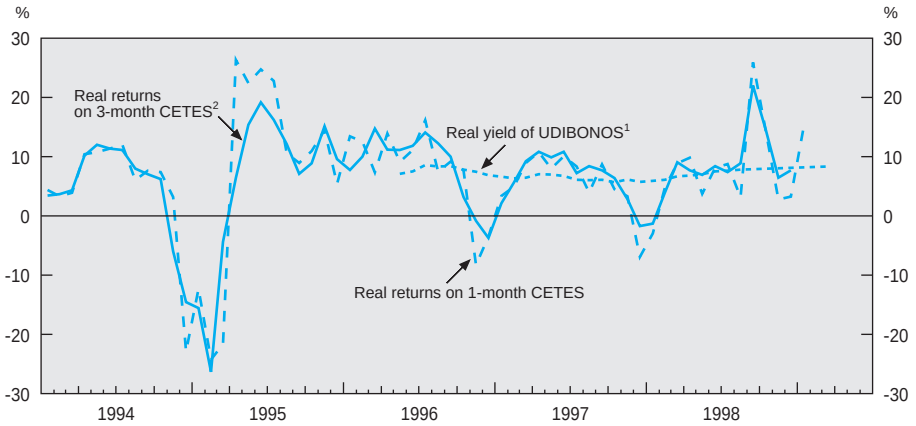
turbulence in international financial markets put pressure on the peso/dollar exchange rate which lost some ground, and short-term interest rates moved up for a few weeks, before easing back. In early 1998, the fall in oil prices gave rise to concern about the Mexican balance of payments; at the same time, inflation expectations started to rise, mainly as a result of the weakening of the exchange rate and hikes in government-set prices and tariffs. With the peso resuming its slide in the foreign exchange market, the central bank tightened monetary policy in several stages: in March, it moved from a neutral stance to leaving the banking system in "short" position; and further restrictive steps were taken in June and August. But the impact on short-term interest rates was rather modest and the peso continued to depreciate, albeit at a slow pace.

At end-August, investors concerns about emerging markets intensified following continued weakness in Asia and the outbreak of the Russian crisis. The risk of contagion to several Latin American countries was compounded, in the case of Mexico, by domestic uncertainties linked to the public debate about the financial reform. Most capital inflows dwindled or turned negative, putting the peso under severe pressure. This, together with the upturn in inflation expectations, led the central bank to react forcefully. But to ensure that its action would be effective in the new unsettled context, new instruments had to be developed. Since the money market was still in a situation of excess liquidity, putting the banking system in "short" position was not sufficient to engineer a major increase in interest rates and prevent a depreciation of the peso which could jeopardise the fight against inflation. Hence, among other mechanisms, the central bank introduced a remunerated mandatory deposit for commercial banks which, in combination with open market operations, would allow it to mop up excess liquidity and regain control of short-term interest rates (Box 1, item *iii*). At the same time, faced with mounting pressure on the peso/dollar rate and the risk of excessive volatility in thin markets, the Foreign Exchange Commission decided that a discretionary intervention in those markets was needed in addition to the predetermined dollar-auction mechanism (which is triggered when the exchange rate depreciates by more than 2 per cent in one day).[26] In one day (10 September), the Bank sold US$478 million, of which US$200 million through the pre-announced auction.

Outcomes

While until August 1998, most of the pressure arising from dwindling capital inflows was absorbed by the exchange rate, following the introduction of the new monetary instruments pressure was absorbed through a combination of interest rate and exchange rate changes. The three-month *Cetes* rate rose from around 25 per cent in August to 42 per cent in September. It hovered around 35 per cent in October, before falling back somewhat later in the year. In nominal

Figure 11. **Real returns on indexed bonds and money market instruments**

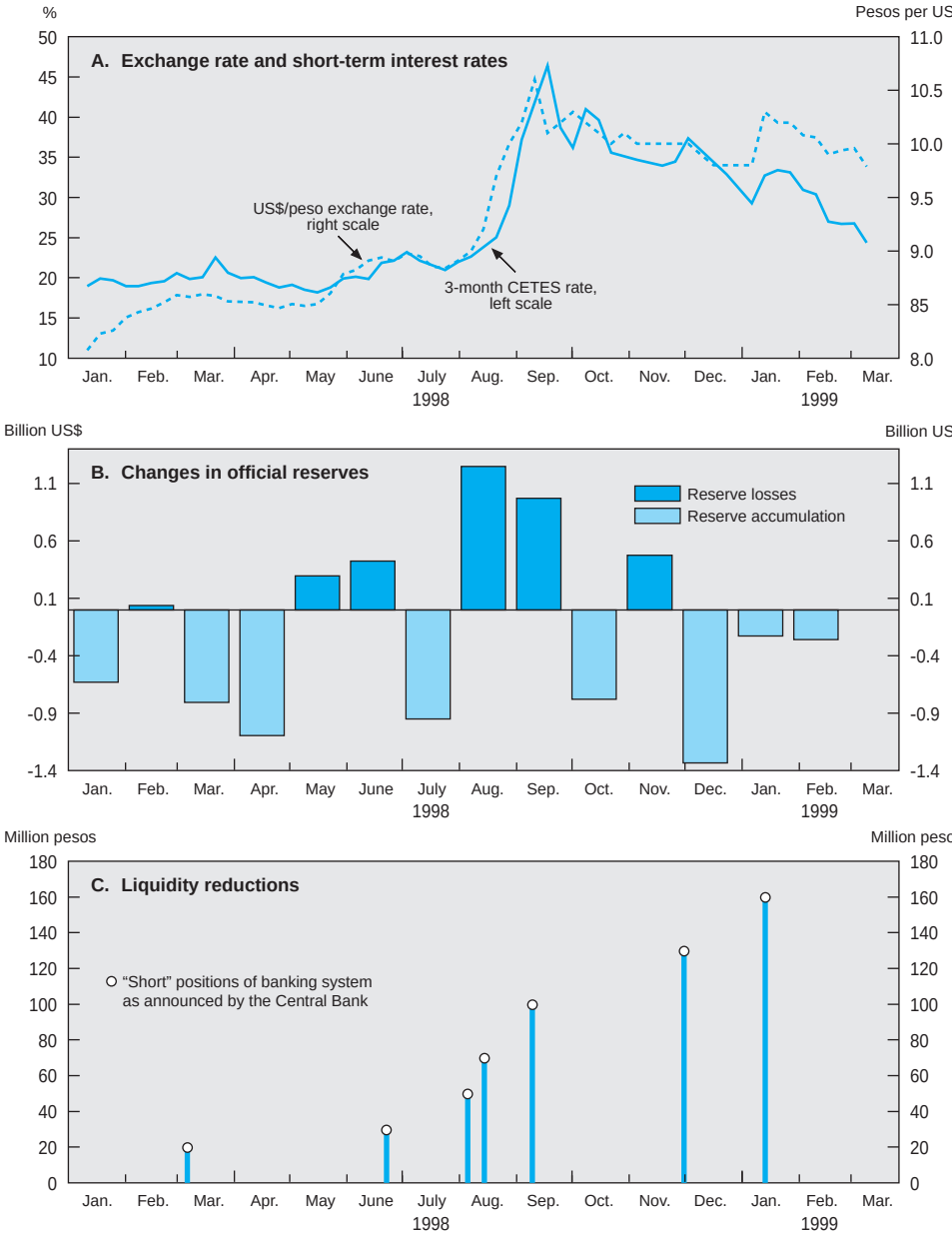


1. UDIBONOS are UDI-linked bonds (inflation indexed). The last observation plotted is March 1999; the previous auction was in August 1998.
2. 3-month CETES rate minus inflation in the current quarter.

Source: Banco de México.

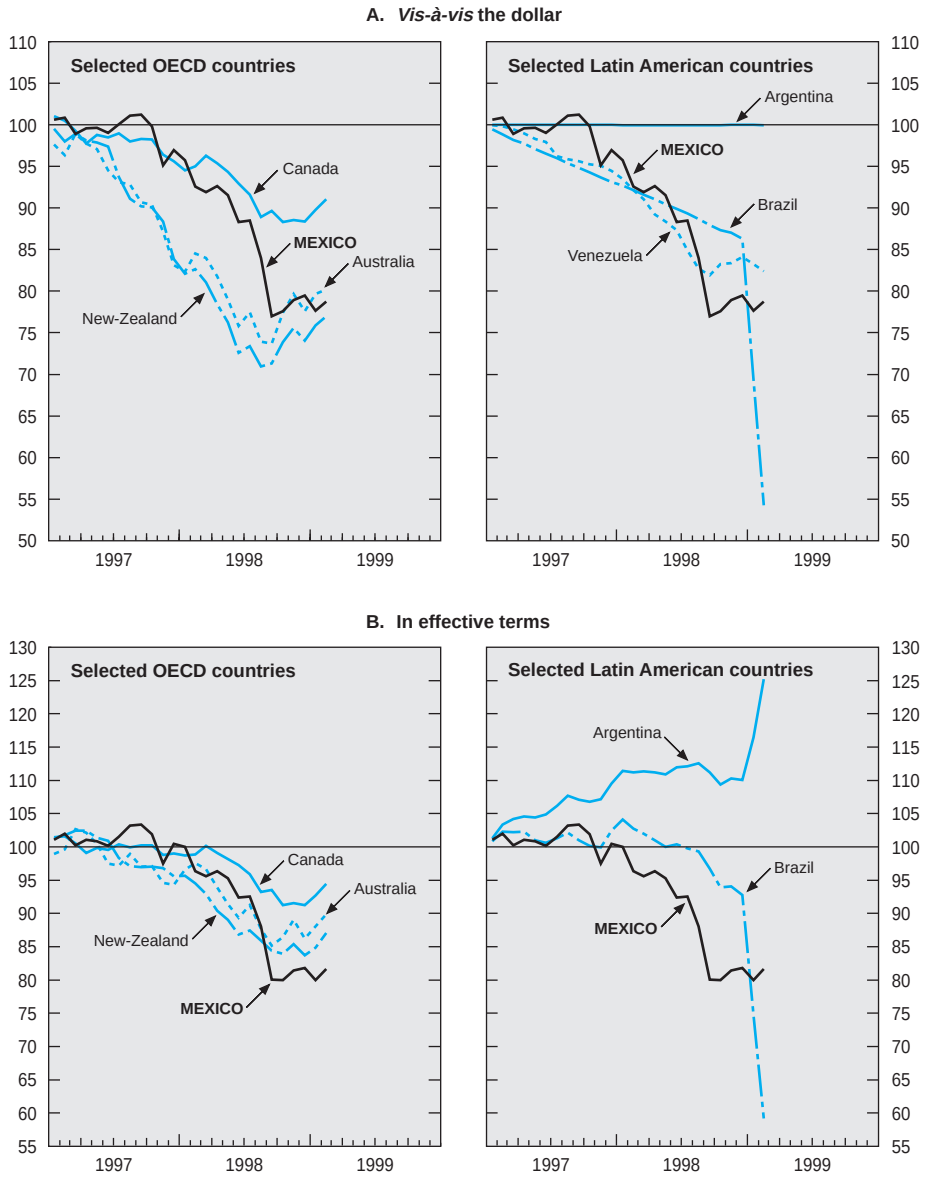
terms, short-term rates thus remained well below the peaks reached in the 1994-95 crisis – some 70 per cent in March-April 1995 – but since inflation is now considerably lower than four years ago, the 1998 increase in short-term rates was higher in real terms than that experienced in the previous crisis (Figure 11). Over most of the period under review, there was a very close "inverse" correlation between the evolution of short-term interest rates and the exchange rate of the peso, with interest rates rising when the peso was depreciating (Figure 12, Panel A). Some of the pressure was also absorbed by using official reserves, mostly through the predetermined dollar-auction mechanism. After increasing until end-April – to a level of over US$30 billion – official reserves fell by US$0.7 billion in May-June, by US$2.2 billion in August-September and by nearly US$0.5 billion in November. However, reserves rebounded by nearly US$1.4 billion in December and continued to increase, albeit at a slower pace, in January and February 1999 (Figure 12, Panel B).[27] In the twelve months to December 1998, the peso depreciated by about 20 per cent both *vis-à-vis* the dollar and in effective terms. The peso thus lost more ground, especially in effective terms, than the currencies of other major Latin American countries – Venezuela, Brazil and notably Argentina which has pegged its currency to the dollar through a strict currency board; on the other hand, its devaluation was more similar to that of other OECD commodity-exporting countries, such as Australia and New Zealand (Figure 13).

Figure 12. **The exchange rate and selected monetary variables in 1998**



Source: OECD Secretariat and Banco de México.

Figure 13. **Exchange rates of selected countries**
End-1996 = 100



Source: OECD Secretariat.

The knock-on effect of the Brazilian crisis in January 1999 was less disruptive than the shock of the summer of 1998, perhaps because it was much less of a surprise. After weakening by about 8 per cent *vis-à-vis* the dollar, the peso quickly recovered, and by early March it was back to a level close to that of end-1998. Short-term interest rates had a similar reaction, rising from around 30 per cent to 36 per cent, before falling back to less than 30 per cent.

The main monetary aggregates, such as money and credit, seem to have been less affected by the external shocks which took place in 1998 than interest rates and the exchange rate. In the twelve months to December 1998, the monetary base expanded by 20.8 per cent. Its evolution during 1998 was close to that projected by the Bank of Mexico's monetary programme – deviations from the expected path falling for the most part within the statistical confidence interval.[28] Both inflation and interest rates were higher than expected, with their effect on the demand for money offsetting each other. In the twelve months to December 1998, the narrow monetary aggregate (M1) expanded by 17.8 per cent, as both cash in circulation and demand deposits increased. High interest rates led to a continuous deceleration of M1 growth during 1998, however. The effect was felt most strongly on demand deposits in domestic currency. The growth of the wider money supply aggregate (M4) also decelerated in 1998: from 28 per cent in December 1997 (year-on-year) to 23.2 per cent in December 1998; with inflation accelerating during 1998, the slowdown in M4 was more pronounced in real terms (Table 7). For the year as a whole, both M1/GDP and M4/GDP remained roughly unchanged (Figure 14).

After a pronounced decline since the 1994 peso crisis, outstanding credit to the private sector picked up in 1998, but the upturn was limited both in scale and duration. The year-on-year real growth of commercial bank lending to the private sector turned positive in July 1998,[29] boosted by the dynamism of consumer credit, in particular credit card purchases. Net borrowing by enterprises and individuals with entrepreneurial activity also started to increase (in real terms) in the second quarter of 1998. Mortgage credit granted by commercial banks on the other hand, continued to decline, although this is likely to have been partly offset by the increase in mortgage lending by specialised financial institutions. The improvement in credit market conditions was interrupted at the end of August. The adverse global and domestic environment that has prevailed since then, with further monetary tightening and rising uncertainties about growth prospects, has increased the risks of loan delinquency, which in turn, has been reflected in higher risk premia. This is likely to have made households and enterprises, as well as banks, more reluctant to engage in credit operations. As noted in Chapter IV, uncertainties were compounded during most of 1998 by the lack of progress in adopting the financial reform package, which left unclear how much of the final cost would be paid by the banking sector.

Table 7. **Bank lending and financial saving**

	Dec. 1994	Dec. 1995	Dec. 1996	Dec. 1997	June 1998	Sept. 1998	Dec. 1998
	End-of-period stocks, billion pesos						
M4	729	880	1 166	1 493	1 636	1 702	1 839
Bank credit							
Total	640	810	910	989	1 058	1 110	1 135
To private sector	613	764	863	918	987	1 022	1 050
Excluding official asset purchases[1]	613	619	511	499	542	561	564
	Percentage year-on-year changes, in real terms						
M4	15.9	–20.6	3.8	10.6	5.7	5.2	3.9
Bank credit							
Total	32.7	–16.7	–12.1	–6.0	–1.4	2.4	–3.2
To private sector	31.9	–18.0	–11.5	–8.1	–2.9	–0.2	–3.6
Excluding official asset purchases[1]	31.9	–33.6	–35.3	–15.6	–1.4	1.7	–4.7

1. Excludes assets that have been sold or transfered to trust funds in the context of debtor and bank support programmes.

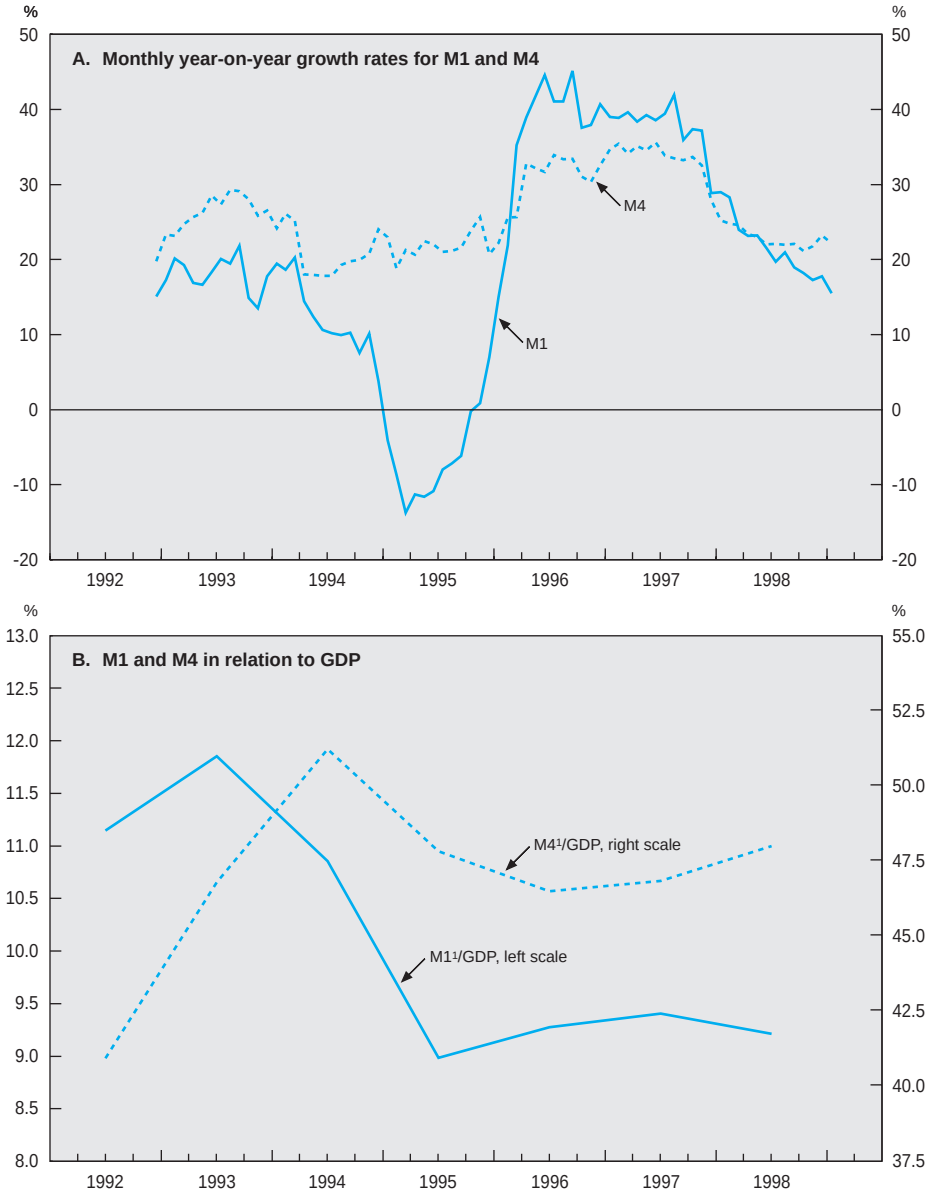
Source: Banco de México.

Development banks – except for Bancomext – have also further cut back lending to the private sector, continuing the trend of the past three years.[30] This has largely reflected discretionary action from the Ministry of Finance to induce these banks to strengthen their balance sheets. The credit restraint has affected more significantly medium-sized enterprises, which now account for only 9.2 per cent of total credit granted by development banks.

Assessment

The international financial crisis of 1998 has been a severe test of the monetary policy framework put in place in Mexico after the 1994 peso crisis. The new framework, and notably the instruments aimed at controlling domestic liquidity and interest rates, as well as the mechanisms for curbing volatility in the foreign exchange market, have proved their usefulness but also their limits, especially in the aftermath of the Russian debt default when pressure on the peso intensified. A floating exchange rate, at first, conveniently absorbs external shocks and limits the loss of domestic output, but it also presents the risk of overshooting through a dangerous interaction between inflation, inflation expectations and depreciation. Hence, when the Bank of Mexico estimated that the depreciation of the peso was inconsistent with the fundamentals of the economy and that the rise in inflation expectations was becoming worrying, it moved forcefully to

Figure 14. **Monetary aggregates**



1. End of year figures.
Source: Banco de México.

complement existing instruments of monetary policy with more powerful and discretionary ones. The introduction of new instruments allowed the authorities to tighten monetary conditions sharply, hike interest rates and provide a powerful signal to the market. In addition, the pre-announced dollar-auction mechanism was once supplemented – in September – by direct intervention in the exchange market to prevent excessive volatility.

This combination of exchange-rate flexibility and policy firmness, was on the whole, well received by the market, and proved an effective way of handling the crisis as well as responding to the fall in international oil prices and the consequent deterioration in the Mexican current account. It helps explain that contagion effects to Mexico were relatively limited, and confidence was largely preserved. One key indicator suggests that in this respect Mexico did better than most other Latin American countries, including Venezuela, Brazil and Argentina. In fact, although it widened to nearly 10 percentage points at the height of the crisis last summer, the spread between yields on Mexican sovereign bonds (in dollars) and US Treasuries consistently remained the lowest among these countries.

But this crisis management did not prevent a pick up in inflation which ended the year at 18.6 per cent, well above the official target of 12 per cent. If, following a new shock, the peso depreciates excessively, inflation will start rising again from an already high level, with the risk of jeopardising the credibility of monetary policy. The top policy priority should thus be to avoid a spiral of depreciation and inflation and put the country firmly back on a disinflationary path. With a floating exchange regime, a credible inflation target is needed to provide an anchor for economic policy, preserve confidence and, ultimately, allow interest rates to fall.

A policy of inflation control is likely to entail some costs, especially in the short term. Real GDP growth is already decelerating – although perhaps not as much as feared, especially in view of the near drying-up of bank credit to the domestic economy; and it is unclear to what extent the still-fragile banking sector will suffer from high interest rates. The protracted slump in the domestic banking credit market has not prevented the economic recovery from continuing at a strong pace, however, at least until recently. Available evidence suggests that enterprises have financed themselves through other sources: there has been a fast expansion of supplier credit and lending by non-bank institutions, while export-oriented companies have been able to obtain foreign-currency credit at relatively favourable terms (Chapter I). Nonetheless, it remains to be seen for how long these alternative sources of financing can support the economy; and over the longer run, a strengthening of the financial system will be required to ensure sustained economic growth (see Chapter IV).

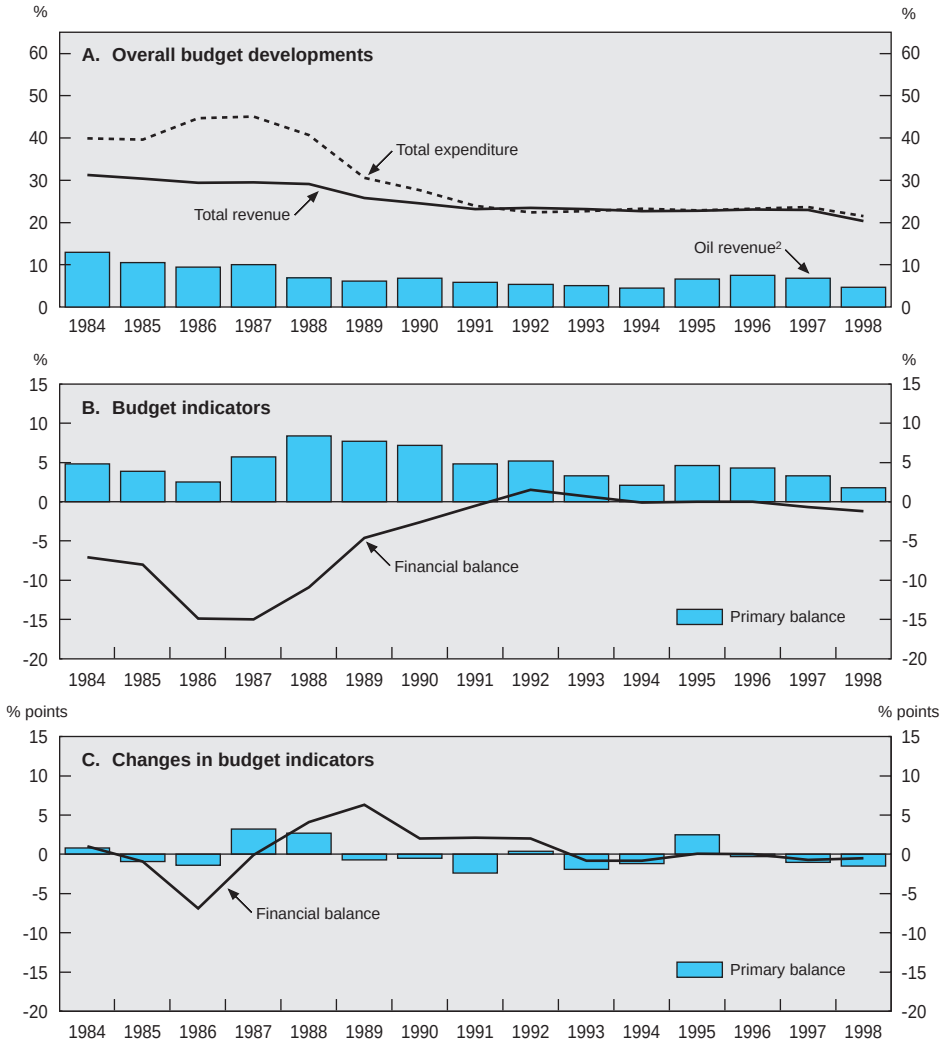
Fiscal policy: adjusting to lower oil prices

General considerations

After the December 1994 peso crisis, fiscal policy was tightened as part of the stabilisation programme. Since then a cautious policy stance has been maintained, aimed principally at consolidating macroeconomic stability, while at the same time promoting investment and reorienting spending towards social programmes (Figure 15). From 1995 to 1997 the budget outturns were close to balance (Table 8); the primary surpluses averaged 4 per cent of GDP, and the debt-to-GDP ratio came down steadily. In 1998, original budget projections allowed for a deficit of 1.25 per cent of GDP, in part as a result of the expected decline in oil revenue.[31] Excluding the cost of the pension and health insurance reform, however, the public sector financial balance was projected to show a small surplus. In both 1997 and 1998, actual revenues differed significantly from initial budget projections; but spending was adjusted by a similar amount, so the budget deficit turned out close to target. In 1997, buoyant activity made for higher-than-expected revenue, which allowed an overrun in actual expenditures of 0.7 per cent of GDP (compared with the original budget). In 1998, on the other hand, the decline in oil-related revenue forced the government to implement budget cuts worth approximately 0.9 per cent of GDP.

In addition to the influence of exogenous variables such as the movements in international oil prices and world interest rates, fiscal developments since 1995 have been affected by a number of structural factors. First, provisions had to be made to cover the cost of programmes put in place to support banks and debtors in the aftermath of the crisis.[32] Of the total estimated fiscal cost – 14.4 per cent of GDP according to early-1998 estimates – the equivalent of 2.9 per cent of GDP has already been disbursed over the period 1995-97. The projected allocations for 1998 amounted to a further 0.3 per cent of GDP. In the coming years, the securities that will replace liabilities of FOBAPROA (the government agency dealing with bank insolvencies) will be fully guaranteed by the government, but the terms and conditions are still to be defined. Since they do not become direct public debt, there is no regular servicing schedule.[33] Second, in the context of increasing decentralisation of spending responsibilities, transfers from the federal government to lower tiers of government have been growing, reaching 3 per cent of GDP in 1998. Third, the implementation of the social security reform in 1997 has started to put pressure on the budget balance: contributions to the IMSS health insurance are lower;[34] furthermore, the transition from the previous pay-as-you-go pension system to a fully-funded capitalisation system implies both lower revenue (as contributions formerly made to the IMSS are channelled to private fund managers), and higher expenditure to fill the gap between accumulated individual savings and pension rights if the worker chooses the old system. As reforms were implemented in July 1997, their impact on a

Figure 15. **Budget indicators**[1]
As a percentage of GDP



1. The public sector comprises federal government and public enterprises under budgetary control (such as PEMEX). Financial intermediation by development banks are not included; the financial balance is the "economic" balance according to the Mexican terminology; the primary balance is the financial balance less net interest payments.
2. Estimates for 1998 are budget projections adjusted for impact the fall in oil prices.

Source: Ministry of Finance, Banco de México.

Table 8. **Public sector financial accounts**

Percentage of GDP

	1995	1996	1997		1998			1999	2000
	Outturn	Outturn	Budget	Outturn	Original budget	Adjusted budget[1]	Outturn[2]	Budget[2]	Official projection
Revenue	22.8	23.1	22.4	23.0	21.2	20.3	20.4	20.4	20.5
Expenditure	22.9	23.3	23.0	23.7	22.5	21.6	21.5	21.8	21.6
Financial balance	**0.0**	**0.0**	**–0.5**	**–0.7**	**–1.3**	**–1.3**	**–1.2**	**–1.3**	**–1.0**
Primary balance	**4.6**	**4.3**	**3.4**	**3.3**	**2.1**	**2.1**	**1.8**	**2.8**	**2.9**
Memorandum items:									
Interest payments, total	4.6	4.4	3.9	3.9	3.6	3.6	2.9	4.0	3.8
of which:									
Provisions for support programmes[3]	0.8	0.8	–	1.3	0.3	0.3	0.3	0.5	0.5
Cost of IMSS reform[4]	–	–	0.7	0.7	1.4	1.4	1.5	1.5	1.5
Revenue sharing	2.7	2.8	2.8	3.0	3.0	2.9	3.0	3.0	3.0

1. Original budget, adjusted for the three fiscal packages of January, March and July 1998.
2. The outturn for 1998 is the Ministry of Finance preliminary estimate as of January 1999. The projection for 1999 is the approved budget.
3. Deposits made to FOBAPROA to cover part of the cost of the support programmes for banks and debtors (see section on banking sector in Part IV below).
4. IMSS is the social security institute for private workers; the total cost of the reform includes both reduced revenues and higher expenditures.

Source: Ministry of Finance.

full-year basis started to be felt only in 1998; the yearly cost is estimated at 1.5 per cent of GDP for about 20 years, declining gradually thereafter.

Budgetary developments in 1997

In 1997, the financial accounts of the broad public sector showed a deficit of 0.7 per cent of GDP, slightly larger than budget projections. Revenues increased faster than expected, reaching 23 per cent of GDP, 0.6 percentage point above projections. Most of this increase came from federal tax revenues (income tax and VAT in particular) as a fast-growing economy boosted the tax base and measures to increase compliance and fight evasion bore fruits (Table 9). On the other hand, non-tax revenues of the federal government (including contributions from the state-owned oil company PEMEX) were down to 6.2 per cent of GDP compared with 6.6 per cent a year earlier, mostly reflecting the fall in oil prices in the latter part of the year. Revenues from state companies under direct budgetary control increased as a share of GDP, reaching 9 per cent, as higher sales of goods and services (especially by the Federal Electricity Commission, CFE) offset the decline in PEMEX's own revenues and in contributions to IMSS[35] (Table 10).

Table 9. **Federal government budget**

Percentage of GDP

	1995	1996	1997		1998			1999
	Outturn	Outturn	Budget	Outturn	Original budget	Adjusted budget[1]	Outturn[2]	Budget[2]
Revenue	**15.2**	**15.7**	**15.2**	**16.0**	**14.9**	**14.1**	**14.1**	**14.8**
Contribution of PEMEX	3.9	4.5	4.0	4.1	3.2	2.2	2.3	1.7
Other non-tax revenue	2.0	2.1	2.1	2.1	1.6	1.7	1.4	1.8
Tax revenue	9.3	9.0	9.1	9.8	10.1	10.2	10.4	11.3
of which:								
Income tax	4.0	3.9	3.5	4.2	4.2	4.2	4.3	4.7
VAT	2.8	2.9	2.8	3.1	3.0	3.0	3.1	3.2
Excise taxes	1.3	1.2	1.7	1.4	1.9	1.9	2.0	2.4
Import duties	0.6	0.6	0.5	0.6	0.5	0.5	0.6	0.6
Expenditure	**16.1**	**16.1**	**16.3**	**17.3**	**16.4**	**16.0**	**15.8**	**16.3**
"Programmable"	9.5	9.3	9.8	10.4	10.3	10.0	10.0	9.5
Current expenditure	7.8	7.6	8.7	8.7	9.1	8.7	8.4	–
Capital expenditure	1.9	1.9	1.7	1.9	1.6	1.7	1.6	–
Delayed payments	–0.2	–0.2	–0.5	–0.3	–0.4	–0.4	–	–
Non-programmable	6.6	6.8	6.5	6.9	6.1	6.0	5.8	6.8
Revenue sharing	2.7	2.8	2.8	3.0	3.0	2.9	3.0	3.0
Interest payments[3]	3.8	3.7	3.3	3.4	2.8	2.8	2.5	3.5
Other[4]	0.1	0.2	0.4	0.6	0.3	0.3	0.3	0.3

1. Original budget, adjusted for the three fiscal packages of January, March and July 1998.
2. The outturn for 1998 is the Ministry of Finance preliminary estimate made in January 1999. The projection for 1999 is the approved budget.
3. Includes provisions made on account of the support programmes for banks and debtors.
4. Includes accounts payable from previous fiscal period and net expenditure of the Federal Government on behalf of state-owned enterprises.

Source: Ministry of Finance.

At the federal level, most of the extra revenue was offset by increased spending on public investment and social programmes. Current expenditure soared by 14.1 per cent in real terms, reflecting enhanced programmes for health, education and nutrition. As a result, total expenditure on social development reached 55.4 per cent of total "programmable" spending, up from 53.2 per cent in 1996 (Table 11).[36] By 1997, social expenditure had recovered to above pre-crisis levels in real terms, with social security leading the increase (partly as a result of the IMSS reform). Non-programmable expenditures increased slightly as a share of GDP, mainly reflecting higher transfers to state and municipalities. Public sector interest payments reached 3.9 per cent of GDP in 1997, significantly below the 1996 outturn, reflecting the favourable impact of lower interest rates on the public debt servicing cost. At the same time, payments made to cover part of the cost of the support programmes for banks and debtors were slightly higher than in the previous year.

Table 10. **Public enterprises under budgetary control: financial accounts**[1]

Percentage of GDP

	1995	1996	1997		1998			1999
	Outturn	Outturn	Budget	Outturn	Original budget	Adjusted budget[2]	Outturn[3]	Budget[3]
Revenue	**8.9**	**8.7**	**8.6**	**9.0**	**7.9**	**7.8**	**8.0**	**7.3**
PEMEX	2.7	2.9	3.0	2.7	2.6	–	2.2	2.1
Other public enterprises	4.8	4.5	4.2	4.4	3.7	–	4.3	3.5
Transfers received	1.4	1.2	1.4	1.9	1.6	–	1.5	1.7
Expenditure	**8.3**	**8.4**	**8.0**	**8.3**	**7.7**	**7.2**	**7.4**	**6.8**
Current expenditure	5.9	5.7	5.7	6.2	5.4	5.2	5.6	–
Capital expenditure	1.6	2.0	1.7	1.6	1.7	1.4	1.4	–
Interest payments	0.8	0.6	0.6	0.5	0.5	0.5	0.4	0.5

1. State-owned enterprises under budgetary control, excluding public financial intermediaries (development banks and development trust funds).
2. Original budget, adjusted for the three fiscal packages of January, March and July 1998.
3. The outturn for 1998 is the Ministry of Finance preliminary estimate as of January 1999. The projection for 1999 is the approved budget.

Source: Ministry of Finance.

Table 11. **Public programmable expenditure by sector**

	1995	1996	1997	1998		1998/1997[1]
				Original budget	Adjusted budget[2]	Real growth
TOTAL peso billion	286.3	397.2	508.1	604.4	574.8	–
(per cent change)	(–15.1)	(7.1)	(7.5)	(5.8)	(0.6)	0.6
	Per cent of total					
Social development	53.9	53.2	55.4	58.5	60.1	9.1
Education	23.8	23.3	23.1	24.6	25.6	11.5
Health	14.2	13.2	13.5	13.8	14.2	5.9
Social security	8.0	8.4	10.8	11.8	12.1	12.4
Labour	0.2	0.2	0.4	0.4	0.4	–2.5
Social assistance and supply	2.2	2.5	1.8	1.8	1.8	–0.5
Urban and regional development	5.4	5.5	5.7	6.2	6.0	5.3
Rural development	7.1	6.7	5.6	5.2	5.1	–9.3
Environment and fishing	0.5	0.3	1.1	1.1	1.1	–1.6
Communications and transport	5.0	5.6	5.1	3.4	3.3	–34.5
Energy	22.8	24.1	22.6	22.3	20.8	–7.6
Justice and security	5.8	6.1	5.6	5.6	5.8	4.5
Administration	4.9	4.0	4.6	3.9	3.9	–15.6

1. Percentage change from 1997 outturn to 1998 adjusted spending.
2. Original budget, adjusted for the three fiscal packages of January, March and July 1998.

Source: Ministry of Finance.

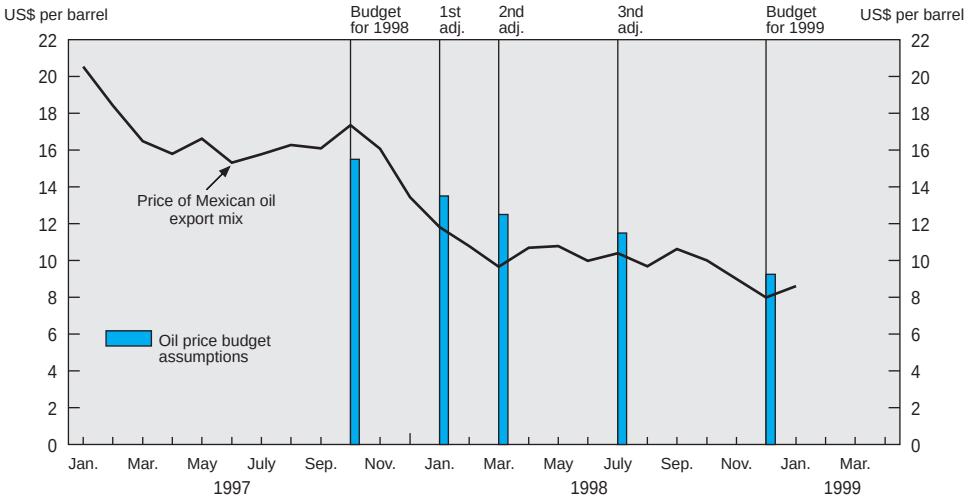
The 1998 budget and its implementation

The 1998 budget was prepared taking into account the pressures coming from social security reform and the cost of the financial rescue package, plus an assumption on the expected decline in oil export prices. Both revenues and expenditures were expected to fall as a share of GDP, and the public sector deficit was targeted at 1.25 per cent of GDP, about 0.5 percentage point above the 1997 outturn. This projected widening of the financial deficit and the decline in the primary surplus (by 1.2 per cent of GDP) implied relatively tight control of public finances, however. Social security reform alone was expected to represent an additional fiscal cost of 0.8 per cent of GDP in 1998, while lower oil prices embedded in the budget projections amounted to a revenue shortfall for the federal government of about 1 per cent of GDP compared with 1997. With real GDP growth projected to slow down to 5.2 per cent, the fiscal stance was therefore cautious, and as such it was meant to be supportive of the disinflation process.

According to initial budget projections, public sector revenue was expected to decrease to 21.2 per cent of GDP, down almost 2 percentage points from the 1997 outturn. On the expenditure side, total disbursements of the public sector were budgeted to decline as a share of GDP (by 1.2 percentage point from the 1997 outturn). The programmable component of federal government spending was projected to increase substantially in real terms. As in previous years, disbursements on health, poverty alleviation and education were given priority so that the share of social expenditure in the total was projected to increase to 58.5 per cent. Non-programmable spending, by contrast, was projected to fall slightly relative to GDP, due to the expected fall in public debt servicing costs and lower provisions for the cost of the bank and debtor support package.

Early in 1998, as oil export prices fell more than expected, it became clear that initial estimates for revenues from the oil sector (which still account for about a third of total federal revenues) were too optimistic. The authorities responded swiftly, making the necessary adjustments to spending with a view to meet the budget target. Three successive fiscal packages were announced – in January, March and July 1998 – to compensate for the expected impact of the decline of oil prices on the public sector revenue. The oil revenue shortfall amounted to approximately 1 per cent of GDP, according to official estimates, as the oil price assumption underlying budget projections was revised down successively by about 4 dollars to US$11.50 a barrel[37] (Figure 16). However, the spending cuts resulting from the three sets of measures were slightly lower – 29.8 billion pesos, about 0.8 per cent of GDP – because non-oil revenue came in higher than initially budgeted. This reflected both proceeds from privatisation and an unexpected good outturn in tax collection (Table 12). The government's corrective action took place against a background of still buoyant domestic demand and booming imports; and it was thus seen as helpful to cooling the economy. Furthermore,

Figure 16. **Oil prices and budget assumptions**[1]



1. Budget assumptions for average price of the Mexican oil export mix embodied in the initial budget for 1998, made at the time of announcement of corrective measures during 1998, and underlying the budget for 1999.

Source: Ministry of Finance.

Table 12. **Budget impact of the decline in oil-related revenue and offsetting measures**[1]

Billion pesos

	Adjustments			Total
	First	Second	Third	
Shortfall in oil revenue	15.3	11.2	9.8	36.2
(per cent of GDP)	(0.4)	(0.3)	(0.3)	(1.0)
Windfall gains on income	0	2.2	4.3	6.5
Higher tax revenue	0	0	1.5	1.5
Higher non-tax revenue[2]	0	2.2	2.8	5.0
of which: Privatization	0	0	1.8	1.8
Cut in programmable spending	15.3	9.0	5.5	29.8
(per cent of GDP)	(0.4)	(0.2)	(0.2)	(0.8)
Net impact on public sector balance	0	0	0	0

1. Expected budget impact; official estimates of July 1998.
2. Includes revenue from sales of public goods and services, and interest received on contingency fund.

Source: Ministry of Finance.

fiscal restraint in response to a negative terms of trade shock was deemed appropriate as it could contribute to enhance confidence in economic policies in the face of global financial turbulence.

The first adjustment involved cuts in investment and, to a lesser extent current expenditures, amounting altogether to 0.4 per cent of GDP. These included the postponement of construction projects, a partial salary and hiring freeze for the public sector and a reduction of the operational expenses in a number of ministries. Most affected were state-owned companies, such as PEMEX and the electricity company CFE, which altogether accounted for around 61 per cent of total cuts (Table 13). The second adjustment, although smaller (0.24 per cent of GDP), was similar to the first, in that it affected investment and state-owned companies to a greater extent. The third package of budget cuts was worth 0.15 per cent of GDP. At the same time, the expected tax and privatisation revenues for 1998 were revised upwards, contributing 0.11 per cent of GDP to the adjustment. Even though PEMEX has borne the brunt of the cuts, mainly through a reduction in investment spending, corrective measures have also applied to the budgets of the health and education ministries. In September, following devastating floods in Chiapas (where damage amounted to more than 0.1 per cent of GDP, according to preliminary estimates), the Government asked its agencies to freeze a number of spending items and to earmark money saved throughout 1998 to finance special relief programmes.

Results for 1998 show the impact of fiscal restraint especially on capital expenditure which, for the public sector as a whole, declined by 8.2 per cent in real terms, relative to a year earlier.[38] Current spending, on the other hand, remained broadly unchanged in real terms. The wage bill of the public sector increased marginally (by 2 per cent) in real terms, as a real decline in staff costs of the central government and public enterprises was more than offset by a significant real rise in wages of teachers and medical staff which are paid by the states. In 1998, as in the previous year, interest payments on the public sector debt turned out to be lower than budget projections. As Mexico's public debt (both internal and external) is increasingly long-term,[39] these interest payments were only moderately affected by the high level of interest rates prevailing in the latter part of 1998. Provisions made on account of the support programmes for banks and debtors amounted to 0.3 per cent of GDP. Finally, spending on pensions increased significantly in 1998 relative to 1997, reflecting the cost of the IMSS reform over a full year. On the revenue side, oil prices declined further in the fourth quarter of 1998, averaging about US$10 a barrel for the year as a whole. This was only partly offset by higher-than-expected tax revenues, so that total revenues turned out to be lower than budget projections, by 0.8 percentage points of GDP. Overall, action taken through the year has succeeded in keeping the public sector deficit on target, with a primary surplus slightly lower than budgeted (1.8 per cent of GDP instead of the original projection of 2.1 per cent of GDP).

Table 13. **Adjustment to 1998 budget expenditures**

	Original allocation (bn pesos)	Final allocation[1] (bn pesos)	Cumulative cut (bn pesos)	Cumulative cut (%)	Share of total cut (%)
Total programmable expenditure	**509.3**	**479.6**	**29.6**	**5.8**	**100.0**
Central public administration	**239.0**	**227.6**	**11.4**	**4.8**	**38.3**
Presidency	1.1	1.1	0.0	0.9	0.0
General administration	6.4	6.3	0.0	0.5	0.1
Foreign affairs	2.4	2.4	0.0	0.0	0.0
Finance	16.4	15.1	1.4	8.2	4.6
Defense	14.2	14.2	0.0	0.2	0.1
Agriculture	18.5	17.9	0.6	3.0	1.9
Communications and transport	13.7	12.6	1.1	8.0	3.7
Trade	1.9	1.8	0.1	4.2	0.3
Education	61.7	60.3	1.5	2.4	4.9
Health	13.6	12.8	0.8	5.9	2.7
Navy	5.9	5.7	0.2	3.2	0.6
Labour	2.7	2.5	0.2	7.5	0.7
Agrarian reform	1.4	1.4	0.0	0.7	0.0
Environment	12.8	11.8	1.0	7.5	3.2
Comptroller-general	3.4	3.4	0.0	0.6	0.1
Energy	0.7	0.7	0.0	0.0	0.0
Social security contributions	17.8	17.0	0.9	4.8	2.9
Social development (SEDESOL)	9.4	8.1	1.3	13.6	4.3
Tourism	1.1	0.9	0.1	12.3	0.4
Wage provisions	10.4	8.1	2.3	22.1	7.8
Basic education fund	19.1	19.1	0.0	0.0	0.0
Poverty alleviation	3.4	3.4	0.0	0.0	0.0
Other	1.2	1.2	0.0	0.8	0.0
Organisations and enterprises under budgetary control	**270.3**	**252.0**	**18.3**	**6.8**	**61.7**
PEMEX	78.5	69.9	8.6	11.0	29.1
Federal Electricity Commission (CFE)	54.3	47.4	6.9	12.7	23.3
Luz y Fuerza[2]	8.9	8.9	0.0	0.2	0.1
CONASUPO	7.4	6.5	0.9	11.6	2.9
IMSS	85.6	84.5	1.1	1.3	3.7
ISSSTE	25.1	24.6	0.5	2.0	1.7
Other	10.5	10.2	0.3	2.7	0.9

1. Adjusted for the budget cuts of January, March and July 1998.
2. Power and Electricity Company of Central Mexico.
Source: Ministry of Finance.

The 1999 budget

The 1999 budget was prepared at a time of significant uncertainties, related both to global financial turbulence and domestic developments, such as the lengthy debate in Congress about the financial reform proposal (Chapter IV).

By late 1998, the peso/dollar exchange rate had depreciated substantially; domestic interest rates were much higher than a year earlier; inflation had picked up and output growth had slowed down. Monetary policy was significantly tightened during the year, with a view to cooling down domestic demand and putting a lid on inflation. In that context, it was important to continue to pursue a prudent fiscal policy in order to preserve confidence and take some of the burden off monetary policy, so as to create conditions favourable to moving interest rates back down. Moreover, given the fragility of the banking sector, restraint coming from a tight fiscal stance was seen as less damaging for the economy than persistently high interest rates. The task of drawing the budget was further complicated by declining oil prices and the expected deceleration in tax revenue mirroring the slowdown in activity.

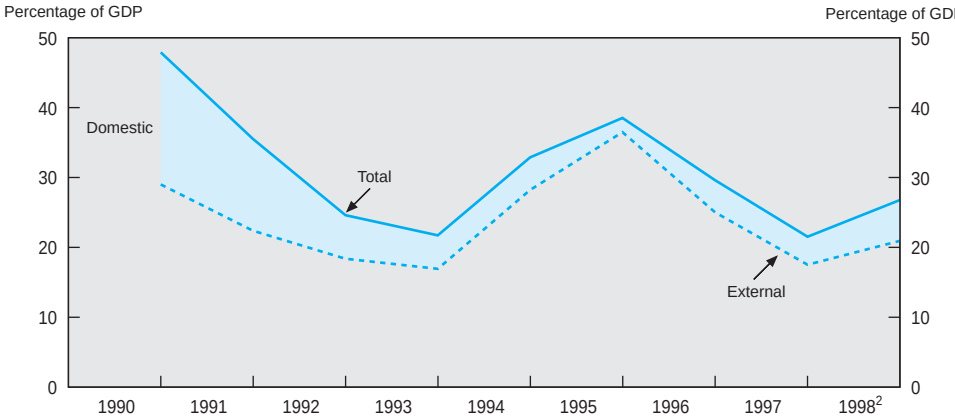
In response to these challenges, the 1999 budget remains cautious, targeting a financial deficit of 1.25 per cent of GDP, the same as the 1998 outturn.[40] The primary surplus is projected to widen by 1 per cent of GDP. Budget assumptions include an average price for the Mexican oil export mix of US$9.25 per barrel (below the average for 1998, but above the US$7.50 reached at the end of 1998, reflecting prices implicit in the futures market), GDP growth of 3 per cent, and inflation of 13 per cent by year-end (by then the exchange rate is assumed to be at 11.1 pesos/dollar, implying a further depreciation of some 11 per cent from end-1998). Several measures have been taken with a view to strengthen tax revenue, including an increase in the top marginal rates for personal income tax and maintaining a tax on alcohol and beverages, which are expected to increase federal tax revenue by 1 per cent of GDP. On the spending side, programmable expenditure of the federal government has been set to decline slightly in real terms, so as to partly compensate the expected rise in the cost of public debt servicing. Given the projected slowdown in economic activity in 1999, the widening of the primary surplus that is targeted in the budget would imply some degree of fiscal tightening.[41]

Debt management

After their success in refinancing short-term dollar linked securities (*Tesobonos*) following the 1994 peso crisis, the authorities have focused on lowering the funding costs and lengthening the average maturity of the public debt. On the external front, early payments were made in 1996 and 1997 on obligations to the IMF and the United States government, and on bank notes issued at floating rates and due in 2001. These payments were financed by the issuance of securities placed in international financial markets at more favourable terms. This strategy was greatly facilitated by the generation of significant primary surpluses which, combined with privatisation revenues, led to an overall declining trend of public debt, both in absolute terms and as a percentage of GDP. Also helped by

Figure 17. **Total net debt of the public sector**[1]

Broad economic debt, stocks outstanding, end of period

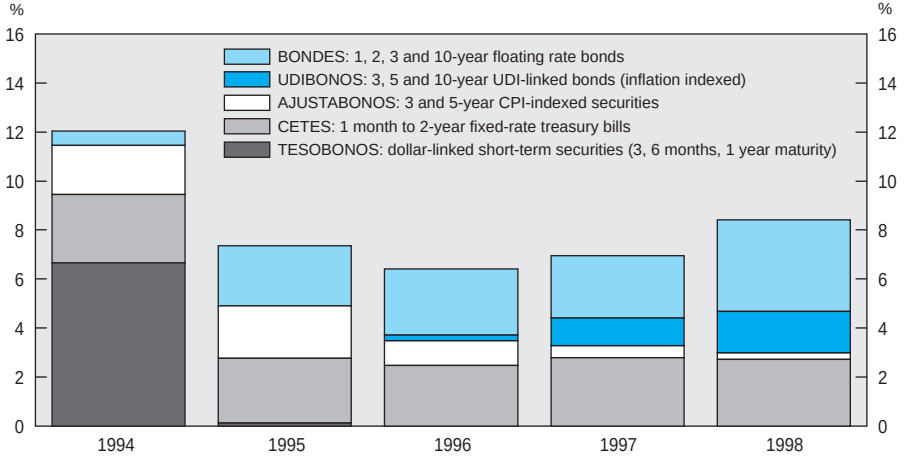


1. Total gross debt less deposits and financial assets of the public sector (comprising the Federal Government, public enterprises, development banks and official trust funds).
2. December 1998. These data differ slightly from Ministry of Finance data, which do not include official trust funds.

Source: Bank of Mexico.

Figure 18. **Public sector domestic debt**[1]

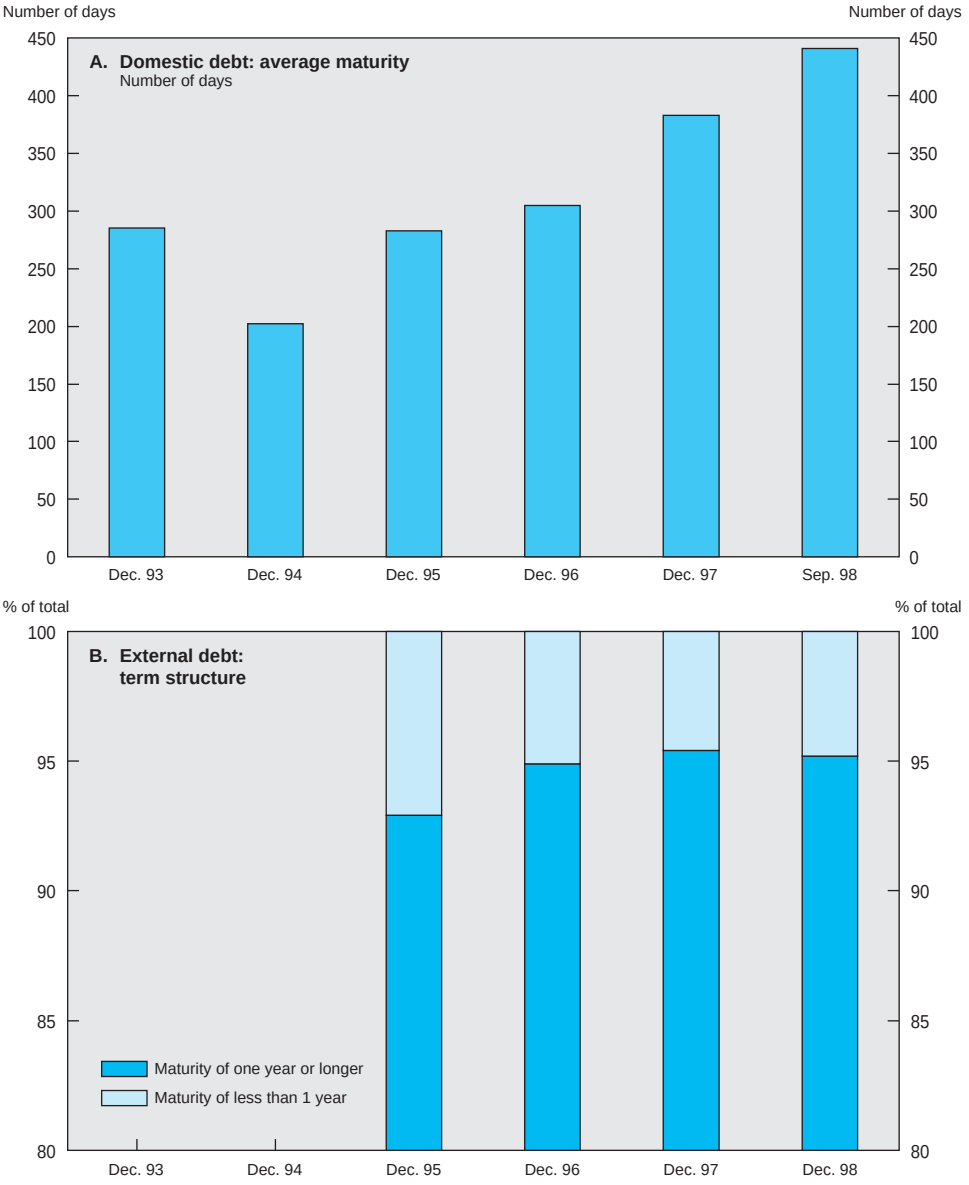
Stocks outstanding, end of period, as a percentage GDP



1. Excludes repurchase agreements and deposits with monetary regulation purposes.

Source: Ministry of Finance.

Figure 19. **Structure of public debt**
End of period



Source: Ministry of Finance.

valuation effects, the total net debt of the public sector fell from 38.5 to 26.8 per cent of GDP between December 1995 and December 1998 (Figure 17).[42] The fall was more pronounced for net external debt, in part because some of this debt was replaced by domestic debt through foreign reserve accumulation and its sterilisation.

The domestic debt management strategy has centred on improving maturity schedules and limiting interest payments. Since 1996, the authorities have stepped up the issuance of one-, two- and three-year treasury bonds (*Bondes*) and three- and five-year indexed bonds (*Udibonos*).[43] By December 1998, these two instruments already accounted for two-thirds of the total public sector domestic debt outstanding, up from a third at the end of 1995 (Figure 18). As a result, the average maturity of domestic public debt has lengthened (Figure 19).

Mexico's public debt in relation to GDP is low by comparison with most other OECD countries, and the external component – at about 20 per cent of GDP – is similar in size to that of other Latin American countries and has a maturity composition which currently implies rather limited refinancing needs. Short-term (less than one year) obligations accounted for only about 7 per cent of Mexico's external public debt in December 1998 (or 1.5 per cent of GDP). Amortisation of the external public debt is expected to amount to a little more than US$10 billion a year in 1999 and 2000, of which more than US$8 billion *vis-à-vis* official lenders. Nonetheless, during 1998 debt management was complicated by global financial turbulence. In the first half of the year, Mexico was able to place a total of US$3.1 billion of bonds in international capital markets – including a 10-year bond for US$1 billion with interest rates 288 basis points above the US equivalent (compared with 335 basis points paid in the January 1997 issuance). In the second half, however, with risk aversion hitting emerging market debt and interest rate spreads exceptionally large, the average cost of new issues by Mexico's public sector rose again, and for 1998 as a whole, it reached 323 basis points. In October, the Mexican authorities decided to draw on a credit line of US$2.66 billion from commercial banks which had been put in place in 1997 as a contingency measure. This credit line can be repaid and replaced by new bond issuance in case market conditions improve. In early 1999, faced with less favourable market conditions in the wake of the Brazilian crisis, the Mexican authorities announced that they were negotiating a stand-by loan from the IMF to help refinance IMF borrowings coming due in 1999 and 2000.

Economic policy for 1999-2000 and longer-term considerations

Together with the draft budget for 1999, the authorities presented an outline of their objectives for 1999-2000, including budget projections and a macroeconomic scenario.[44] Since the previous set of multi-year projections

included in the National Programme for Financing Development, 1997-2000 (PRONAFIDE) released in June 1997 has been overtaken by events, it is useful that the government provides revised information on where the economy seems to be heading and the concomitant policy stance. As argued in OECD *Economic Surveys* since 1995, by providing information on policy strategy over more than one year, the authorities can help stabilise private sector expectations. The PRONAFIDE was aimed at playing this role;[45] the policy actions sketched out for 1999-2000 are intended as an update of this programme.

The economic policy strategy for 1999-2000, which was presented to Congress in November 1998, reiterates the objectives of the PRONAFIDE, *i.e.* to foster sustainable output and employment growth and to lower inflation. Structural reform is to be continued along the lines followed in recent years (Chapter IV). It is also stated explicitly that policies will aim at creating conditions for an orderly transition into the next administration: by providing the economy with the required fiscal and financial discipline it will seek to avoid a repeat of the economic crises that up to now have characterised changes in administration (every six years). These concerns underlie the government's fiscal strategy for the two years ahead. The document stresses in particular the importance of maintaining a substantial level of public saving – the only component of saving that can be directly influenced by government action – to avoid excessive reliance on foreign saving and keeping the current-account deficit at a sustainable level.

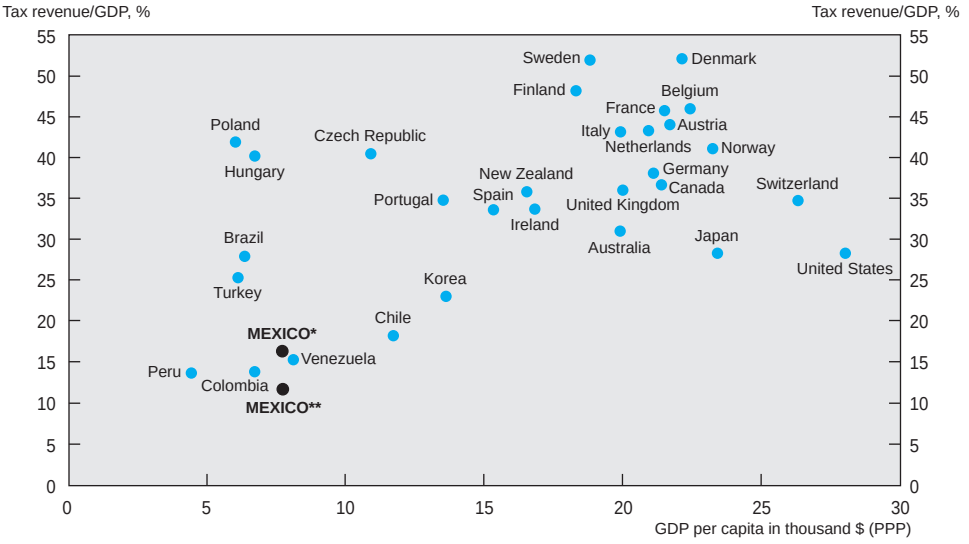
The vulnerability of the budget to oil prices and, to a smaller extent, interest rates requires that strong public finances be maintained to provide some margin of manoeuvre in case of adverse developments.[46] In addition, there are large contingent liabilities with respect to the banking sector support package and the cost of the social security reform that, altogether, will continue to put pressure on public finances to the amount of some 2 per cent of GDP per year. It should be noted, however, that by paying the transition cost of the pension reform, the government is actually reducing an implicit public debt (*i.e.* reducing contingent liabilities corresponding to the IMSS pension obligations over the longer run). On the other hand, the new scheme to finance infrastructure programmes is building up contingent liabilities for the public sector, as no expenditure is recorded in its financial accounts until payments start, upon completion of the projects; these payments will be netted out from the revenue stream generated by the projects. An increase in tax revenue would help loosen some of the constraints on fiscal policy. Mexico's tax in relation to GDP is very low even considering the country's per capita income, while there are large and urgent needs for developing human capital and physical infrastructure, in addition to meeting the above-mentioned contingent liabilities. A detailed review of Mexico's tax system and proposals for reform are presented in Chapter III.

Increasing the share of tax revenue in total revenue, thus reducing oil dependency, would give more predictability to fiscal policy over the short-term and improve conditions for strong sustainable growth in the longer run. There are two separate issues at stake. One is the volatility of oil-related revenue which makes public finances excessively vulnerable to exogenous shocks from one year to the next; this problem could be addressed by using a contingency, or stabilisation, fund.[47] Another issue is related to uncertainties about longer-term trends in the oil market. Medium- and long-term forecasts are particularly hazardous. According to one view, an upward trend in oil prices can be expected over time, because of scarcity of supply in the face of increasing demand from developing countries and China in particular. Long-term forecasts by the International Energy Agency, IEA, and by the US Energy Information Agency tend to assume that growing world demand will lead to world prices in the neighbourhood of US$20 per barrel for reference prices, because marginal supplies would have to come from non-conventional, relatively high-cost, sources by the period 2010-2020. Another view holds, to the contrary, that technological progress is such that it can dramatically reduce production costs, while allowing significant advances in the use of energy-efficient engines (for cars in particular). A number of forecasters expect price weakness in the medium term, with assumptions for the coming decade in a range of US$14 to 18 per barrel for reference prices – which would be equivalent to a price of the Mexican crude of US$10-14 per barrel. Furthermore, in the longer run, if the international effort to address the climate-change issue proves successful and leads to a reduction in fossil-fuel consumption, this could also result in a decline in oil prices.

III. The tax system: a need for strengthening the revenue raising capacity

The Mexican tax system encompasses a number of commendable features, and disincentives to work, save and invest appear less severe than in most other OECD countries. However, it still contains major deficiencies, which contribute to the fact that Mexico has by far the lowest level of tax revenues in relation to GDP among the OECD countries. As expenditures are also relatively low the budget outcomes are in general reasonably satisfactory – but the low capacity for

Figure 20. **Tax revenues and the level of income, 1996**[1]



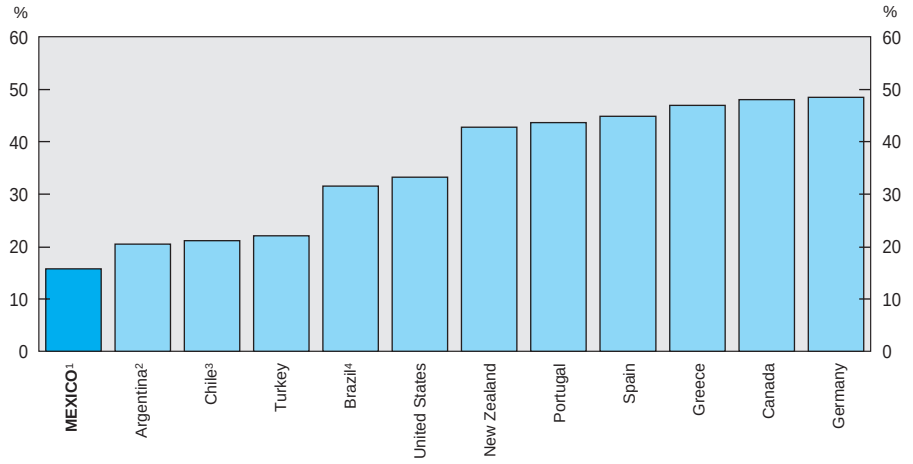
1. General Government including social security contributions. 1996 or nearest year available.
* Mexico including PEMEX contributions.
** Mexico excluding PEMEX contributions.
Source: OECD, *Revenue Statistics;* IMF, *Government Finance Statistics Yearbook,* 1995 and 1996 issues.

raising tax revenue severely limits the scope for public spending, including in areas where potential social returns are high. The experience of other countries also suggests that there is considerable scope for strengthening the potential revenue raising capacity of the Mexican tax system despite the wide distribution and low level of income. A number of countries, including Turkey and several Latin American countries, which have similar or even lower incomes per capita than Mexico, thus manage to raise substantially higher tax revenues in relation to GDP (Figure 20). Furthermore, as much as 30-40 per cent of government revenues in Mexico stem from oil taxes, mainly in terms of royalty payments from the state-owned oil company PEMEX, underlining the modest amount of revenue collected from non-oil sources.[48]

The economic and social context influencing the design of the tax system

The Mexican tax system has to be evaluated against a number of economic and social features that distinguish Mexico from most other OECD countries. First, the relatively low level of government expenditure in Mexico (Figure 21) is mainly a reflection of the low tax revenues and the need for prudent

Figure 21. **General Government expenditure – Average 1990-1996**
As a percentage of GDP



1. Mexico: federal government only.
2. Argentina: average for years 1990 to 1992.
3. Chile: central government only.
4. Brazil: average for years 1991 to 1995.

Source: OECD, *Economic Outlook, Economic Surveys* various issues; IMF, *Government Finance Statistics.*

budget outcomes. However, more public spending to enhance capacity and investment in areas like education, health, physical infrastructure and poverty alleviation, which are key to further economic and social development of the country, would clearly be desirable, provided it is well-designed and effectively implemented.[49] In addition, the social security reform and the cost associated with the banking sector support package (FOBAPROA) are expected to raise annual government expenditure by more than 2 per cent of GDP over the next 10 years (as noted in Chapter II). Financing these expenditures will require increases in the tax base, given the need to limit government borrowing.[50] Some theoretical and empirical analyses suggest that higher tax levels could be counterproductive in terms of their effect on economic growth, whereas other recent studies raise serious doubts about the robustness of this proposition (Box 2). When taking into account the way additional revenues are spent as well as the initial tax levels, growth effects from increased taxation might even be positive – in particular for a low tax country like Mexico. Furthermore, the structure of the tax system may be more decisive than the overall level of the tax burden in determining the balance between economic and social returns from

Box 2. Taxation and economic growth

The direction of tax effects on the level and growth of income is not always clear. Taxation may, in fact, be beneficial for the economy if it provides the financial basis for the provision of public goods that improve average living standards and social welfare. More and better public goods and services may also increase the productivity of private fixed and human capital and hence increase economic growth. This is in particular true in countries with underdeveloped human and physical capital. In addition, government transfers may reduce poverty and improve social cohesion. On the other hand, higher taxes increase distortions and may reduce saving, investment and work incentives. The adverse effects on economic efficiency may grow disproportionately with the increase in the tax level. As the net effect of taxation on economic performance depends on the level and structure of taxation, and whether tax revenue is spent in a productive or unproductive way, the benefits and costs of taxation are difficult to disentangle empirically. The Secretariat has previously examined the relationship between taxation and growth for a sample of OECD countries (not including Mexico, Turkey and other relatively low-income countries) and found that a 10 percentage point increase in the tax/GDP-ratio is accompanied by 0.5 percentage point lower growth (Leibfritz *et al.*, 1997). This is more or less in line (at least, in qualitative terms) with the findings of Barro (1991), Plosser (1992) and King and Rebelo (1990). However, the robustness of the negative relationship between taxation and growth has been questioned by several other studies, including Levine and Renelt (1992), Slemrod (1995) and Easterly and Rebelo (1993), who generally find non-significant or even positive correlation. In particular, there may be non-linearities implying a positive growth effect if taxes are increased from a low level and a negative growth effect if they are increased from a high level.

more expenditure and the costs of higher taxation. However, it is important to stress that any move to raise tax revenues will be sustainable only to the extent that the population at large perceives it is getting sufficiently value for money in terms of the quality of public spending.

Second, the dependency on oil-related revenues in government budgets implies that volatility in oil prices – besides imposing volatility on the real economy – significantly influences fiscal policy due to the strict adherence to announced budget targets. The consequence is that discretionary fiscal policy decisions tend to exacerbate the effects of oil price changes on the economy. For instance, three discretionary spending cuts were made during 1998 following the sharp drop in oil prices, allowing the deficit target of 1.25 per cent of GDP to be roughly achieved. If more revenues could be raised from non-oil sources, and at least some of it being used to improve the average budget position, this would allow the government to absorb a larger part of the oil price changes, thereby reducing the frequency of disruptive discretionary changes to expenditure and/or taxes.[51] This is also a pertinent concern in more general terms. Mexico has shared the experience of many Latin American countries where fiscal policy has been pro-cyclical more often than not – and this has contributed to the considerable overall macroeconomic volatility of the Mexican economy. The variation in the growth rates of real GDP and real private consumption has thus exceeded the OECD average by a factor of three and five, respectively, since 1980. Another macroeconomic factor shaping the design of the tax system has been a history of high and volatile rates of inflation that Mexico shares with most other Latin American countries.

Third, income is considerably lower and much more unequally distributed in Mexico than in the other OECD countries, except for Turkey. When measured by the Gini coefficient, post-tax income inequality in Mexico is only slightly lower than in Brazil and Chile. One illustration is that the most affluent 10 per cent of households receive nearly 40 per cent of total income, while less than 10 per cent of total income is earned by the 30 per cent of households with the lowest income levels.[52] Moreover, distribution has become slightly more unequal over the past 10 to 15 years according to household surveys. The low level and wide distribution of income give rise to economic and social problems, not least in terms of poverty (including malnutrition, high infant mortality, illiteracy and lack of basic health care) that are virtually absent in most other OECD countries. Furthermore, the wide income distribution in practice limits the base for individual income taxes to a relatively small group of people, who then face rather high tax payments in absolute terms. As the most affluent individuals typically have access to instruments of avoidance or evasion, this acts to limit the scope for raising revenues through direct taxation of incomes.

Table 14. **Estimates of informal sector size in selected countries**[1]

Informal sector employment in per cent of non-agricultural employment, mid-1990's

	Per cent	
	ILO estimates[2]	OECD estimates[3]
Mexico	**57**	**20-44**
Argentina	53	–
Brazil	56	–
Chile	51	–
Turkey	–	21

1. These estimates are subject to considerable uncertainty.
2. ILO, "World Employment 1996/97; National Policies in a Global Context".
3. Estimates published in OECD, *Economic Survey of Mexico* (1997) and OECD, *Economic Survey of Turkey* (1996).

Source: ILO; OECD.

Fourth, the informal sector in Mexico is large compared with other OECD countries, except perhaps for Turkey, although it does not appear to be out of line with that of other Latin American countries (Table 14). By several estimates, employment in the informal sector could be as much as one-third to one-half of total urban employment,[53] but since incomes are relatively low in the informal sector (as compared with the formal sector) its share in GDP is probably somewhat lower. Precise estimates are inherently difficult to obtain, but a recent study by the Ministry of Finance, based on micro-data from the INEGI household survey, estimated the size of the informal sector to be around 10 per cent of GDP.[54] The informal sector in Mexico is primarily a by-product of widespread poverty, lack of basic skills of a large proportion of the population and absence of a social safety net and is thus not comparable with the "undeclared" economy of more advanced economies (Box 3). It is a characteristic feature of the informal sector that it expands in economic downturns because it provides the only fall-back option for people being laid off in the formal sector. Integration of the informal economy is thus a complex issue that goes well beyond tighter administration and control and even beyond tax structure. From a taxation point of view, most activities in the informal sector are of a subsistence nature and would thus potentially represent a very limited, or even negative, income tax base in the short to medium term (as a result of the negative income tax for workers earning less than 3.2 times the minimum wage). However, the informal sector also facilitates evasion by higher income individuals and larger corporations, thus contributing to erosion of the formal sector tax base. It is thus important to institute mechanisms whereby these activities can be included in the tax base so as to mitigate problems of horizontal inequity, non-neutrality between economic activities and a sense of unfairness among taxpayers. Nevertheless, this challenge should not divert attention from more urgent problems related to the income and value added tax systems.

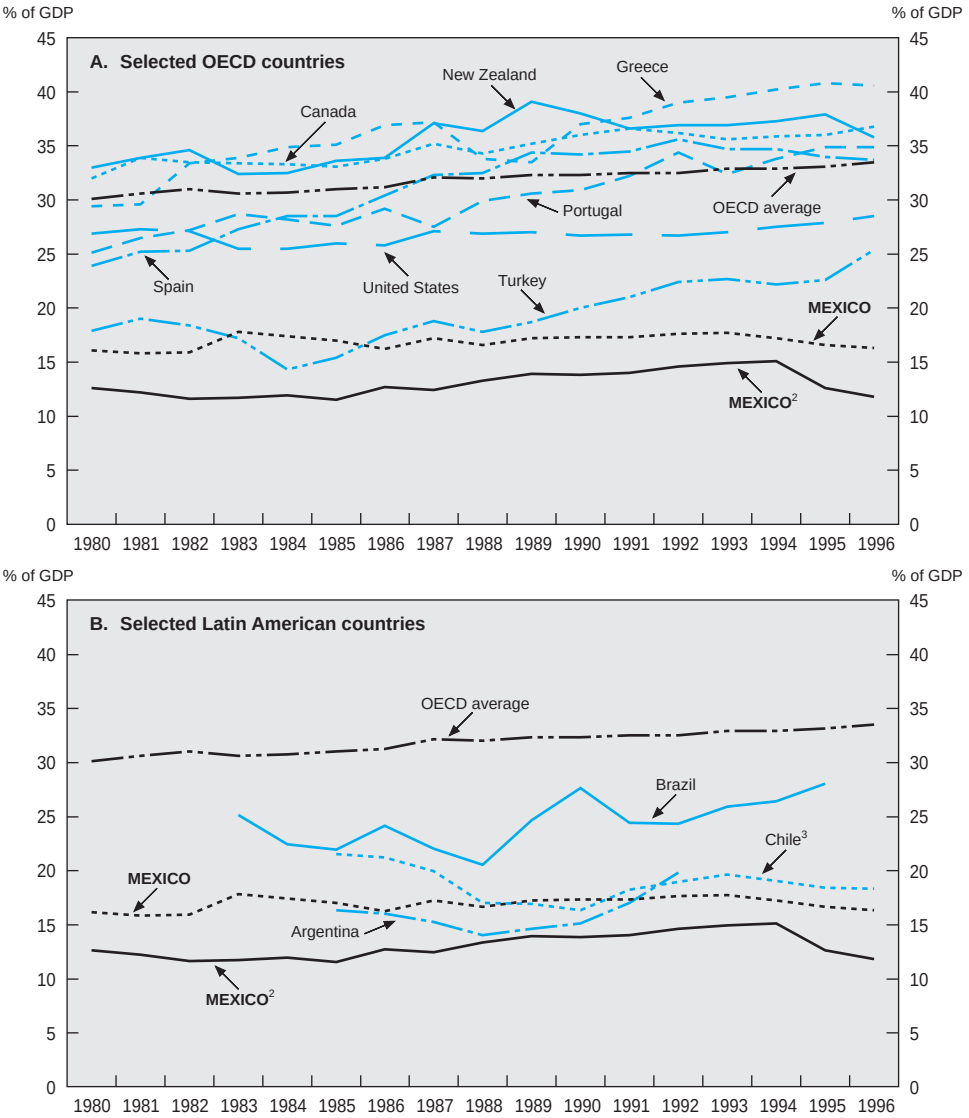
Box 3. The informal sector

The informal sector in Mexico covers a very diverse and heterogeneous set of economic activities, individuals and enterprises, and is thus a much wider concept than the limited "grey sector" in more advanced OECD economies. Although most of agricultural workers in Mexico operate without any formal contract, informality as such is often associated with non-agricultural activities and is thus seen as having emerged with urbanisation. It is particularly widespread in domestic services, retail trade and cleaning services. Operating on a very small scale, informal units very often have no accounts and premises, produce or sell low-quality/low-cost products and services, rarely comply with either various registration obligations or labour regulations, and are generally short-lived. They also tend to rely on family ties for labour contracting. However, part of the informal sector operates on a larger scale and it is, for instance, not uncommon for firms registered in the formal sector to supply part of their production into the untaxed informal sector. It also appears that street vendors and other retailers in the informal sector, at least to some extent, are supplied by non-registered and non-taxed imports and sometimes also stolen goods. Unfortunately, reliable quantitative estimates of these problems are not available. See also the *Economic Survey of Mexico* (1997).

Broad features of the Mexican tax system[55]

The Mexican tax system has undergone a number of major reforms since 1980. Primary features of these reforms have been the introduction of a general value-added tax (VAT); indexation to neutralise the effects of inflation on taxation; and integration of personal and corporate income taxes to ensure more neutrality between retaining and distributing profits. Since the late 1980s, the reforms have had two main objectives: first, to compensate the fall in revenue caused by the collapse in oil prices in 1986; second, to reduce distortions and inefficiencies, including the incentives to switch between different kinds of income and between tax systems of different countries. In particular, in line with the gradual opening of the economy which ultimately led up to the NAFTA (1994), there has been some convergence of corporate and individual top income tax rates with those of the United States and Canada. The reforms of the tax system (which are summarised in Annex I) made the system more equitable and efficient: marginal income-tax rates were lowered for corporations and individuals, and the tax base widened in line with the principles guiding tax reforms in several other OECD and Latin American countries.[56] Despite the improvements in the tax structure, the reform measures did not produce any lasting and significant increase in tax revenues (Figure 22).[57] The main reasons for this have been the continuation, extension or

Figure 22. **Total tax revenues**[1]
As a percentage of GDP



1. General government, including social security contributions.
2. Mexico, excluding PEMEX contributions.
3. Chile – Central Government only.

Source: OECD, *Revenue Statistics*; IMF, *Government Finance Statistics*.

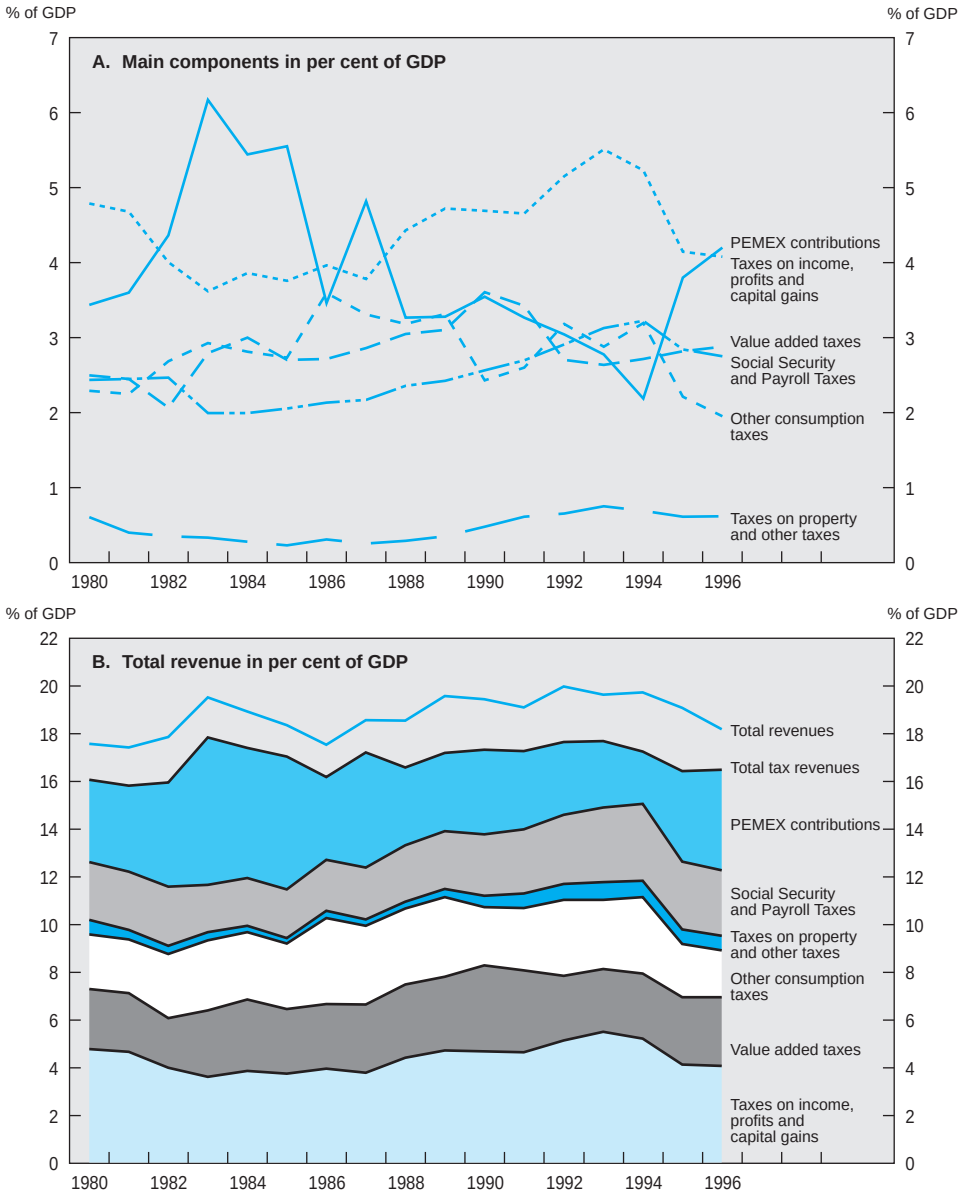
introduction of preferential regimes, various types of tax relieves, and – closely interlinked – the fact that reforms were apparently not successful in ensuring stronger compliance with the tax law.

Mexico's tax-to-GDP ratio of around 15 per cent is by far the lowest among OECD countries, although it should be noted that such ratios may not be fully comparable between countries.[58] Mexico, for instance, uses tax credits instead of social transfers to compensate low-income wage earners in the formal sector implying – all else being equal – lower levels of taxes and expenditure than in countries using transfers as compensation (although this effect should not be overstated as it amounts to less than half a percentage point of GDP). However, it is striking that Mexico collects nearly 10 percentage points of GDP less than Turkey, which at 25 per cent of GDP has the second-lowest ratio in the OECD area. Excluding royalty payments from PEMEX, the gap is even larger. Furthermore, there are few signs that Mexico is catching up – on the contrary, revenues as a percentage of GDP have fallen in recent years and the difference between Mexico and the rest of the OECD has increased slightly since 1980. Compared with other Latin American countries, Mexico's tax revenues (excluding PEMEX royalties) as a percentage of GDP lag behind those of Argentina, Chile and in particular Brazil.

All major sub-components of total tax revenues, excluding PEMEX royalties,[59] displayed a slowly increasing trend as a per cent of GDP from the early 1980s until 1994 (Figure 23). Then, the economic crisis in 1995 triggered a sharp decline in revenues from income taxes, social security contributions and other consumption taxes. VAT revenues remained almost unchanged as the rise in the standard VAT rate from 10 to 15 per cent that year offset the impact from the sharp drop in real private consumption and real imports. Any comparison of the mix of collected tax revenues with that in other OECD countries must be made cautiously, given the special features of the Mexican economy which have affected the collected tax mix. These include the low and unevenly distributed income, the importance of oil as a revenue source, the informal sector, as well as the severe economic crises of the past two decades. Given this caveat, the tax mix (excluding PEMEX royalties) has in many respects followed trends in many other OECD countries. There has been increasing reliance on social security contributions and general sales taxes, whereas import duties have become less important – in part, due to the trade liberalisation that started in 1986-87 and culminated with NAFTA.[60] Significant variations exist in the tax mix across OECD countries, but Mexico's current tax structure, excluding PEMEX, does not stand out as unusual (Figure 24).

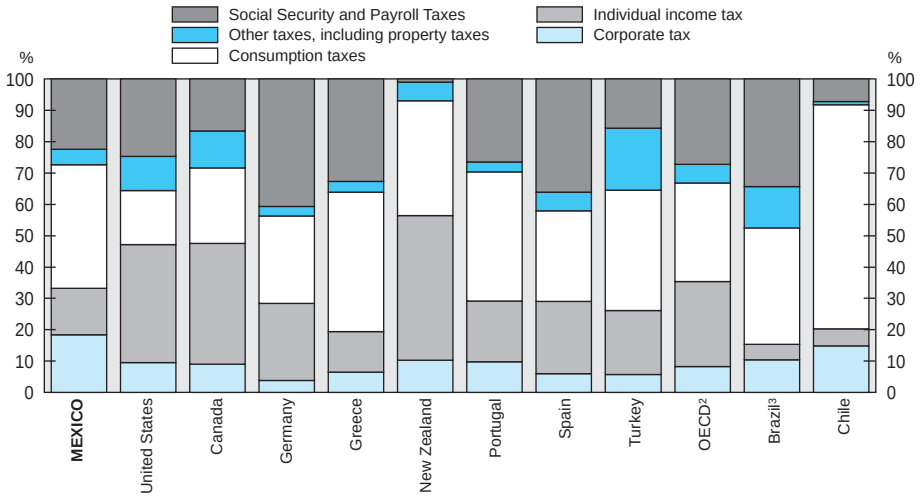
Cross-country comparison of the share of income taxation from, respectively, individuals and corporations is blurred by diverging accounting practices. In particular, Mexico includes in corporate taxes payments from self-employed as

Figure 23. **Tax mix in Mexico, 1980-1996**[1]



1. General Government.
Source: OECD, *Revenue Statistics.*

Figure 24. **Tax mix in selected countries, 1996**[1]



1. General Government. Contributions from fiscal monopolies are not included *(i.e.* PEMEX contributions are excluded for Mexico).
2. Unweighted average.
3. Figures for Brazil are for 1995.

Source: OECD, *Revenue Statistics;* IMF, *Government Finance Statistics.*

well as some withholding taxes (*e.g.* on payments to foreign residents and on interest income from bank deposits), which are allocated to individual income taxes in most other countries. This implies a tendency to report a higher share of corporate taxation in Mexico than elsewhere. Indeed, on a comparable basis the share of income taxes from corporations may be roughly on level with, or even below the OECD average.[61] Another salient feature is that the share of capital income in GDP is much higher in Mexico than in most other OECD countries, implying that the effective taxation of capital is very low (see below). The reported revenue share from individual income taxation and payroll taxes is below the average OECD share, but similar to that of Greece, Turkey and Brazil. However, this result is also affected by the differences in definitions.

Leaving aside preferential regimes in corporate and individual income taxation as well as in the VAT system, the statutory rates and features of the general tax regime in Mexico compare favourably with those of most other OECD countries (Annex III). The integration of personal and corporate income taxes

Box 4. PEMEX and subsidiaries' contributions to the federal government budget

PEMEX and its subsidiaries effectively contribute to the federal government budget through four channels:

- a hydrocarbons fee of 60.8 per cent of gross revenues (excluding VAT);
- ordinary VAT payments and customs duties;
- an "excess profit fee" on windfall gains from oil exports (for 1998, the levy was programmed to be 39.2 per cent of the excess of oil prices above US$15.5 per barrel).

According to Article 4 of the Federal Revenues Law for 1998, the hydrocarbons fee paid by PEMEX is formally made up of contributions from three different sources: an oil extraction royalty (which comprises an ordinary fee, an extraordinary fee and an additional fee), the regular income tax of 34 per cent and excise taxes (special tax on products and services, IEPS). As changes in excises are determined endogenously by the market price of oil and the regulated price at the pumps (Annex III), the oil extraction fees act as a residual to ensure that the aggregated contribution from PEMEX is equivalent to the budgeted 60.8 per cent of gross revenues.

The oil extraction royalty paid by PEMEX has fluctuated sharply since 1980, *cf.* Figure 23 above, contributing between 10 and 30 per cent of total federal revenues. In most recent years, *i.e.* 1993-98, the contribution has varied between 10 and 25 per cent of total revenue (2-4.5 per cent of GDP). Including the special gasoline tax, payments from PEMEX have been more stable, contributing annually 25-30 per cent of total revenue (4-5 per cent of GDP) during 1993-97. Net VAT payments from PEMEX have amounted to 0.3-0.6 per cent of GDP during the same period.

implies equal treatment of corporate financing through issuing equity or using retained earnings, while indexation to inflation ensures that tax distortions from high and volatile inflation rates are kept to a minimum (Box 5). The top marginal statutory tax rate on individual wage-income is among the lowest in OECD. The marginal corporate tax rate is close to the median of OECD-countries, but significantly higher than in some Latin American countries, including Brazil and Chile. The special asset tax on the net worth of enterprises serves as an important control device without creating any significant disincentive for investment.

The progressivity of the individual income tax system is ensured by the tax schedule itself, as well as by the progressive salary tax credit and the so-called tax subsidy. The various tax reliefs pull in the other direction. The tax credit, which is refunded in cash for low-wage earners with negative tax payments, was introduced in 1994 (with retrospective effect from October 1993) and extended in 1995 to alleviate adverse distributional consequences of the VAT increase from 10 to 15 per cent.[62] Unlike the system in some other countries, for

Box 5. Indexation to inflation

High inflation has several potentially adverse consequences for non-indexed tax systems: bracket creep (although it raises revenue, it is normally considered a non-transparent and not necessarily efficient way to finance government expenditure); high taxation of investment when depreciation rules do not take into account the effects from inflation (which gives incentives to invest in assets with low duration); distortion of saving and investment decisions when only nominal interest payments are taxed or deductible; overvaluation of capital gains which could lead to "lock-in effects" and thereby lower investment; and eroded revenues from specific excises. Moreover, high inflation is often accompanied by high volatility, which adds to tax uncertainties and an inefficient resource allocation if the system is not indexed. These costs of non-indexation have to be weighed against the administrative difficulties and complexities of introducing accounting for inflation, which is why many low-inflation OECD countries choose not to index their tax systems.

For Mexico, the semi-annual adjustment of brackets, subsidies and credits in the individual income-tax system effectively limits bracket creep. The full indexation of assets and liabilities ensures that only real interest income is included in taxable income and indexation of realised and accrued foreign exchange gains and losses render the tax system neutral regarding expatriation or repatriation of capital. Purchases of goods and services by corporations are immediately expensed so as to eliminate any effect of inflation on the calculation of taxable income arising from the treatment of inventories.

instance Canada, the tax credit is independent of family status and number of children, which tends to lessen its impact on progressivity as low-income families typically have more children. The tax subsidy has been implemented to mitigate problems of horizontal inequality imposed by the non-taxation of fringe benefits.[63] Combining the statutory rates, the tax credit and the tax subsidy gives the theoretical effective tax rate for each income category of wage earners in the formal sector (*i.e.* taxes expected to be paid under full compliance with the tax law). The tax system is progressive throughout all income categories (Table 15). Progressivity is highest at the lower-income intervals, whereas the rate structure flattens somewhat after ten times the minimum wage as the top marginal effective rate rises to just below 25 per cent. The theoretical effective tax rates points to another salient feature of the system, *i.e.* that the revenue-generating base of the system is extremely narrow. Indeed, by December 1997, formal sector employees with earnings less than 3.2 times the minimum wage[64] – amounting to some 70 per cent of all dependent employees – would in theory be paying negative income taxes, *i.e.* they would receive cash payments from the tax authorities. It should be noted, however, that the theoretical rates does not reflect actual amounts paid in tax by different income groups and that they do not apply to self-employed and

Table 15. **Tax progressivity and distribution of wage-earners**[1]

Income expressed as a multiple of minimum wage	Theoretical effective tax rate[2]	Number of workers (thousands)	Number of workers (per cent of total)	Cumulated number of workers (per cent of total)
1	–22.3	1 980.7	18.7	18.7
2	–7.4	3 740.3	35.2	53.9
3	–1.7	1 785.2	16.8	70.7
4	3.4	920.2	8.7	79.3
5	6.4	590.5	5.6	84.9
7	11.0	597.3	5.6	90.5
10	15.6	407.4	3.8	94.4
15	19.3	274.6	2.6	96.9
19	21.3	111.0	1.0	98.0
20	21.7	18.5	0.2	98.2
24	23.1	62.4	0.6	98.7
25+	23.4	133.3	1.3	100.0
Total	**. .**	**10 621.4**	**100.0**	–

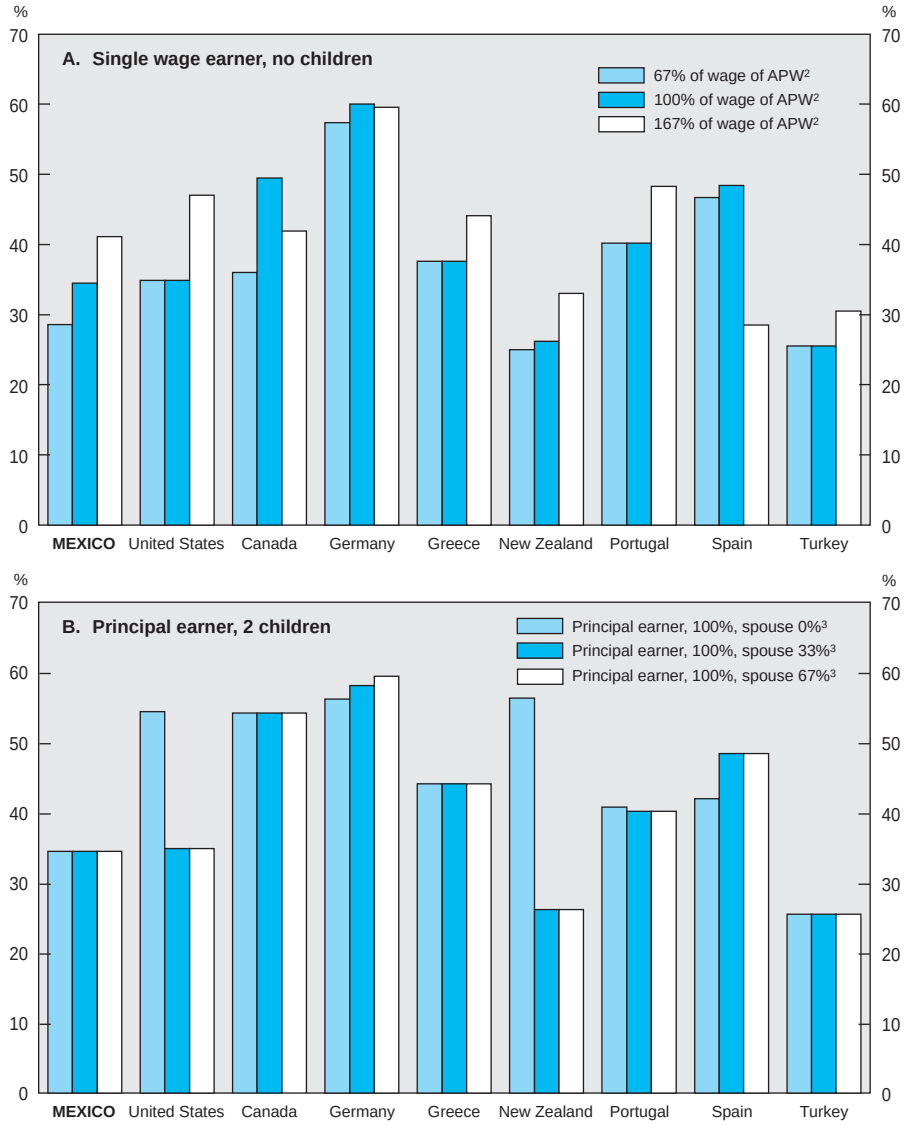
1. Based on IMSS data as a proxy for number of workers by tax bracket.
2. The theoretical effective tax rates combine for each income category the statutory rate, the tax credit and the tax subsidy, and thus measure the taxes expected to be paid under full compliance with the tax law. Fringe benefits are assumed to account for one-third of total remuneration.

Source: Ministry of Finance.

individuals working in the informal sector. Tax avoidance and evasion erode the revenue-generating capacity of the system and actual progressivity may differ from the theoretical in a non-negligible way.

Measured by statutory marginal tax wedges, distortions in the labour market are generally lower in Mexico than in most other OECD countries, including the United States and Canada, for most family types (Figure 25). Of the selected countries shown, only New Zealand and Turkey generally apply lower marginal tax wedges than Mexico.[65] Higher marginal rates affect both labour supply and demand by raising the wedge between what workers receive from pay increases and what firms pay, and empirical analysis suggests that employment falls as a result, thereby lowering potential output. The extent of the fall in employment depends on labour-market institutions and the wage-bargaining framework. In countries with flexible real wages, such as Mexico, taxes tend to get shifted back onto labour, reducing the take-home wage, while leaving labour demand relatively unaffected. Labour-supply elasticities, however, could be somewhat higher in Mexico than in most other OECD countries, at least with respect to wage earners with incomes above three times the minimum wage, *i.e.* those paying positive taxes, and self-employed, for whom informal sector employment appears to be a close substitute to the formal sector.[66]

Figure 25. **Marginal tax wedges by family type and wage level, 1996**[1]



1. Marginal tax rates covering employees' and employers' social security contributions and personal income tax, with respect to a change in gross labour costs.
2. APW: Average production worker in manufacturing.
3. Refers to proportion of wage of APW.

Source: OECD, *Tax/Benefit Positions of Employees.*

Incentives to save and invest do not appear highly distorted in Mexico compared with most other OECD countries. In particular, when measured by statutory marginal effective tax rates on investment in various physical assets and different sources of financing, Mexico stands out among the group of OECD countries as having one of the most neutral tax systems.[67] The required pre-tax rates of return of different investments, taking into account both corporate and personal taxation, are thus relatively uniform across various assets and financing sources (Table 16). The relative neutrality reflects mainly the integration of individual and corporate taxation[68] and the indexation to inflation. As a summary measure of neutrality, the standard deviation of these returns (amounting to 0.3) is lower in Mexico than elsewhere in the OECD, except for New Zealand. Moreover, the required level of pre-tax returns of each instrument in Mexico is not far from the OECD average, indicating that no excessive tax disincentives exist for each of these investments and financing sources. Income from private savings are receiving favoured treatment in terms of a tax rate of only 17-20 per cent for all levels of income, and only on the first 10 percentage points of nominal interest rates. Some non-neutrality across savings instruments is imposed by the tax

Table 16. **Required real pre-tax rates of return, 1996**[1]

	Sources of financing[2]			Physical assets[3]			Overall weighted average[4]	Standard deviation
	Retained earnings	New equity	Debt	Machinery	Building	Inventories		
Mexico[5]	**4.0**	**4.0**	**4.0**	**4.1**	**4.2**	**3.3**	**4.0**	**0.3**
United States	3.9	7.6	3.1	3.5	4.8	4.1	4.0	1.6
Canada	6.7	7.3	3.8	4.6	6.4	7.4	5.7	1.5
Germany	3.2	2.4	2.7	2.6	3.3	3.3	3.0	0.4
Greece	6.7	6.7	2.2	4.6	3.5	8.4	5.2	2.3
Portugal	5.7	5.3	3.1	4.7	4.7	4.9	4.7	0.9
Spain	3.9	3.4	2.7	2.8	4.0	4.0	3.4	0.6
New Zealand	4.5	4.5	4.5	4.4	4.3	5.0	4.5	0.2
Average	4.8	5.1	3.3	3.9	4.4	5.1	4.3	1.0

1. The required real pre-tax rate of return an investment has to earn in order to provide a personal investor with the same real post-tax return as a bank deposit earning a pre-tax 5 per cent real rate of interest.
2. Calculated using the following weights: Machinery 50%, buildings 28%, inventories 22%.
3. Calculated using the following weights: Retained earnings 55%, new equity 10%, debt 35%.
4. The weighted average uses weights indicated in footnotes (2) and (3).
5. Assuming full indexation to inflation and immediate expensing of inventories. The real interest rate is assumed to be 5 per cent, implying a 34 per cent tax on real interest income. No account is taken for the possibility of immediate deduction of certain kinds of investment (in practice amounting to around 15 per cent of total investment).

Source: Gordon, Kathryn, and Tchilinguirian, Harry (1998). "Marginal effective tax rates on physical, human, and R&D capital", *OECD Economics Department Working Paper No. 199* and OECD Secretariat's calculations.

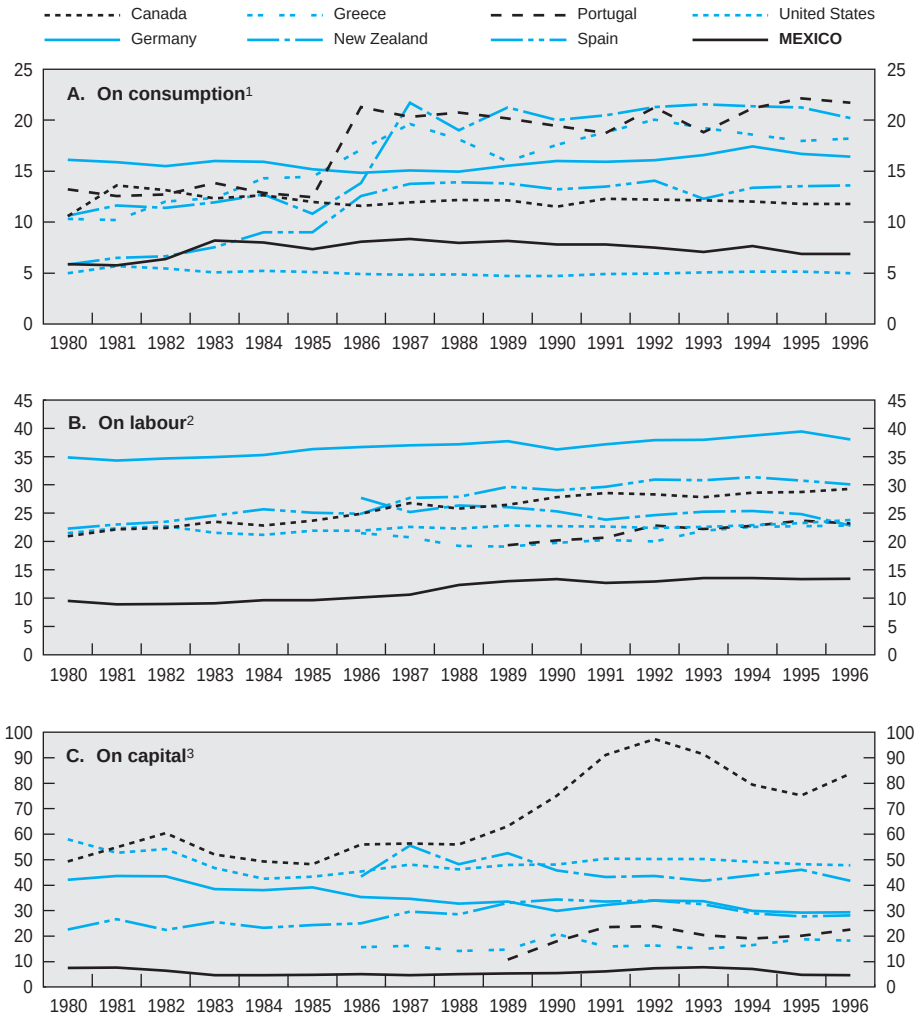
Box 6. Measuring effective tax rates

The estimates for effective tax rates on capital income, labour income and consumption are based on a modified version of the Mendoza *et al.* methodology, which has also been used by the European Commission. The three tax rates are measured as *ad valorem* estimates by classifying virtually all forms of tax revenue at the general government level into one of the three taxes. Each measure of tax revenue is then expressed as a fraction of the corresponding tax base, *i.e.* consumption, wages and the overall operating surplus in the economy. The method seeks to take into account net effect of tax rules and also incorporate those effects from taxes that are not filed by individual returns. A major caveat to the methodology is that all sources of household income are assumed to be taxed at the same rate (the estimates turn out to be quite sensitive to changes in this assumption, for instance if Mexican households are assumed to pay zero tax on capital income, it raises the effective tax on labour by 1 to 1.5 percentage points and lowers that on capital by the same amount). Another pertinent issue is whether to apply gross or net operating surplus (*i.e.* operating surplus *less* capital consumption) in the base for capital taxation. In theory, the net operating surplus should be used, but problems of measuring capital consumption sufficiently accurately in the national accounts could favour using the gross concept in some circumstances. Here, however, the net operating surplus is used in the base, implying that effective tax rates on capital are very high, and probably overestimated, for some countries, *e.g.* Canada. Finally, the numbers are not taking into account tax bases not measured by the national accounts – such as imputed rental income from owner-occupied housing, work at home, etc. A particular problem is the informal sector or "non-declared" economy. Income in this sector is to some extent covered by national accounts data, and is therefore included in the income bases used in the calculations, but it does not generate income tax revenues. This implies that actual tax rates on formal sector income are to some extent underestimated depending on the size of the undeclared economy and how much of this activity is included in official GDP figures. The methodology used here (described in more detail in Carey, *et al.* forthcoming) explicitly treats imputed wages to the self-employed as labour income, whereas the Mendoza *et al.* methodology attributes the entire operating surplus of unincorporated enterprise to capital. The estimates use revenue data collected by the OECD and national accounts data. Due to incomplete national accounts data for Mexico, assumptions have been made from the 1994 Household Survey, with respect to self-employment income and household property income.

exemption on interest income from various financial instruments, including government securities and instruments with maturity exceeding one year, as well as the tax deductibility of pension savings up to a certain limit.

Actual average tax rates collected on consumption, labour income and capital income are somewhat lower in Mexico than in most other OECD countries (Figure 26). The taxation of capital income is particularly low, but in general the numbers show that the low overall tax revenue to GDP ratio for Mexico reflects relatively low actual tax payments on all three bases. The advantage of using actual average rates, as opposed to statutory rates, is that they capture effects

Figure 26. **Average effective tax rates**



1. General taxes on goods and services *plus* excises, divided by private and government non-wage consumption excluding these taxes.
2. Individual income taxes paid on wages (including tax on self-employed labour income) *plus* Payroll Taxes and Social Security contributions, divided by wages and salaries (including imputed wages of self-employed), and including employer's contribution to social security and to private pension schemes.
3. Individual income taxes paid on self-employment and property income received by households *less* imputed wages, *plus* corporate taxes and taxes on property divided by the net operating surplus of the economy *less* wages for self-employed.

Source: OECD, *Revenue Statistics, National Accounts,* OECD Secretariat's calculations.

from preferential regimes and other tax-relief as well as overall compliance, which are not reflected in the effective statutory rates. The levels of actual average rates are not adequate for comparing overall tax distortions across countries, however, since low rates may primarily reflect preferential treatment of selected groups of taxpayers and high taxation of others. Moreover, average levels of taxation are not necessarily good approximations of the incentives created by the tax system, which are primarily determined by marginal rates facing the individual taxpayer. Finally, besides the methodological problems discussed in Box 6, a major complicating factor in analysing tax effects is determining who finally bears the incidence of the tax – since taxes levied on a given base may end up being shifted along the production and distribution chain.[69]

Main issues for strengthening the tax system

The structure of the general Mexican tax system in terms of statutory rates and features appears satisfactory overall, but the system does not seem to be robust enough to meet prospective financing needs and there are a number of areas where improvements would contribute to both better equity and efficiency. The key challenge facing the tax system is to increase the potential revenue raising capacity, primarily by broadening the base and improving collection, while keeping the rates near the current low levels. This would strengthen revenue significantly without increasing distortions much, and at the same time it would alleviate problems of horizontal inequity and help reducing budget volatility. The main concerns are the preferential regimes and the closely related deficiencies in tax administration. The preferential regimes, besides diluting revenue and complicating administration *per se*, create significant loopholes and facilitate evasion. This, in turn, hampers the effectiveness of the tax administration and creates major problems of horizontal inequity (some pay taxes, others do not). On the other hand, inefficiencies in the tax administration itself may compound negative revenue consequences arising from preferential regimes to the extent that inadequate surveillance and enforcement induces taxpayers to falsely register their activity or transactions in these regimes. Lack of administrative capacity at the sub-national level could also be a major impediment for decentralising more expenditure responsibilities and taxing powers to states and municipalities. Finally, considerations related to the informal sector, income distribution and the impact from oil price volatility on the budget are also relevant with respect to reforming the tax system.

Broadening the base: preferential regimes and the informal sector

Preferential regimes (or tax expenditures in a broad sense) are major contributors to the low level of revenues. Not only do they lead to low revenues

per se, they also open multiple possibilities for non-compliance through tax evasion and avoidance. Moreover, preferential regimes may introduce considerable non-neutralities in terms of incentives to various economic activities. The exemptions and zero-rated goods and services in the VAT system, as described in Annex III, have resulted in a significant reduction of the base of taxed consumption[70] and also contributed to low VAT compliance, as some transactions are falsely attributed to the zero-rated base. Moreover, the system of refundable VAT credits exacerbates the incentives to fraud and complicates administration further.[71] According to the authorities' estimates, VAT compliance in Mexico is only around 63 per cent – *i.e.* only 63 pesos were collected out of every 100 pesos that should have been collected (Table 17). This is below compliance levels in other OECD countries for which comparable data are available, including Canada and Portugal, as well as some Latin American countries. The narrow base and the induced low compliance implies that Mexico ranks lowest among the OECD countries in terms of VAT productivity.[72] This is also the case for the effective VAT rate[73] in per cent of standard rate – again indicating that only a small part of the potential base is actually being taxed.

Table 17. **Indicators of VAT compliance and productivity**

Per cent

	VAT compliance[1] (1993)	VAT productivity[2] (1996)	Effective VAT rate[3] in per cent of standard rate (1996)
Mexico	**63**	**19**	**29**
Canada	77	37	62
Germany	..	45	77
Greece	..	51	61
Portugal	86	47	79
Spain	..	34	59
New Zealand	95	68	123[4]
Turkey	..	40	61
Argentina	69	10	..
Chile	77	51	..

1. VAT compliance is VAT actually paid in per cent of theoretical VAT collected under full compliance with existing tax laws.
2. VAT productivity is VAT revenue in per cent of GDP divided by standard VAT rate.
3. Effective VAT rate is VAT revenue divided by base (*i.e.* consumption).
4. The general sales tax at standard rate is levied on 60 per cent of the value of the supply for long term stay in a commercial dwelling which may partly explain why the number exceeds 100.

Source: Ministry of Finance; OECD *Revenue Statistics*; *National Accounts*; IMF, *Government Finance Statistics*; OECD, *Economic Survey of Turkey* (1997); and OECD Secretariat's calculations.

Preferential treatment in the corporate income scheme concerns primarily the simplified regime for certain sectors and the small taxpayers regime (Annex III). In addition, until the 1999-tax package tightened consolidation rules and eliminated the immediate deduction scheme for specific kinds of investment, these two features represented important preferential regimes and correspondingly significant tax losses.[74] The sectors benefiting from the *simplified regime* – *i.e.* agriculture, fisheries and land transportation, which together represent around 10 per cent of GDP and 25 per cent of total employment – pay only little or no corporate income tax, as taxation is based on cash flows, and only on resources withdrawn from the enterprise. These sectors are thus *de facto* liable to the assets tax only.[75] A large part of taxpayers in these sectors are small family-run businesses that would be lightly taxed in any circumstance (under the regime for small taxpayers), but also larger corporations are avoiding income taxation under the simplified regime. There seems to be little economic justification for the special treatment of medium- and large-sized corporations in agriculture and land transportation. The *regime for small taxpayers*, on the other hand, serves the useful purpose of making formal sector economic activity more accessible – in terms of less complicated rules and requirements – for self-employed and micro-businesses who would otherwise have an incentive to operate in the informal sector.

Finally, the non-taxation of fringe benefits in the individual income-tax system leaves around one-third to one-fourth of total labour income in the formal sector untaxed. Moreover, the non-taxation has necessitated the introduction of a tax rebate – the fiscal subsidy – for those receiving a relatively small share of remuneration as fringe benefits, in order to mitigate adverse consequences for horizontal equity. This, in turn, means that tax revenues and the base of taxpayers are lowered further and adds to the complexity of the tax system. Fringe benefits are included in the income-tax system in most OECD countries,[76] but the tax treatment of different types of fringe benefits varies significantly across countries. Only New Zealand and Australia have separate fringe benefit taxes.

In sum, Mexico applies preferential regimes on a broad range of economic activities and incomes – a practice which causes a major drag on tax revenues, uneven incentives across activities and severely complicates tax administration. A rough calculation would suggest that the direct effect of completely eliminating the major preferential regimes described above could provide as much as 3 per cent of GDP in additional tax revenues.[77] This estimate does not take into account additional positive revenue effects from increased compliance if preferential regimes were removed. Nor does it consider any potential dilution of revenues due to substitution between formal and informal activity, increased tax avoidance or the possibility of increased substitution between consumption and savings, as well as between leisure and work. On balance, however, these indirect effects are likely to be small compared with the significant increase in revenues

that would certainly arise as a direct consequence of the elimination of preferential regimes.

The informal sector would appear to be another target for base widening. Individuals and enterprises in the informal sector do not pay income taxes and retail sales from vendors in the informal sector are, to some extent, escaping consumption taxes (although often only in the final transaction). However, the informal sector may not have a large direct impact on tax revenues as the larger part of it seems to be related to micro-business and low-skilled/low-income employment. By some estimates as much as 90 per cent of the workers in the informal sector earn less than three times the minimum wages and would thus be eligible for a negative tax payment were they registered as wage earners in the formal sector.[78] Moreover, most of the corporate activity would be taxed under the small taxpayer system. It is thus doubtful whether incorporating larger parts of the informal sector into the formal sector would provide much additional tax-revenues – in fact, it could even be a drag on tax revenues, at least in the short term, due to the tax credits. However, the non-taxation of higher-income individuals and firms in the informal sector and the size and visibility of informal activities have negative effects. They contribute to horizontal inequity, lower tax morale and complicate the political process of raising revenue from those already paying taxes.[79] The task of integrating the informal economy is complicated by the fact that a major proportion of the labour force is self-employed,[80] and that self-employment in the formal sector and informal activity are often combined or close substitutes. The same problem relates to the large – and growing – number of small and micro-enterprises.[81]

Two other potential areas for increasing tax capacity would be excises and taxation of property. On both accounts, Mexico falls behind most other OECD countries and also some of its Latin American neighbours (Table 18). With respect to excises, the recent step to increase the tariff rates on gasoline and diesel should be seen against the fact that fuel prices by international standards are very low in Mexico, although not lower than in the United States and Canada. In contrast to most other OECD countries, which view inherited wealth as a legitimate base for redistributive taxation, Mexico does not levy taxes on individuals' wealth or on inheritance and gifts. Such taxes imply double taxation of capital and may thus have some adverse impact on investment, although this effect is likely to be small in most countries. Moreover, they are often difficult to enforce, and in general they do not provide significant revenues in countries where applied. Taxation of immovable property is administered by the municipalities with the tax being levied on the value of property. A main barrier for raising revenue from this tax appears to be inadequate valuation practices and lax local administration and enforcement of collection.

Table 18. **Revenues from property taxation and excises, 1996; gasoline prices, 1998**

	Taxes on property per cent GDP (1996)	Excises per cent GDP (1996)	Unleaded gasoline price including taxes US$/litre (1998 Q2)
Mexico	**0.3**	**1.2**	**0.40**
United States	3.1	1.5	0.33
Canada	3.8	2.0	0.35
Germany	1.1	2.8	0.91
Greece	1.4	6.1	0.70
New Zealand	2.0	2.0	0.44
Portugal	0.9	5.1	0.90
Spain	1.9	2.7	0.75
Turkey	0.5	0.3	0.73
Argentina[1]	1.4	1.0	..
Brazil[2]	0.4	..	..
Chile	0.1	2.0	..

1. Figures for Argentina are for 1992.
2. Figures for Brazil are for 1993.

Source: OECD, *Revenue Statistics*; IMF, *Government Finance Statistics*; IEA, Energy prices and taxes, 2nd Quarter 1998.

Tax administration

Significant efforts have been made in the last few years to improve the effectiveness of tax administration, but there is still scope for improvement in several areas. A major complicating feature for tax administration is the existence of numerous preferential regimes and loopholes, and to a lesser extent, the significant informal sector. The attempts made in recent years to improve administration have covered a broad range of measures. Control of corporate income taxes has been strengthened through the assets tax, which was introduced in 1989 (and later amended on several occasions with the aim of reducing negative effects on investment decisions). Tax administration of small enterprises has been made easier by the introduction of a special regime for small taxpayers, which should also facilitate integration of informal enterprises into the formal economy. Furthermore, the reform of the pension and health-care system is expected to provide better incentives for tax compliance. Various administrative measures have also been taken: to enhance control more widely; to specify and improve the legal framework, including sanctions and penalties in case of fraud (which now include the possibility of prison sentence); and finally, in 1997, to modernise the administration itself by establishing an autonomous administrative body under the Ministry of Finance,[82] and creating a professional "career" for employees of the tax administration. Box 7 summarises the measures implemented in the last few years to improve collection and control. The reform strategy mainly relies on

Box 7. Recent initiatives to improve tax administration and control

Several measures to *modernise and simplify the tax administration* have been taken in recent years, including:

- creation of the Tax Administration Service (SAT) from 1 July 1997 as a decentralised institution with budget and management autonomy, to administer tax and customs policy designed by the Ministry of Finance and to issue norms to accomplish its function;
- establishment of a professional career for taxemployees;
- changes in the regulation of consolidated enterprises;
- changes in definitions and sanctions for tax fraud including strengthening the taxpayers' legal certainty;
- new measures against the use of tax heavens;
- reduction of information requirements in tax forms and requirements to keep accounting records (from 10 to 5 year);
- electronic file returns and payments (*e.g.* through Internet).

Main developments to *improve surveillance and enforcement* have been:

- introduction of a special regime for small enterprises;
- tightening of transfer pricing rules;
- shortening the delays for initiating the surveillance process by updating, on a permanent basis, the federal tax registers to be able to identify more rapidly the non-submission of a tax declaration;
- improved incentives for states to carry out audits more efficiently, as they are allowed to keep additional revenues;
- monitoring changes in taxpayers' behaviour and differences in contributions over time;
- identification and registration of the 8 000 largest taxpayers in a special database. These taxpayers contribute some 60 per cent of all tax revenue. The idea is to develop a close relationship with these taxpayers in order to minimise resources devoted to them, thereby freeing up personnel to increase the effort towards other taxpayers;
- improved auditing of small and medium-sized enterprises' issuance of invoices. In the first nine months of 1998 nearly 100 000 small and medium-sized enterprises were audited (out of a stock of some five million) compared with only 6 000 in all of 1997. Sanctions were imposed on nearly 70 000 enterprises and some 6 000 were closed temporarily. The sample has mainly been chosen in typical "problem areas" like: medicine, restaurants, jewellery stores, construction, material retailers, liquor stores, department stores and shopping malls;
- more control of VAT evasion and the informal sector by enhanced import registration. Under-valuation and non-registration of imports are important sources of VAT evasion. Besides being a tax collection problem the under-valuation of imports creates an unfair competition for the formal private business sector. A database has been constructed to register importers and exporters by tariff codes. The database contains information of the volumes and unit prices, making it easier to identify "suspiciously low" unit prices. Exchange of information with US customs, US Internal Revenue Service and the Canadian authorities is increasing, but it is still far from being fully effective.

tighter supervision of the largest taxpayers, more – and better targeted – auditing, as well as tighter surveillance of imports, which typically constitute a "grey area" in terms of taxation.[83] With respect to filing requirements, corporate entities in the general regime are obliged to submit detailed information to tax authorities on a regular basis. Individual taxpayers with income from more than one source (and above a certain limit) are obliged to file an annual tax form. Wage income is subject to monthly withholding of income tax at source, and corporate taxes are paid as monthly advance payments (or quarterly for small enterprises). Taxes on interest payments are withheld at source.

Although it is still too early to evaluate the full achievements of the administrative initiatives of recent years, some indicators point to improved effectiveness. Fiscal presence, *i.e.* number of audits per taxpayer (corporate and self-employed), has thus increased dramatically since 1996 and is now only slightly below the level in *e.g.* Spain. Moreover, there has been a shift in the structure of auditing towards specific audits, which have proven to be relatively efficient in terms of revenue collected per auditor. But, even though the reform of administration seems to be moving in the right direction, there is still scope for increasing effectiveness further – in terms of integrating registers in the social security system with that of tax authorities; improving import registration and control, for instance by extending even further the already close co-operation with US and Canadian customs and tax authorities; improving third party reporting requirements and the processing of tax information, *e.g.* through simplified individual tax forms, electronic processing etc.; increasing skills at the local tax administrative levels; and strengthening enforcement and the legal framework for collecting taxes, including legal certainty for tax payers. In the longer-term it would also be appropriate to integrate the municipalities' registration of street vendors and real estate valuation registers into the central tax registers.

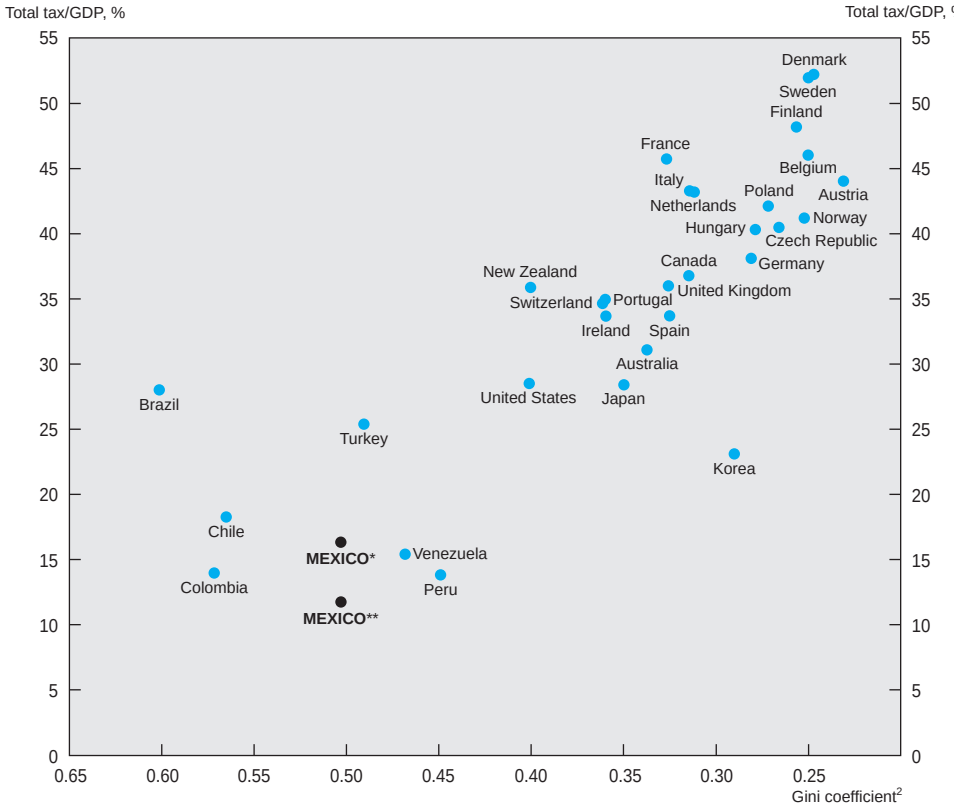
The tax system and income distribution[84]

Even though distributional objectives to some extent can be achieved through progressivity in the tax system, it is widely recognised that such objectives are more efficiently handled from the expenditure side. The issues that Mexico will have to address concerning income distribution and taxation thus primarily involve how much progressivity is needed to raise sufficient revenues in order to allow for more public spending – including on improving distribution. Related concerns would be how incentives are affected, how the progressivity should evolve along the income scale and to what extent distributional objectives can be achieved through progressivity. A major consideration in this respect is how changes in marginal statutory rates for the most affluent income groups would affect avoidance and evasion, and thus influence the actual progressivity of

the system (*i.e.* taxes paid by income deciles).[85] The wide income distribution in Mexico significantly limits the base for individual income taxation. First, 70 per cent of wage earners in the formal sectors are not paying income taxes at all, but receive negative tax payments from the tax credit. Out of 15 million registered individual taxpayers in Mexico, only some four to five million are thus paying income tax – which should be seen against the needs of a population of 90 million people. Second, although tax rates for high-income individuals are still relatively low in Mexico, there may be limited scope for significantly increasing their effective tax rates as these income groups may have access to instruments for shifting income into domestic or foreign low-tax regimes. Mexico is thus in line with an international pattern showing a distinct relationship between income distribution and tax revenue, whereby higher income inequality (after tax), measured by the Gini coefficient, is associated with lower tax revenue as a percentage of GDP (Figure 27).

The tax system influences the distribution of income, primarily through its overall progressivity, but also indirectly through effects on relative wages and employment.[86] The statutory progressivity of the Mexican tax system ranks roughly in the middle compared with other OECD countries (Figure 28), when measured as the difference between the top statutory rate and the average tax rate for an average production worker.[87] However, as is evident from Figure 29, progressivity in theoretical effective rates (combining the statutory rates, the tax credit and the tax subsidy, as defined above) is much steeper at the lower end of the income scale than for higher incomes, in particular those above ten times the minimum wage. Moreover, progressivity has increased quite sharply since 1993 for incomes between two and four times the minimum wage, due to a proportionally larger reduction in effective tax rates for these income groups than for incomes above ten times the minimum wage. This should be seen against a decline in real wages of around 18 per cent since 1994, implying that problems of poverty at lower levels of income, if anything, have increased.[88] The increased progressivity at lower income levels is also likely to have exacerbated problems of under-declaration of income.[89] A flattening of the progressivity at the lower income levels could, for instance, be achieved by extending the interval over which the tax credit is withdrawn and widening the brackets of the income-tax system, although this measure would probably involve substantial cost in terms of revenue foregone. Increased progressivity at the higher income levels could be obtained through higher rates[90] or by replacing tax reliefs, which are deducted at the highest rate, with tax credits. The latter option is not likely to be very effective, however, since Mexico only allows for rather limited tax deductions at the individual level. In order to achieve higher actual progressivity for high-income earners it would also be necessary to improve enforcement to make the recent increase in top statutory rates more effective. Another relevant aspect of taxation and progressivity is the allocation of income between monetary and

Figure 27. **Tax revenues and the distribution of income after tax, 1996**[1]

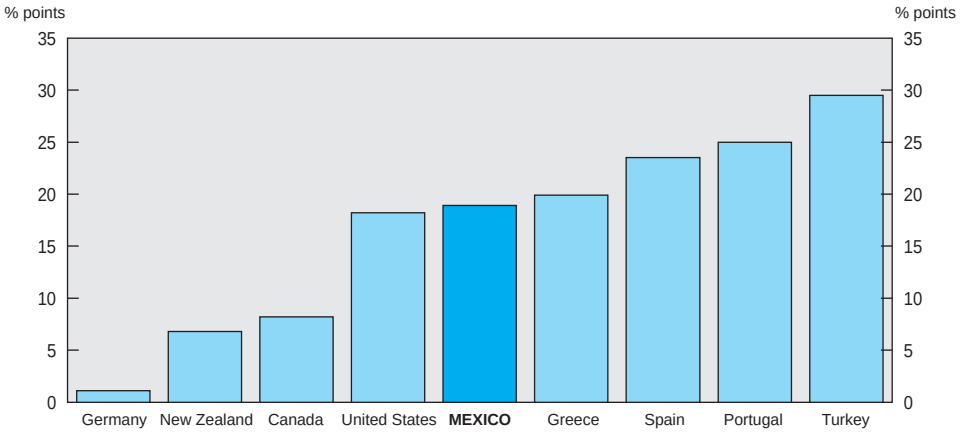


1. General Government, including social security contributions. 1996 or nearest year available.
2. The Gini coefficient is a measure of income inequality: the higher the coefficient, the wider the income distribution.
* Mexico including PEMEX contributions.
** Mexico excluding PEMEX contributions.
Source: OECD, *Revenue Statistics;* IMF, *Government Finance Statistics Yearbook,* 1995 and 1996 issues.

non-monetary income. In the case of Mexico, this appears to contribute to a higher actual progressivity (Box 8), although the effect should probably not be overestimated.

Another channel of redistribution is the taxation of consumption. Many OECD countries have traditionally applied lower rates or exemptions to basic staples like food and medicine in order to introduce some progressivity into the VAT system. This typically implies a considerable dead-weight loss as people at

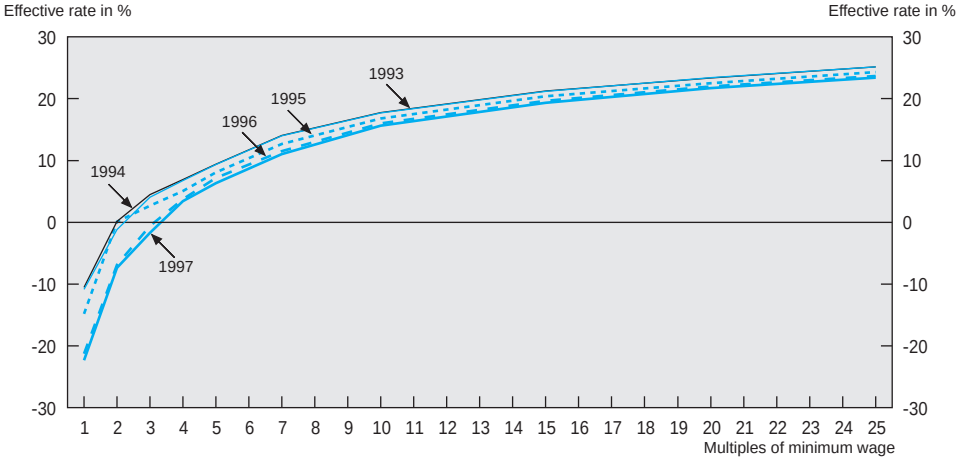
Figure 28. **Overall progressivity of individual income tax systems in selected OECD countries,[1] 1996**



1. Progressivity is measured as the difference between the top marginal personal income tax rate (including social security contributions) and the average statutory tax rate for an average production worker.

Source: OECD, Tax Database, *Tax/Benefit Positions of Employees,* Tax Equations and OECD Secretariat's calculations.

Figure 29. **Theoretical effective tax rates for different levels of income[1]**



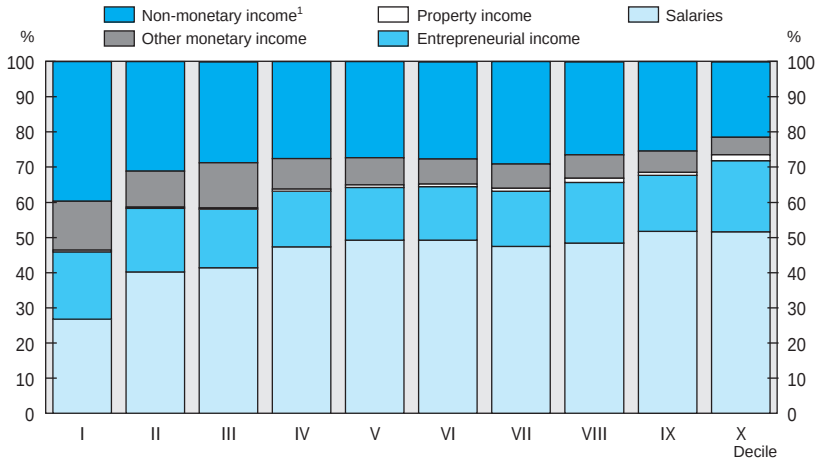
1. Statutory tax rate stemming from the combined effects of the income tax schedule, the fiscal subsidy, the tax credit and assuming fringe benefits are one third of total remunerations for all income levels.

Source: Ministry of Finance.

Box 8. Income allocation for different income levels

The Mexican after-tax income distribution shows a pattern in which the share of wage income increases with overall income, and the share of non-monetary income is highest among the lower-income deciles (Figure 30). Although the share of property income increases with income, it is still a relatively small part of total income at the highest income decile – amounting only to around 2.2 per cent of total income. The non-monetary income consists, to a large extent, of own-consumption (for example, of home-grown farm products) at the lower deciles, whereas it almost entirely reflects imputed rent of owner-occupied housing at the higher end of the income scale. Progressivity in the individual income scheme is thus compounded by the differences in monetary versus non-monetary income, and consumption taxes are more progressive than otherwise (*ceteris paribus, i.e.* even without considering the incidence of the zero-rated base), due to the distribution of own-consumption.

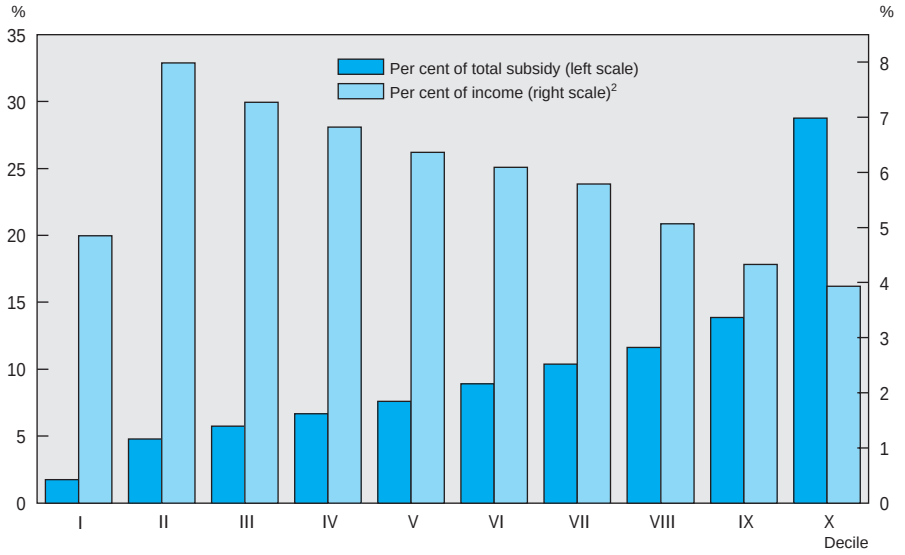
Figure 30. **Income sources by decile, 1994**



1. Including consumption of self-production, imputed rent of owner-occupied housing, etc.
Source: INEGI, *Encuesta nacional de ingresos y gastos de los hogares 1994,* México, 1995.

higher income levels are compensated most in absolute terms since they consume more of all goods and services. In the case of Mexico, the total implicit subsidy due to the zero-rating on food can be estimated at some 1.8 per cent of GDP.[91] The distribution by income deciles shows that the highest decile captures nearly 30 per cent of this amount, while the lowest three deciles together receive

Figure 31. **Distribution of implicit vat subsidy, 1996**[1]



1. The implicit VAT subsidy at each income decile is estimated by total expenditure on food multiplied by the standard VAT rate of 15 per cent. The consumption of food is approximated from total household expenditure according to the 1996 household survey and the relative weight on food and non-food per decile from the 1994 survey. The estimate of subsidy does not consider substitution effects between consumption and savings or within the consumption basket when prices change.
2. Income per decile is from the INEGI 1996 Household Survey.

Source: INEGI, *Encuesta nacional de ingresos y gastos de los hogares 1994 and 1996*, Mexico; OECD Secretariat's calculations.

only 12 per cent of the value of the subsidy (Figure 31). Measured as a per cent of household income, the implicit subsidy amounts to nearly 4 per cent for the highest decile and 5 per cent for the lowest decile. If the zero-rate on food were fully eliminated, all else being equal, total tax revenues would increase by the value of the subsidy, *i.e.* 1.8 per cent of GDP, or nearly 15 per cent.[92] A politically more viable option would probably be to retain the zero-rating on basic staples, like milk, bread, tortillas and eggs. Applying the zero-rating only to these goods would imply an implicit subsidy of only 0.4 per cent of GDP. Moreover, compensating, for example, the first four income deciles for the loss of subsidy on the rest of their food consumption (via direct income transfers or an extended tax credit) would only amount to some 0.25 per cent of GDP,[93] leaving a total budgetary gain of more than 1 per cent of GDP. The extent to which compensation should be given through retaining zero-rating on a few basic staples or through expenditure schemes, or both, is a political question tightly linked to the overall distributional

Box 9. Taxing the affluent more

Given the highly unequal income distribution and low revenue raising capacity in Mexico, taxing the highest-income individuals more heavily have advantages, although this involves questions about both value judgements and economics.[1] One possibility would be to raise income-tax rates for the highest-income individuals. The tax revenue effects of such a move are often uncertain due to increased avoidance and evasion incentives for the high-income groups – in particular for countries where top marginal tax rates are already high. This uncertainty, in turn, points to the importance of a strong enforcement of tax compliance to accompany any increase in top marginal rates. There is also a risk that increased taxation could entail a reduction in "entrepreneurial" and other socially productive activity among these tax payers.[2] Another pertinent concern, especially for countries with a structural need to increase private savings, is the effect that higher income taxes on the richest individuals would have on total private savings. It is often assumed that saving ratios increase with income, thus implying a trade-off between income distribution and private savings. However, this assumption could be called into question both empirically and theoretically: in a life-cycle framework, for instance, there would be a positive savings effect in redistributing income from the elderly to the young through a more progressive tax system (Slemrod, 1998).

Mexico has recently, in its 1999 tax package, opted for increased statutory rates for the highest income individuals. However, the new top marginal rate for individuals – at 40 per cent on both wage income and dividend income – is still among the lowest in OECD and it remains considerably below the top marginal rates on wage and dividend income in the United States and Canada (including state/provincial taxes). Nevertheless, the rise in rates increases the importance of further strengthening of enforcement in order to ensure compliance. Other ways of taxing more the affluent include abolishing some of the exemptions related to taxation of capital income, as well as introducing gift and inheritance taxes (although this would imply introducing some double taxation of capital). One alternative could be taxation of (large) real-estate holdings. Real estate taxes are less easy to evade than taxes on financial assets and with a sufficiently high minimum limit, it would be possible to achieve a progressive profile. Finally, the massive exemptions and zero-rating in the VAT system subsidise the top income decile households by an average of 4 per cent of their disposable income. Therefore reducing the preferential treatment of consumption to a minimum would be one way of taxing the most affluent households at higher effective rates.

1. Simple theoretical models of optimal income taxation do not necessarily point to sharply progressive tax structures even if society places a relatively high weight on the welfare of the less well off individuals. This conclusion, though, depends critically on the sensitivity of labour supply to the after tax wage-rate. With a low elasticity, redistributing income through the tax system could take place at low marginal cost, thus implying highly progressive tax structures (Slemrod, 1998).
2. In the United States, where the top 1 per cent of individual tax payers contributes nearly 30 per cent of total federal income tax revenues, there is no solid empirical evidence showing measurable macroeconomic effects of the substantial fluctuations in income taxation of the rich that has taken place over the 1950-97 period (Slemrod, 1998). Evidence from the Tax Reform Act of 1986, suggests considerable responsiveness in the

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timing of economic decisions (for instance, the pattern of realisation of capital gains) as well as financial and accounting responses (for instance corporations shifting status), whereas the evidence of the response of real activities by firms and individuals, *i.e.* with respect to savings, labour supply and investment, is more mixed. Evidence from the 1993 marginal tax rise in the United States (OBRA93) also suggests that the response to tax changes is to a large extent a question of short-run shifts in the timing of compensation rather than a permanent reduction in taxable income. The dead-weight loss of this particular tax increase, based on short-term elasticities, may thus have been considerable (as much as twice the revenue generated according to Feldstein *et al.*, 1996), but when shifts in the timing of *e.g.* exercising stock options are taken into account, a much lower estimate is found. Correcting for such transitory effects, Goolsbee (1997) thus arrives at an estimated dead weight loss of 15-25 per cent of the revenue generated.

objectives. However, it should not divert attention from the significant budgetary gains that are likely to arise in any circumstance from eliminating, or reducing drastically, zero-rating and exemptions in the VAT regime.

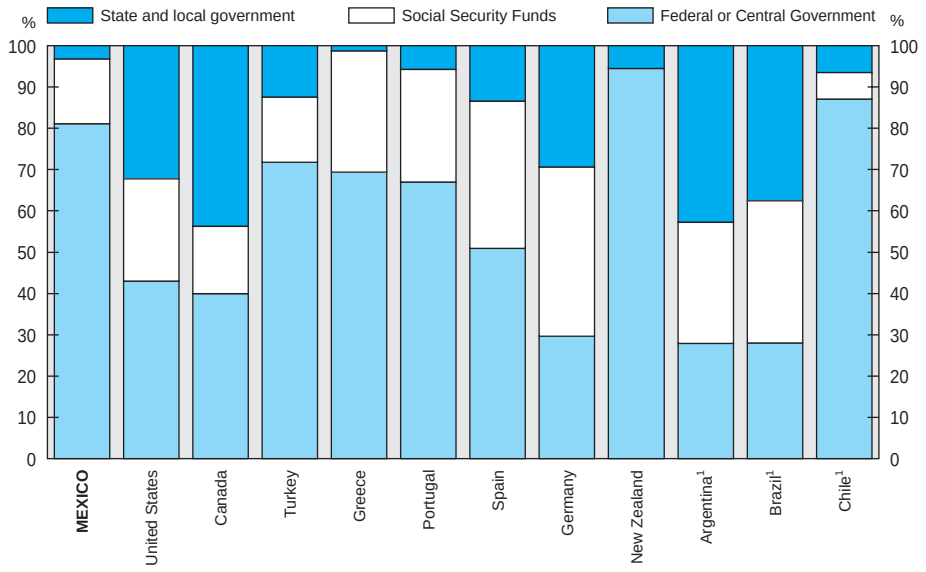
If zero-rating and exemption of goods and services were to be completely eliminated, or at least concentrated on a few basic staples, the question would arise as to how in practice to provide compensation to the lower-income groups. One way would be to extend the tax credit for these groups by an amount based on cost-of-living indices for reference low-income families. However, such compensation would only benefit wage earners who work in the formal sector, *i.e.* a minor part of the population. Another potential problem is that the tax credit payments are administered by the employers, who might be tempted to try to shift higher credit onto lower wages. Fraud could also be a potential problem. A better option might be to use the individual pension savings accounts (AFOREs) for workers included in the social security system into which the government could make direct transfers. The problem would remain, however, of non-coverage of individuals outside the formal sector. The poorest families could be compensated by expanding the targeted poverty-alleviation programme (PROGRESA) to cover a much larger number of families. It would probably be more difficult to reach those who are poor, but not poor enough to be covered by PROGRESA, and not working in the formal sector. For these groups, a separate compensation mechanism would have to be created and eligibility criteria chosen (for instance income below three minimum wages). Still, a number of problems arise besides the political difficulties involved in overhauling the VAT system. First, the compensation scheme will necessarily increase effective marginal tax rates for those not included (even though a gradual phasing out of the compensation could mitigate this problem). Second, there are no instruments in place to ensure that individuals covered by PROGRESA do not, at the same time, have an

AFORE account, thus being compensated twice – although the risk for overlap is presumably small.[94] More significant is the risk that some individuals would have more than one AFORE account, as is currently the case due to still imperfect registration. Finally, some of the increased revenue from the VAT would have to be shared with the states, leaving the marginal budgetary gain for the federal government somewhat smaller.

Fiscal federalism[95]

Sub-national governments (states and municipalities) only collect a small part of total tax revenues, *i.e.* around 5 per cent in recent years. This is below the level in most other OECD countries, as well as Argentina and Brazil, but not much different from the situation in for instance Greece, Portugal and Chile (Figure 32).[96] On the expenditure side, however, the role for sub-national governments is greater with around 20-25 per cent of total expenditure being administered at the state and municipality level. Against this background, the federal

Figure 32. **Tax collection by level of government in selected countries, 1996**



1. Data for Argentina refer to 1992, data for Brazil and Chile refer to 1995.
Source: OECD, *Revenue Statistics*; IMF, *Government Finance Statistics.*

government is engaged in a process of strengthening sub-national fiscal autonomy by the devolution of additional spending and taxing responsibilities, increasing transfers and improving administrative co-operation between the federation and lower levels of government. In most recent years, the spending responsibilities of lower levels of government have increased, in particular, within the areas of education and health care.

A key issue is how to ensure spending efficiency and accountability given that the local governments are only faced with a small part of the burden of financing. In particular, if the marginal financing of spending is (near) zero for the local authorities, moral hazard problems might occur. Thus, while potentially beneficial in terms of allocative efficiency of public expenditure (by bringing the spending decision closer to the needs of the citizens), increased decentralisation also carries considerable risk: possible loss of macroeconomic and budgetary control by the central government; insufficient administrative capacity at the local level; higher overall administrative and compliance cost; excessive tax and expenditure competition between different regions; and concessions to local political pressure groups. A major challenge for the Mexican authorities over the coming years would thus be to establish mechanisms whereby further decentralisation can be achieved while limiting the scale of such problems. This may involve careful targeting of grants based on objective expenditure needs (a nationally agreed minimum cost per pupil, per patient etc.), improved information and accounting systems as well enhanced incentives for local governments through tighter links between spending and financing decisions.

Financing of sub-national government in Mexico mainly takes place through a revenue-sharing scheme, in which the federal government transfers around 25 per cent of its so-called "sharable" revenue to the states and municipalities. Most of the transfers are given as unconditional grants. The "sharable revenue" consists of most federal taxes except for the additional and extraordinary fees from extraction of petroleum,[97] the federal tax on new motor vehicles and the tax on usage of cars, which are directly appropriated by the states, and a percentage of excise taxes on tobacco and alcohol. The share of total transfers going to each state is primarily determined by its revenue collection capacity (*i.e.* a sort of compensation from the federation for revenue "foregone" by the state) and population size. A third element, amounting to less than 10 per cent of the total sharable revenues, is used as an equalisation mechanism through which poorer states receive more.

Mexican states face few constitutional limitations as to their taxing powers, but they are in practice severely restricted by the Law on Fiscal Co-ordination that regulates tax responsibilities between different levels of government. At present, the states levy a small number of minor taxes, including a payroll tax, a tax on transfer of property, a tax on old motor vehicles, a tax on hotel

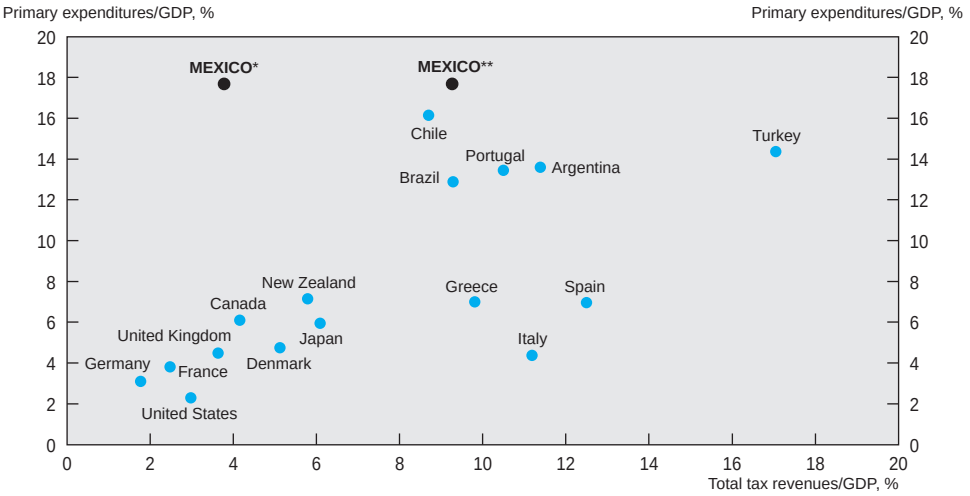
accommodation and a tax on lotteries. In addition, they may levy fees for some public services. The states are also allowed to keep any additional revenue collected as a result of audits conducted in the area of income tax, assets tax and VAT.[98] The municipalities have a constitutional right to tax local real estate and to levy local fees. Even though it would be desirable to increase revenues from immobile property, perhaps even significantly, the administrative and political barriers may be prohibitive in the short term. However, in the longer-term, a more adequate valuation and tighter administration of property taxes could turn property taxation into a more important source of revenue at the local level. Apart from its benefits in terms of strengthening the autonomy and revenue base of municipalities, this would probably have the effect of placing more of the tax burden on the income groups with most capacity to pay taxes. In the medium term, other revenue mechanisms could include the possibility of income or consumption (final sales) taxation at the sub-national level, although problems of administration and mobility may hamper the effectiveness of such taxes. In the process of delegating more responsibility for taxation to sub-national levels of government, it is imperative that the efforts of states and municipalities to raise revenue are consistent with the overall aims of the tax system (such as minimising non-neutralities) and that negative externalities on other regions or the federal government are minimised.

Budgetary volatility

The budget in Mexico is more exposed to volatility than it is in most other OECD countries. Historically, a major source of this volatility has been macroeconomic instability, partly reflecting macroeconomic policies in Mexico, but partly also a characteristic of emerging market economies generally. A more specific source of volatility has been movements in world oil prices, given the importance of the oil sector for the budget and the overall economy. The linkage between oil price changes and the budget is non-trivial. For instance, economic slowdowns or crises in Mexico have often been associated with a depreciation of the peso against the US dollar, in which case oil revenues (*i.e.* royalty contributions from PEMEX) have increased in peso terms at the same time as revenues from other tax sources were declining. This has historically, *i.e.* since 1980, resulted in a negative correlation between oil and non-oil revenues. Indeed, the total tax revenue in per cent of GDP (including PEMEX contributions) has been less volatile than the tax revenue excluding PEMEX contributions (Figure 33). However, in general there is not a straightforward and fixed relationship between the developments in oil prices and the real economy, and hence between oil and non-oil revenues (Figure 34).[99]

Another salient feature of Mexico's budget experience is the high degree of volatility in primary expenditure as a percentage of GDP. This is an indication

Figure 33. **Coefficient of variation in tax revenue and primary expenditures**[1]



1. The coefficient of variation is the standard deviation divided by the average of the tax/GDP ratio (including social security contributions) and primary expenditures in per cent of GDP of General Government (Mexico is Federal Government only). Sample period is 1980 to 1996 for all countries except Argentina (1985 to 1992), Chile (1985 to 1994 for primary expenditures, 1985 to 1996 for tax revenues – figures for Central Government only), and Brazil (1983 to 1993 for primary expenditures and 1983 to 1995 for tax revenues).

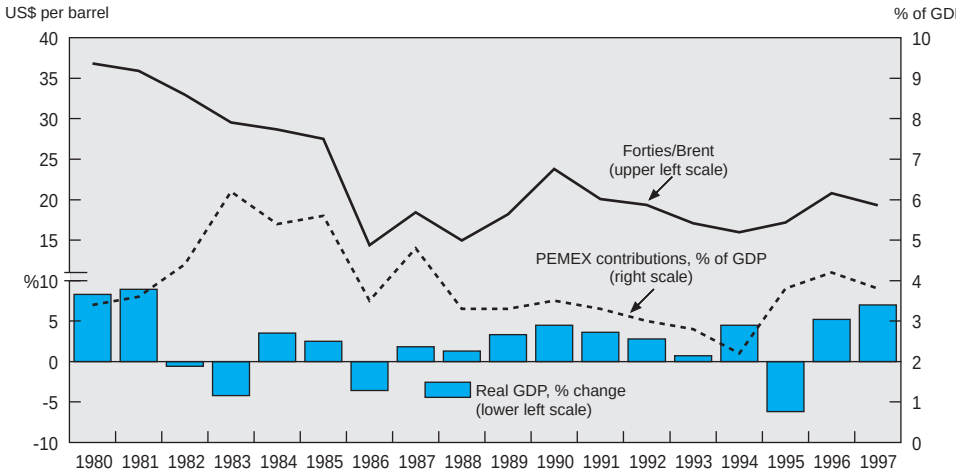
* Total tax revenues including PEMEX contributions.

** Total tax revenues excluding PEMEX contributions.

Source: OECD, *Revenue Statistics*, Analytical Data Bank; IMF, *Government Finance Statistics*; OECD Secretariat's calculations.

that the impact from revenue shocks on the budget has, to a large extent, been absorbed through discretionary spending adjustments in order to meet deficit targets. As a result, pro-cyclicality of government spending has contributed to macroeconomic volatility.[100] The high volatility in expenditure is undesirable from a macroeconomic smoothing point of view and in terms of its impact on the effectiveness of spending directed at programmes that are essential for the longer-term development of the country – *e.g.* infrastructure, basic education and health care and poverty alleviation. This points to the need of promoting a more stable development in both oil and non-oil revenues. With respect to the latter, broadening the tax base and improving administration as discussed above, would help to stabilise non-oil revenues – and thus lessen the budgets dependency on oil revenues and thereby the vulnerability to exogenous shocks. This, in turn, could pave the way for a sustainable long-term solution to dampen the impact on the budget from fluctuations in oil revenues.

Figure 34. **Oil revenues, oil prices and real GDP growth**[1]



1. The reference oil price is a continuous series based on Forties (1980-1984) and Brent (1985-1997). The Mexican oil price is lower than the world reference price but shows broadly identical fluctuations.

Source: OECD, *Revenue Statistics*; Banco de México; British Petroleum.

There are, in principle, several ways of fully or partially insulating the budget from the direct effects of sharp fluctuations in oil-prices and thus royalty contributions from PEMEX. One solution could be to privatise major parts of the oil sector and design the transaction in such a way that it allows a substantial reduction of the risk borne by the government. However, major privatisations in the oil sector are probably not an option in the short and medium term. Another solution could be the creation of a fund outside the government accounts to absorb the direct budgetary effects from short-term fluctuations in oil prices. Although such a fund would not avoid the government sector's exposure to oil-price volatility, it would spread it over longer time horizons, thereby making it less costly for society.[101] Norway and Chile are among the countries which have opted for some revenue stabilisation via the creation of funds for commodity-related revenues.[102]

The key issue for setting up an oil stabilisation fund in Mexico would be to determine the reference price of oil that would trigger payments to or from the fund. On the one hand, the reference price would have to be set sufficiently low so that the fund would on average be accumulating assets. This would allow the fund to build up and to be available to cushion government finances when oil prices are temporarily low. On the other hand, total revenues are already too low

to finance current and future expenditure needs, so cutting the contributions from oil would require a corresponding increase in tax collections. Until the revenue raising capacity of the main broad-based taxes is strengthened, this puts a practical limit on how low the reference price could be. Finally, considerations would have to be given to future oil-price trends, an issue taken up in Chapter II.

Conclusion and options for further reform

Further reform of the Mexican tax system in order to raise its revenue generating potential is one of the main keys to dealing properly with many of the challenges facing the economy, including improving the physical and human capital stock, *i.e.* developing the social infrastructure and the health care system. The tax system contains many commendable features that contribute to limiting non-neutralities and distortions to incentives. Effective average tax rates on consumption, labour and capital appear to be low compared with other OECD countries and statutory marginal tax wedges on labour income are also relatively modest for most wage levels and family types. The adverse effects from high and volatile inflation are largely neutralised through indexation and the personal and corporate income taxes are integrated so as to avoid double taxation of dividends and distributed profits. This helps to make the tax system reasonably neutral in terms of its impact on different types of investment and financing instruments.

However, with respect to practical implementation there is still ample scope for improvement. In particular, the extensive preferential regimes that exist in the taxation of consumption, individual income and corporate income should be reduced to a minimum and the administrative surveillance and enforcement should be strengthened further. The administrative improvements should go beyond those that would automatically derive from a more comprehensive tax system. Besides increasing the revenue generating capacity of the tax system *per se* such measures taken together would also reduce the dependency of the budget on oil-related income.

Reforms to the tax system would have to take into account considerations relating to income distribution, expenditure needs and the quality of spending as well as the issue of how to integrate the informal sector into the formal economy. Another pertinent issue would be the scope for granting local governments more taxation and expenditure responsibilities. Many of these elements are stressed in the Government's agenda as laid out in the *National Development Plan 1995-2000* and reiterated in 1997 in the PRONAFIDE, the medium-term programme for financing development. However, the changes made to the tax system in recent years have generally not succeeded in increasing the tax-to-GDP ratio. The budget for 1999

Box 10. Changes to the tax system contained in the budget for 1999

The individual tax rate is increased from 35 per cent to 37.5 per cent for annual incomes above 1.5 million pesos and 40 per cent for annual incomes above 2 million pesos. Taxes on interest payments are modified accordingly and the exemption on interest income from long term bonds is eliminated. The corporate income tax rate is lowered to 32 per cent in 1999 and to 30 per cent in 2000 for retained earnings, while the rate for dividends is increased from 34 to 35 per cent. The immediate deduction scheme for investment outside the metropolitan areas of Mexico City, Monterey and Guadalajara, as well as for small and medium-sized firms, is abolished and consolidation rules are tightened. An increase in the tax on alcohol and beverages, which was introduced for one year in 1998, is maintained. Import tariffs have been raised for countries which do not have a free trade agreement with Mexico (the surcharge is 3 per cent for capital, intermediate and basic consumption goods; 10 per cent for other consumption goods).

Raising the top statutory rates to 37.5 and 40 per cent is warranted from a revenue and distributional point of view, although increased avoidance and evasion at higher levels of income to some extent could reduce the effect of this change. The taxation of interest income from long term bonds is a step forward in reducing distortions between capital and debt financing.

The changes made to corporate taxation have both benefits and drawbacks in terms of neutrality and their revenue-raising capacity. The tightening of consolidation rules is a step forward as it improves equity and reduces complexity and distortions (such as the incentive to preserve economically inefficient enterprises for tax purposes) as well as problems of tax segmentation and avoidance. The reduction of the statutory rate on retained earnings and the elimination of the immediate deduction scheme could be considered as a package aiming at keeping the overall effective tax rates for corporations roughly unchanged while reducing tax expenditure and distortions implied by the immediate deduction scheme. These distortions arose because only firms with a tax liability would benefit from the scheme, implying that the scheme was procyclical and that it introduced a bias against new firms and firms in high growth sectors. Moreover, it enhanced the possibilities for tax avoidance through transfer pricing. The lowering of the corporate rate should also be seen against the tendency in other OECD-countries and Latin American countries to reduce taxation of capital. However, given the relatively light taxation of capital in Mexico and the level of statutory corporate rates, it is not evident that the elimination of the immediate deduction scheme would necessitate a lowering of the rates – on the contrary, the needs for raising revenues would suggest that the corporate rate should not have been lowered. Moreover, by introducing a wedge between dividends (taxed at a marginal rate of 40 per cent) and retained earnings (taxed at 30 per cent), a bias in favour of financing through retained earnings is created. The adverse effects of this change on capital allocation may be small though, as the share of distributed profits is relatively low in Mexico and because debt financing in any circumstance may be difficult to obtain for small and medium-sized enterprises.

Finally, the 1999 tax package contains a number of measures to curb evasion and informal sector activities, including: additional powers of the tax authority to

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incorporate new taxpayers (and thereby immediately initiate a tax audit), a change in the alcohol excise tax structure (*i.e.* the excise will only be applied at the producer level), a change in tax subject for specific operations in VAT (for certain operations between enterprises and individuals – *e.g.* consulting – the firm will retain the VAT and pay it directly to the government instead of paying it to the provider) and the elimination of the zero VAT rate for "indirect exporters".

aims to increase federal government tax revenues by around 1 percentage point of GDP from the 1998 outturn (see Box 10). Although this is significant compared with previous efforts, it will still be insufficient for longer-term spending needs. Additional measures in the area of taxation would thus be necessary in light of the challenges facing the tax system.

Substantial changes to existing tax structures – be they formal or informal, administrative or legislative – almost inevitably raise political difficulties. However, efforts to improve the system should benefit from the fact that the general tax regime is in many ways already adequately designed. In any case, the reforms carried out during the past decade cannot achieve their potential unless further steps are taken. A number of specific options, which follow from the above analysis, are listed in Box 11 below. Some of these could be seen as medium-term recommendations, rather than immediate priorities, and in general changes will have to be implemented gradually and with due consideration to balancing the interests of those being affected. Even though the recommendations could in principle be implemented on an *à la carte* basis, it should be emphasised that the package of measures, if implemented as a whole would be much more efficient and balanced than individual measures taken in isolation. If all of these initiatives were implemented together, they could raise tax revenue by some 3 percentage points of GDP, which seems to provide a reasonable benchmark for Mexico in the medium term.

In addition to the specific recommendations, two more areas have been identified where action might be appropriate – at least in the longer-term. First, in attempting to manage the informal economy, the government should use a "carrot and stick" approach that involves providing services in the formal sector that will attract potential taxpayers into the formal sector, *i.e.* law enforcement, in-work benefits, adequate social insurance, etc. The stick would then be coercive tax enforcement and control. Besides the beneficial social and economic consequences from integrating the informal economy, there would also be positive long-run effects on government tax revenues. The steps taken in terms of the

Box 11. Strengthening tax revenue: a synopsis of policy recommendations

The VAT regime: the extensive zero-rating of goods should be limited to a few basic staples; the zero-rating for agriculture and fishing activities and the exemptions for land transportation should be eliminated; and the preferential border rate at 10 per cent should be changed to the standard rate (since the gains from a uniform rate in all likelihood will significantly exceed the short term losses that might arise from possible harmful impacts on local Mexican business in border areas). VAT is normally easy to administer and control, as two sides of each transaction, until the final transaction, are reported to the authorities. Moreover, it is an easy and efficient way to ensure that at least some tax revenues are raised from the informal sector. The current regime dilutes the neutrality of the consumption tax, limits the revenue deriving from individuals working in the informal sector and is a very inefficient mechanism for redistribution of income. Moreover, it considerably complicates administration, creating numerous possibilities of evasion and fraud. Compensation to low-income groups – although not a trivial administrative and political problem – could be provided directly through various transfer mechanisms (including the use of individual AFOREs accounts and PROGRESA, for instance), but it could also take the form of in-kind benefits (school-based food programmes for children, food coupons, hospital or doctor-based nutrition programmes for pregnant women, etc.).

Corporate taxes: the simplified regime for agriculture and land transportation should be abolished, as should the special rate regime for agriculture, publishing, etc. The small taxpayers' regime should be preserved as a way to encourage greater integration of informal sector activity and to lower administrative costs. The assets tax should be maintained due to its importance as a controlling device. A further tightening of consolidation rules could be considered.

Individual income taxes: fringe benefits should be taxed and the fiscal subsidy eliminated. Taxing fringe benefits could potentially pose some additional administrative difficulties; on the other hand the elimination of the fiscal subsidy is likely to yield a non-negligible administrative relief. The proceeds of such changes might partly be used to reduce progressivity at the low- to medium-income levels.

Property taxes – in particular real estate taxes – should be increased, primarily by encouraging local government to make more adequate valuations and improve collection efforts. This would enhance the overall revenue base which supports local government finances, and it would probably be relatively progressive in its overall incidence.

Administration and enforcement should be strengthened by: increasing the number of trained personal, in particular at the local level; integrating registers of the social security system with that of tax authorities; enhancing import registration and control; improving auditing by further increasing the incentives for states to audit; strengthening legal certainty and enforcement; simplifying procedures for tax payments; and strengthening the effort against high income activities in the informal sector.

reform of the pension system and the health insurance are important in this respect, as are the recent efforts of improving the tax administration, the introduction of the small taxpayers regime and several of the measures taken for 1999.

However, further initiatives along the lines described above will still be needed in the medium to long term.

Finally, a stabilisation fund for oil revenues from PEMEX could contribute to ensuring greater stability for both revenue and expenditure. More important, however, is the fact that Mexico, as an emerging market economy, is more exposed to shocks arising in international markets than most other OECD-countries. This underlines the need for a firm medium-term reference to support a stability-oriented macroeconomic policy.

IV. Implementing structural reform: a review of progress

This chapter reviews recent initiatives in major areas of structural reform, with a focus on issues that have high policy priority. First, the banking sector, even four years after the 1995 recession and despite vigorous action by the authorities, continues to be an area of concern. Until banks have been strengthened and credit relations between debtors and creditors fully normalised, the sector is likely to remain an impediment to growth. Second, as highlighted in the Development Plan for 1995-2000, social development is an important element of the government's strategy to durably increase living standards. Hence, the social security reform of 1997 established a capitalised pension system for private sector workers and introduced changes to the financing and management of the health insurance fund. These measures, which were taken to improve the financial sustainability of the social security system and address efficiency concerns, can also be expected to enhance labour market outcomes, for instance by facilitating job creation in the formal sector. As part of the same strategy, efforts to strengthen human capital through education and training have continued, as well as measures designed specifically to address the needs of the poorest segments of the population. Finally, further steps have been taken to remove barriers to private initiative, enhance competition in key sectors of activity and improve the regulatory environment for businesses. The following paragraphs report on recent reforms in each of these areas, highlighting progress accomplished and the need to pursue action forcefully.

Developments in the banking sector

Support programmes

In the early 1990s, the Mexican banking system was characterised by inadequate capitalisation and a continuous deterioration of asset quality, manifest in a growing level of non-performing loans. These problems were masked however, by a rapid expansion of financial markets (financial intermediation, as

Box 12. The financial rescue package

To shore up the financial system in the aftermath of the 1994 peso crisis, the administration acted on three main fronts: *a)* debtor support programmes; *b)* take-over of some banks, putting them under direct supervision; and *c)* the introduction of measures to strengthen remaining banks, including both a temporary capitalisation programme and a loan purchase mechanism to promote permanent capitalisation.

a) Debtor support programmes

Debtor support programmes mainly involved incentives for debt restructuring. Loan maturities were lengthened and the real interest rate was fixed through an indexation mechanism based on UDIs (a unit of account indexed to inflation). The government is bearing the risk of mismatch in interest rates, but the banks continue to carry the credit and funding risks. The UDI programme originally applied to mortgages, loans to state and local governments and commercial loans, but was later extended to foreign currency commercial loans. Another programme (ADE) was designed for small debtors to provide them with an incentive to continue to service their debt. It involved an immediate interest rate reduction and the possibility of restructuring in UDIs. The banks and the government are sharing the interest rate subsidy. More recent programmes targeted the agriculture and fishing industries (FINAPE), state and local governments, small and medium enterprises (FOPYME) and mortgage debtors. They also provided discounts on debt payments and in some cases, incentives for the granting of new loans, with the cost being shared between the government and banks.

b) Take-over of some banks

Since 1995, 13 banks (some of which were capital deficient, some management deficient) were taken over by the government, the last one in mid-1997. By late 1998, many of them had already been closed or re-sold after being re-capitalised and "cleaned-up" of their bad loan portfolio. These loans were purchased by the government agency responsible for dealing with bank insolvencies, FOBAPROA.

c) Strengthening remaining banks

One of the first measures implemented to cope with the emergency in 1995 was the temporary capitalisation programme (PROCAPTE), which injected capital into the banking system in the form of mandatory convertible subordinated debentures. These were temporary loans (5-year maximum term) convertible into equity (if not repaid). By June 1997, the five banks which had received support under PROCAPTE had settled their debts and the programme was discontinued. Another programme involved "permanent capitalisation" for viable banks and consisted mainly of net loan purchases by FOBAPROA. In effect, the loan purchase by FOBAPROA is a swap in which the flows on a 10-year government bond are exchanged for the flows on part of the banks' loan portfolio. Portfolios were evaluated to be worth between 70 and 99 per cent of book value, depending on the bank. Several incentives were created for banks to adequately recover and collect interest on these loans, including the possibility of "repurchasing" them and provisions for loss-sharing in case there would be a shortfall between loan recovery and the bonds' value.

measured by the M4 to GDP ratio, grew from 30 to 50 per cent in the five years to 1994) and by increasing profits from financial operations (without which banks' net income would have been negative since 1991). The 1994 peso crisis hit the banking sector through three main channels: *a)* currency depreciation; *b)* a liquidity crunch in foreign currency; and *c)* the sharp contraction in economic activity and credit demand resulting from policy tightening. These factors increased the cost of funds available to banks, while reducing borrowers' capacity to repay loans.[103] As profitability and capitalisation ratios plummeted,[104] the risk of a systemic crisis led the administration to launch a series of programmes to ease the pressure on the financial system and ensure deposit safety, while reinforcing supervision and regulation, for instance by tightening provision requirements. The support programmes for banks and debtors included incentives for debt restructuring, the take-over of some banks – which were put under direct government supervision – and measures to strengthen remaining banks, through a temporary capitalisation programme and a loan purchase mechanism (Box 12).

From 1995 to 1997, capital injections through the various programmes totalled nearly twice the banking system's equity before the crisis,[105] with loan purchases affecting a quarter of the system's total outstanding loans at that time (December 1994). The total fiscal cost of these programmes will depend *inter alia* on the recovery rate of purchased loans and on real interest rates over the long-term. Under the hypothesis of a 30 per cent average recovery rate across all categories of loans and of an average real interest rate of 6.5 per cent, authorities estimate the final cost of the rescue package at 14.4 per cent of 1998 GDP,[106] of which 2.9 per cent had already been paid by the end of 1997 (Box 13).

Restructuring

Since early 1995, a process of restructuring of the banking sector has been underway, with mergers and acquisitions of domestic institutions by foreign players leading to significant changes in the financial system.[107] In 1997, as the economic recovery strengthened and credit growth and profitability slowly picked up, this restructuring started to bear fruit. By mid-1998, however, the effects of world-wide financial turbulence began to be felt more strongly in Mexico, exposing the persistent fragility of the banking system (see below). Even though solvency indicators are comfortably within international standards, capitalisation ratios are still weak when adjusted for asset quality. Furthermore, profitability indicators, albeit improving, remain well below OECD averages (Table 20). This low profitability reflects in large part the need to maintain high levels of provisions for non-performing loans as well as the continuing decline in net interest income, as a consequence of sluggish credit expansion by most of the largest institutions. The supply of credit is still feeling the adverse effects from the development of a culture of "no debt repayment" by non-financial domestic

Box 13. Estimated fiscal cost of the financial rescue package

An official estimate of the fiscal cost of the financial rescue programme was presented in February 1998, amounting to 14.4 per cent of 1998 GDP (Table 19). This compares with an earlier estimate (October 1997) of 11.9 per cent of GDP and with a total public debt (excluding FOBAPROA's liabilities) of 24.2 per cent of GDP at the end of 1997. The February 1998 estimate was based on the assumptions of a 6.5 per cent real interest rate and a recovery rate across loans of 30 per cent. The estimate may have to be revised depending on the longer-term impact of global financial turbulence of 1998 on: the real interest and recovery rate assumptions; and the size of support programmes that may be required to meet further loan default.

Table 19. **Fiscal cost of support programmes for banks and debtors**[1]

	Per cent of GDP
Debt restructuring in investment units (UDIs)	2.1
Original UDI programme	0.9
Additional programme for mortgage restructuring	1.2
Agricultural and fishing sector programme (FINAPE)	0.5
Small and medium-sized firms support programme (FOPYME)	0.2
Support to small debtors (ADE)	0.2
Debtor support programmes	**3.0**
Loans purchases for capitalisation schemes	2.6
Bank interventions and direct FOBAPROA support[2]	8.3
Restructuring of toll roads	0.5
Bank support programmes	**11.4**
Total	**14.4**

1. February 1998 estimates (as per cent of 1998 estimated GDP).
2. Includes the cost of assuming liabilities of banks that have been put under administration of FOBAPROA.

Source: Ministry of Finance.

FOBAPROA's liabilities do not carry an explicit government guarantee, as is the case for regular public debt; and as long-term obligations with capitalised interest, these liabilities do not have a regular servicing schedule. Instead, the debt is amortised by direct budget allocations. Between 1995 and 1998, budget allocations to the financial sector support programmes amounted to a total of 3.2 per cent of GDP.

The February 1998 estimate of the cost of the financial rescue package (14.4 per cent of GDP) was divided into four main components:

– Debtor support programmes; estimated at 3.0 per cent of GDP (unchanged from earlier estimates), spread over 30 years. By fixing real interest rates, lengthening maturities of restructured debt and giving discounts to debtors as they service their debt, these programmes have a built-in subsidy (shared between

(continued on next page)

(continued)

government and banks) which increases if the real interest rate on the market exceeds the initially-set rate.

- Bank intervention (for capital- or management-deficient banks) was put at 8.3 per cent of GDP, up from 5.9 per cent estimated in October 1997, mostly as a result of a lower-than-expected value for the banks' assets and an increase in the number of banks joining the programme. These FOBAPROA liabilities arising from intervention consist mostly of 10-year indexed government securities, while assets include mainly the intervened banks' loan portfolio (now estimated at 2.5 per cent of GDP).
- Permanent capitalisation programme for other (not taken-over) banks; was put at 2.6 per cent of GDP (up from 2.4 per cent in earlier estimates). As this programme also includes loan purchases through FOBAPROA, it is sensitive to assumptions on interest and loan recovery rates, which have been revised. Contrary to banks under the authorities' direct control however, these banks are committed to a loss-sharing provision as an incentive for recovering loans, with the effect of reducing the total fiscal cost of the programme. The temporary capitalisation programme (PROCAPTE) on the other hand, was funded by the central bank, with no fiscal cost to the government, and its monetary impact was fully offset by a system of compensating reserve deposits.
- Bail-out of the road operators: for a total amount of 0.5 per cent of GDP (unchanged in peso terms from earlier estimates). Liabilities arising from this rescue programme are indexed 5-, 10- and 15-year bonds that are not sensitive to interest rates.

Table 20. **Banking system indicators**[1]

Per cent

	Dec. 1994	Dec. 1995	Dec. 1996	Dec. 1997	Dec. 1998[2]
Profitability, productivity					
Net profits/equity	12.5	7.4	–16.3	5.1	6.5
Net profits/total assets	0.5	0.3	–0.6	0.4	0.5
Operational income/total assets	4.1	3.9	2.8	4.8	5.0
Solvency, capitalisation					
Capitalisation ratio[3]	9.8	12.4	13.2	13.6	14.6
Capital/total assets	5.5	6.8	6.0	8.5	8.4
Non-performing loans/total loans	7.3	6.9	5.9	11.3[4]	11.4
Provisions/non-performing loans	48.6	72.6	119.9	62.8[4]	66.0
Current assets/current liabilities	1.07	1.08	1.02	1.06	1.09

1. Refers to banks in "normal" situation, *i.e.* excluding the balance sheets of banks that have been placed under control of the authorities. Data reflect the situation of privately-owned banks after the clean-up.
2. Preliminary.
3. Net capital over risk-weighted assets. Since December 1997, this ratio also includes market risk.
4. Break in data, reflecting the transition to the new accounting system (based on internationally accepted accounting principles).

Source: National Banking and Security Commission (CNBV).

borrowers over the past three years. Stricter prudential regulations have also contributed to the trend.[108]

Past-due loans remain a significant problem for the banking system as a whole, although their share in total loans has declined steadily since the beginning of 1997 (the doubling of recorded non-performing loans during 1997 is explained by changes in accounting rules which were introduced in January that year).[109] Even though many banking loans have been restructured, problems have persisted, especially with mortgage loans, leading banks to design individual programmes, starting in mid-1997, offering additional relief to debtors. Moreover, the possibility of further debtor support by the government and higher interest rates in the recent period have affected delinquency rates, as some debtors have stopped servicing their debt in the expectation of more favourable terms.

The entry of foreign banks in the market has increased competition on Mexican banks. Foreign banks have played a significant role in solving the capitalisation problem following the crisis, in part because their financial strength reduced the risk of a systemic crisis.[110] The foreign presence helped reduce margins and foster the growth of market segments, such as the retail deposit market, which were underdeveloped historically. Banks with foreign majority ownership (defined as more than 50 per cent of capital) accounted for 20 per cent of the system's total assets in June 1998, up from 4 per cent in 1994, as the reform of the banking law and NAFTA implementation eased barriers to entry in the sector. In addition, foreign banks have acquired minority stakes (defined as 15 to 49 per cent of total capital) in a number of banks, including three of the four largest Mexican banks, accounting for a further 43.4 per cent of total assets. Foreign equity in financial institutions other than banks also increased sharply in the past few years. By mid-1998, firms with significant foreign participation accounted for 27.8 per cent of total assets in the leasing market, 16.8 per cent in the factoring market and 53 per cent in the specialised financial market (consisting mostly of firms dealing with mortgages, also known as "non-bank banks"). The latter, together with savings and loans institutions, have experienced very rapid growth in the past two years and now account for almost 2 per cent of total domestic credit.

Remaining problems and new initiatives

The financial rescue package started to take final shape on 13 December 1998, when Congress approved a series of financial reforms, which were a part of the proposal sent to Congress by the administration in March of that year (Box 14). A decentralised entity has been created, the Institute for Bank Deposit Insurance (IPAB). Its objectives are to administer the support programmes for banks, to dispose of assets and collect overdue loans and to manage the deposit guarantee scheme. FOBAPROA liabilities will be transformed into securities of

Box 14. The financial sector reform

In March 1998, the administration sent to Congress a proposal for a package of reforms concerning the financial sector, including laws to resolve some of the pending problems related to the banking sector's rescue package. Part of the package was approved in December 1998:

- The original proposal to split FOBAPROA into two independent agencies under the authority of the Finance Ministry was modified. A new law created a decentralised entity, the Institute for Bank Deposit Insurance (*Instituto para la Protección del Ahorro Bancario*, IPAB), which will be set up in June. Among its objectives are: administration of the financial support programmes in the banking sector, management of the deposit guarantee scheme, and disposal of the assets and collection of overdue loans. After a transition period, current FOBAPROA assets will be transferred (following completion of audits) to the IPAB.
- The deposit guarantee fund managed by the IPAB will gradually reduce its coverage: by the end of 2005, the cap on deposits would be brought down to 400 000 UDIs (approximately US$100 000).
- The securities of IPAB that will be exchanged for the FOBAPROA liabilities will be fully guaranteed by the government. However, their maturity rates and conditions in general will be determined over the coming months. As FOBAPROA debt will not be directly converted into public debt, it will be incorporated in the yearly budget.
- Foreign ownership restrictions applying to financial institutions have been eased. In the past, foreign ownership was limited at a minority share of stocks in domestic institutions which had more than 6 per cent of total capital in the banking system; these included the three largest banks: Banamex, Bancomer and Serfin. These restrictions have been replaced by non-discriminatory restrictions on individual holdings, to be decided from a competition policy perspective.

The part of the March 1998 proposal which was not approved in December of that year includes:

- Changing the Bank of Mexico Basic Law, to give it full autonomy for exchange rate policy, rather than having this responsibility shared with the government.
- Transferring most functions pertaining to regulation, supervision and control of the financial system to the supervisory agency for banks and securities (*Comisión Bancaria y de Valores*, CNBV), which would thus become a technically-autonomous organism within the Bank of Mexico instead of being linked to the Finance Ministry, as is currently the case.

IPAB, which will be fully guaranteed by the federal government.[111] Finally, Congress agreed to the easing of foreign ownership restrictions applying to financial institutions. At the time of approval, a new debt relief programme was introduced for debt restructuring, with discounts on mortgages, mainly, as well as on small business and agricultural loans.

The part of the package not approved in December 1998 includes the transfer of most functions pertaining to regulation, supervision and control of the

financial system to the supervisory agency for banks and securities (*Comisión Nacional Bancaria y de Valores*, CNBV), which would cease to be linked to the Ministry of Finance, and changes to the Bank of Mexico's Basic law, giving it full responsibility for exchange rate policy (at present the responsibility is shared with the government). Furthermore, the December approval of part of the reform package did not clarify how the cost of the rescue package is to be split between banks and the government, as the nature (and thus the market value) of the new tradable bonds replacing FOBAPROA liabilities is not yet known. Finally, the new agency in charge of asset disposal has been mandated to conduct an audit on FOBAPROA loan purchases, some of which may be returned to banks.

In addition to the domestic uncertainties, the Mexican banking system suffered from the spillover effects of global financial turbulence in 1998. Even though some of the transmission mechanisms were similar, the magnitude of the shock was not comparable with that of the 1994 peso crisis. Furthermore, Mexico has in the meantime adopted a floating exchange rate regime. Hence, currency depreciation in 1998 was more gradual and much smaller than in 1994-95; while liquidity, credit demand and economic activity were less affected. Furthermore, by the time financial turbulence spread to Mexico in 1998, banks had recovered somewhat and were in better health than in 1994. Solvency and capitalisation ratios – although still weak – were higher than in 1994.[112]

Nonetheless, the banking system remains fragile and prolonged turmoil in global financial markets could have serious effects. Until mid-1998, the economy kept growing strongly and the impact of higher interest rates had not been felt on loan delinquency ratios. However, just when banking credit was beginning to recover from the 1995 slump, Mexico was hit, first, by the international financial crisis and then by the Brazilian crisis. The resulting increase in interest rates and the expected slowdown of the economy are likely to aggravate the banks' situation: credit demand could relapse, while asset quality could worsen,[113] leading to an increase in required loan-loss provisions and a decline in profitability. This impact would be compounded by contingent liabilities arising from loss-sharing provisions of FOBAPROA's loan purchase programme, which are sensitive to delinquency rates. Summing up, the banks' situation will be strengthened durably only when interest rates come down permanently, contagion risks from global financial turbulence evaporate, a viable financial reform is fully implemented, and capital inflows resume in line with investors' confidence.

Progress in reforming the social security system

Two years ago, when discussing structural reforms that could contribute to improving labour market performance in Mexico (see next section), the OECD noted that initiatives aimed at making the tax and transfer system more effective,

in addition to serving useful social goals, could play a welcome role in lowering some of the barriers that keep many activities in the "informal sector". Several aspects of the social security reform of 1997 go in this direction. By creating individual accounts for pension savings, the new pension system strengthens the link between workers' contributions and benefits, thus helping to reduce incentives for workers to evade or under-declare income. Furthermore, efficiency gains are expected from the increased competition and transparency brought about by the private management of pension funds; and finally, the possibility of adding voluntary savings on these accounts may in the longer run foster private saving.[114]

The health segment of the 1997 social security reform, although attracting less attention, was another key element of the authorities' overall strategy in this area. The 1998 *Economic Survey of Mexico* discussed the health care system in detail, including the reform of the IMSS health insurance financing. By reducing employers' contribution rates (except for very low-income workers), it should have a favourable impact on employment creation in the formal labour market.[115] Another important objective is to increase coverage of health services to Mexican workers and their families, whether they work in the formal or the informal sector, thereby helping to increase living standards and productivity.

Implementing the new pension system

A fully funded capitalisation pension system has been operating since July 1997, replacing the former pay-as-you-go system (PAYG) for private sector workers affiliated to the IMSS. Pension contributions are channelled to individually-owned accounts which are invested in financial markets by private administrators, AFOREs. By December 1998, the AFOREs had registered 13.8 million workers, more than 90 per cent of the potential number of workers in the IMSS, to whom the scheme is applicable. The resources deposited in individual accounts totalled 105.9 billion pesos, or 2.7 per cent of GDP. However, only about half of that corresponds to retirement contributions, and as such are invested in specialised mutual funds (SIEFORES). The rest is made up of contributions to the housing fund and managed directly by the specialised agency, INFONAVIT.[116] In the twelve months to December 1998, the retirement funds managed by the AFOREs yielded a real return of 5.6 per cent, a rate comparable to the return on other savings instruments.[117]

The AFOREs portfolio is almost entirely (96 per cent) invested in government debt – half in fixed rate one- and two-year bonds and the rest in short-term instruments and indexed bonds.[118] Since mid-1998, AFOREs have been entitled to further diversify their portfolios. However, financial market turbulence has created disincentives to a substantial increase in the share of private sector instruments. Commissions charged by AFOREs tend to be lower than those applied on other savings instruments in Mexico;[119] they are also lower than those

charged in other Latin American countries with similar pension systems – Argentina, Chile, Peru and Uruguay. One important factor explaining the difference in administrative costs among countries is the regulation on how many times a worker is allowed to change fund manager per year: in Mexico, the limit is set at one change per year only (starting after one year of operation, *i.e.* mid-1998), whereas it is twice a year in Argentina, Peru, Uruguay, and Chile (where it was reduced in early 1998, from three times). The AFOREs are in a generally healthy financial situation in terms of asset/liability ratios, and capital ratios are generally adequate; but there is some on-going restructuring which has reduced the number of AFOREs (from 17 when the system started to operate to 14 by the end of 1998).

The new pension system has a number of flaws that need to be corrected. There are some administrative difficulties, in particular, in the identification process that makes the allocation of funds deposited by employers to individual accounts possible, although the amounts involved are small. Also, there are workers who contribute under more than one identification number. Moving forward in connecting social security (IMSS) registers with those of the retirement system (CONSAR) – through a unique identification number – would be one option. Ideally, the information base from the tax administration services should be integrated with these registers as well, so as to allow better cross-control.

The pension reform is a major step towards improving the tax and transfer system, although time will be needed for the reform to yield its full benefits. In the longer run, by strengthening the links between contributions and benefits it can help reduce evasion incentives. Enhanced transparency and competition in the management of funds should bring efficiency gains. The new system can also contribute to a deepening of financial markets, and by giving the option of adding voluntary savings to individual pension accounts, it can stimulate private savings over time. Although it is too early to assess the effectiveness of the reform in accomplishing its objectives, it is clear that complementary measures have to be undertaken. First, a reform of INFONAVIT is necessary. Contributions to the housing fund bring negative returns as the fund cannot fulfil its two objectives simultaneously – providing housing to low-income families and giving positive return on workers' saving. Second, the pension reform which has been introduced only for private workers in the IMSS, should eventually be extended to the public sector (both government employees insured by ISSSTE and those from public enterprises, such as PEMEX, have their own pension system). Among other benefits, such an extension would allow portability of pensions across sectors of activity, thereby enhancing labour mobility. Third, in order to eliminate distortions, voluntary savings in AFOREs should be subject to the same treatment as savings managed by other types of investment funds (tax dispositions were harmonised in the tax reform for 1999, but there remains a difference in the management of these savings for they can only be retrieved once every six months).

New developments in health care reform[120]

During the past year or so, the reform of the health component of IMSS has been progressing along two main lines: widening the coverage of services; and increasing efficiency through structural change including improvements to the budgeting system. The reform introduced in mid-1997 made voluntary affiliation for individuals less costly. By June 1998, close to 100 000 families (meaning approximately 250 000 beneficiaries) had been registered under the "health insurance for the family" scheme, which provides medical insurance to households – but not pension benefits – for a modest fixed fee. Two state governments, several local small merchants' associations and the National Confederation of Agricultural Workers have signed agreements with the IMSS. The latter alone implies that, over time, 3 million new salaried workers and their families will have a right to IMSS medical services, pensions and other benefits.

The IMSS is also undergoing a number of structural changes. First, allowing free choice of family doctor is under consideration to create some competition and improve efficiency in health service delivery. At the beginning of 1998, the IMSS medical direction established criteria for the selection of the medical units which will participate in pilot programmes. Second, the IMSS budgeting system is being revised. Currently, the IMSS determines its basic allocation to first and second level hospitals through capitated risk-adjusted formulas. Over the past year, some progress has been made in developing a system of Diagnostic Related Groups (DRG) to measure costs. These DRGs are now being tested; they will be used as a main determinant of budgetary transfers to hospitals in the future. Goals are being established on the number of hospitals that will be adopting DRG systems over the 1999-2000 period. Third, the budget Law for 1998 included, for the first time, provisions concerning the IMSS, with a programme for the financial strengthening of the institution. This is seen by the administration as the first step in a process that should lead to more IMSS management autonomy.

Enhancing human capital and other labour market issues

The OECD made a detail review of the Mexican labour market in the 1997 *Economic Survey on Mexico*, following the framework set out by the OECD *Jobs Study*. While recognising the large degree of labour-market flexibility, the analysis revealed deficiencies in performance, and drew specific policy recommendations with priority given to actions aimed at enhancing human capital (Table 21). First and foremost, the OECD emphasised the critical role of education and training to underpin longer-term growth and create more equal opportunities across regions and income levels. The *Survey* also stressed the need to seize opportunities as

Table 21. **Implementing the OECD Jobs Strategy – an overview**

Jobs strategy proposals[1]	Action taken	OECD assessment/recommendations
1. Enhance human capital		
• Introduce standardised competency and skill tests (certification)	– Being implemented: certification has started in October 1998	– Continue implementation
• Strengthen vocational/technical education (short cycle)	– Focus on job insertion has increased, the development of a system of skills certification is expected to improve effectiveness of training	– Continue these efforts, monitor and assess results
• Continue targeted programmes to the poorest	– Coverage of integrated Programme for Education, Health and Nutrition, PROGESA, has been increased	– Continue these efforts, monitor and assess results
• Provide basic health package	– Expanded in 1998	– Continue and step up these efforts
2. Reform the tax and transfer system		
• Increase effectiveness of taxation	– Tax administration is being modernised and strengthened	– Consider options to enhance control over a wider range of tax payers
• Public housing scheme (INFONAVIT)	– Administrative reform under way	– Step up these efforts
• Implement the new pension system	– Pension fund administrators, AFOREs are managing individual accounts	– Explore ways to reform the public sector retirement funds
3. Improve infrastructure and foster small enterprise dynamism		
• Develop infrastructure	– Increased private participation	– Boost investment, both public and private, while ensuring transparency regarding private financing of public investment (PIDIREGAS scheme)
• Further simplify bureaucratic requirements for SMEs and micro-enterprises	– Continued in 1998	
4. Increase the flexibility of employment regulation		
• Review employment protection provisions	– No action	
• Broaden scope for short-term contracts	– No action	– Consider options
• Allow probationary periods	– No action	
5. Modernise the collective bargaining framework	– No action	

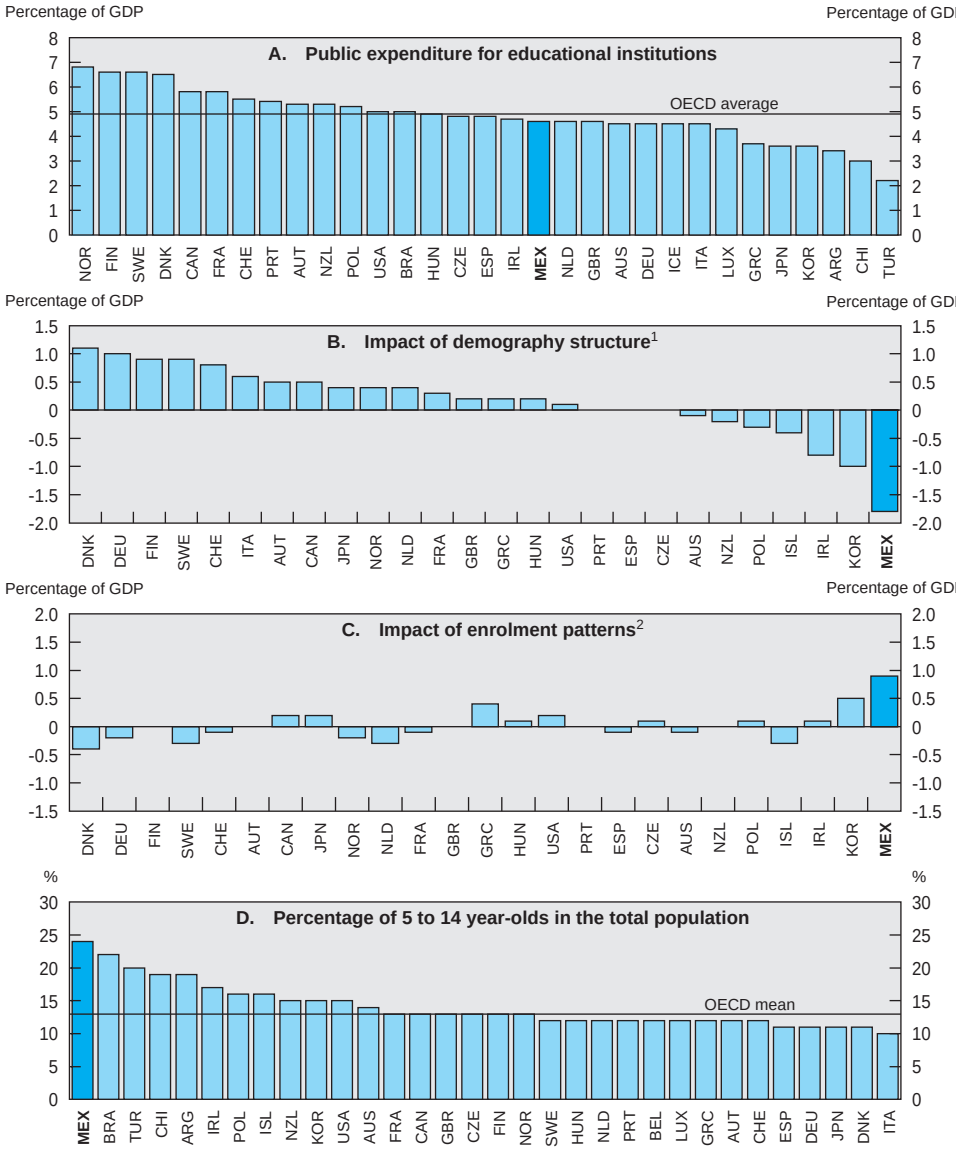
1. Proposals are discussed in detail in OECD (1996), Chapter III and Box 1.
Source: OECD Secretariat.

they become available to undertake a review of institutional arrangements and regulations pertaining to labour markets. The present administration has taken strong action to enhance human capital which, as stressed in the Development Plan for 1995-2000, can contribute to higher growth in productivity and employment over the longer run. In pursuit of this goal, it has been following two main directions: upgrading skills and competences, to facilitate the adaptation of the labour force to a changing environment, and reinforcing measures to fight extreme poverty with targeted actions to marginal groups and remote areas. Compared with other OECD countries, Mexico spends a relatively large share of public resources on education (as is the case for Korea, for instance), an indication that education is considered a priority sector. In relation to GDP, however, education spending is below the OECD average, and when adjusted for the demographic structure of the country (where the proportion of school-age persons is one of the highest among OECD countries), public spending relative to GDP is one of the lowest in the OECD area (Figure 35).[121] Furthermore, private spending on education is also relatively low in Mexico, especially for primary and secondary education. Resources allocated to basic education and training have increased in the past three years, following the severe cuts undertaken in 1995. New teachers have been hired and trained; new buildings and equipment have been put in place;[122] targeted programmes and the use of tele-education have been expanded to improve access to basic services in remote areas.

A formal structure was created to integrate various elements of targeted poverty alleviation measures. The programme for Education, Health and Nutrition, PROGRESA, which started to operate in 1997, includes money transfers to families in extreme poverty, conditional on school attendance by children and regular visits to doctors. By the end of 1997, 400 000 families were receiving support through this programme. A fast expansion in the coverage of PROGRESA was planned for 1998, and further development of basic as well as vocational/technical education was initially programmed. Budget restraint in 1998 and the successive spending cuts that were introduced throughout the year have impeded the full implementation of many of the projected actions; but the coverage of PROGRESA has increased. By September 1998, about 1.6 million families were being reached through it (according to Ministry of Finance estimates).

Training programmes were maintained in 1997-98 at roughly the same scale as in 1996, but their function has evolved. Immediately after the 1994 peso crisis, in view of the dramatic deterioration of the labour market and given the absence of unemployment insurance, some of these programmes were expanded to provide temporary income support to dismissed workers – although the scale of the programmes has always been quite small by comparison with other OECD countries. As labour market conditions improved in the following years, the programmes have increasingly gone back to their original purpose. The provision of

Figure 35. **Education indicators in comparison**



1. Increase/decrease in expenditure on education if the proportion of the population aged 5 to 29 was at the OECD average.
2. Increase/decrease in expenditure on education if enrolment patterns of 5 to 14 year-olds (up to upper secondary) were at the OECD average.

Source: OECD, *Education at a Glance*, 1998.

short-term grants under the training programme for the unemployed (PROBECAT) more than doubled between 1994 and 1996, but was reduced slightly in 1998. On the other hand, training support in small and medium-size enterprises (CIMO) was strengthened in 1998.

The implementation of the national system of competency standards and the establishment of the system of skill certification are expected to yield important benefits in training and technical education. The initiative began in 1995 as a co-ordinated effort of the Education and Labour Ministries and pilot projects are now in place. Concerning the formal certification of knowledge and skills, a national network of 35 institutions is operating. By the end of 1998, the evaluation centres will be providing certificates to about 210 000 workers. By 1999, some 45 norm-establishing Committees are expected to be operating, with business and labour representatives from various sectors of activities. These Committees have already defined some 300 working skills, covering about half of productive functions. The certification of knowledge and skills, regardless of how they were acquired, and the establishment of technical norms for labour skills nationwide, are expected to increase flexibility in training and technical education, which should facilitate labour mobility. In parallel, efforts have been made to train teachers to the new framework, while stimulating demand from enterprises, by providing incentives to promote certification with the support of PROBECAT and CIMO programmes.

When social partners in 1997 signed an "agreement for a new labour culture", this was interpreted as a first step in a process which would lead to a review of labour market practices and policies. Since then, business representatives and union leaders have been holding talks about possible changes to the antiquated Federal Labour law. In September 1998, social partners (including both traditional and new independent unions) have agreed to establish a negotiating table, which would analyse various proposals to try and reach a consensus, although the time horizon for presenting a final proposal has not been defined. Some of the issues that will be considered concern provisions regulating part-time work and temporary contracts, the treatment of productivity gains in wage formation and resources allocated to on-the-job training and technical education – issues which were discussed at some length in the 1997 *OECD Economic Survey*.

In sum, the past year and a half has seen progress in a number of policy areas of key importance for enhancing sustained employment and output growth, including welcome initiatives to strengthen human capital. Nevertheless, problems remain and corrective efforts need to be pursued, particularly in areas such as education, health care and poverty alleviation. Given that these measures yield their benefits with long lags, action should not be delayed, despite current financial constraints. The authorities should also move ahead with the implementation of the indicators that are being developed to evaluate the performance

of the public sector. This should improve the effectiveness of public spending, thereby helping to better achieve policy objectives concerning the development of infrastructure and social conditions.

Increasing product market competition

Privatisation and sale of concessions

For more that a decade, the Mexican authorities have been pursuing a forceful policy of withdrawal from the state production of goods and services. The bulk of the privatisation effort (in terms of output, employment and revenues yielded from the sales) was taken by the previous administration, although this effort largely excluded the liberalisation of public services – a process which, to yield its full benefits, requires the introduction of a comprehensive regulatory framework. In that area, however, the current administration has made significant progress, with the opening of key sectors of activity (such as transportation, telecommunications, natural gas and electricity) to private initiative and competition, and the development of relevant regulatory competences. The revenue from these operations is relatively small, but the impact on the economy is significant because of the efficiency gains expected in the sectors themselves and the incidence they can have on other sectors.[123]

The administration is moving forward with further privatisation and liberalisation of the services sector. Airport privatisation is advancing: bids for the first of the four regional units regrouping Mexico's profitable airports have been put forward and a final decision is expected before the end of 1998, leaving three groups to be auctioned in 1999. The privatisation of the four main railroad lines was completed in 1998, with the sale of the concession for the south-eastern line in October. The purchasers of the concessions (joint ventures, generally with an American railroad company interested in integrating lines across the border) are planning to invest about US$1.7 billion over the next five years, of which about US$1 billion in the north-east line, which carries almost two-thirds of Mexico's rail freight. To facilitate the operations, the government set aside part of the proceeds to cover pension liabilities for the national railroad workers, and some of the needed adjustment of the workforce was made beforehand. Divestiture of smaller lines is programmed for 1999.

The Federal Telecommunication Law, which establishes relevant regulations, was passed in 1995 and full competition in the long-distance market was introduced at the beginning of 1997. Since then, new players have gained about 30 per cent of the market and prices for long-distance calls have come down significantly. In addition to the drop in the average price paid by households, businesses and the government, the most tangible benefit for the average

consumer has been the simplification and increased transparency of the new rates for national long-distance service.[124] The average consumer is not reaping the full benefit of lower prices, however. Many customers make primarily local calls, the price of which increased significantly, reaching a level closer to prices prevailing in the United States and other deregulated markets. Under new regulations, Telmex (*Telefonos de Mexico*), which still has a monopoly on basic services, had to eliminate cross-subsidies which were benefiting local service to the detriment of long distance. During the first year of long-distance competition, Telmex was able to offset a 25 per cent drop in long-distance revenue with a 30 per cent increase in local service revenue (and a substantial increase in revenue from interconnection fees).

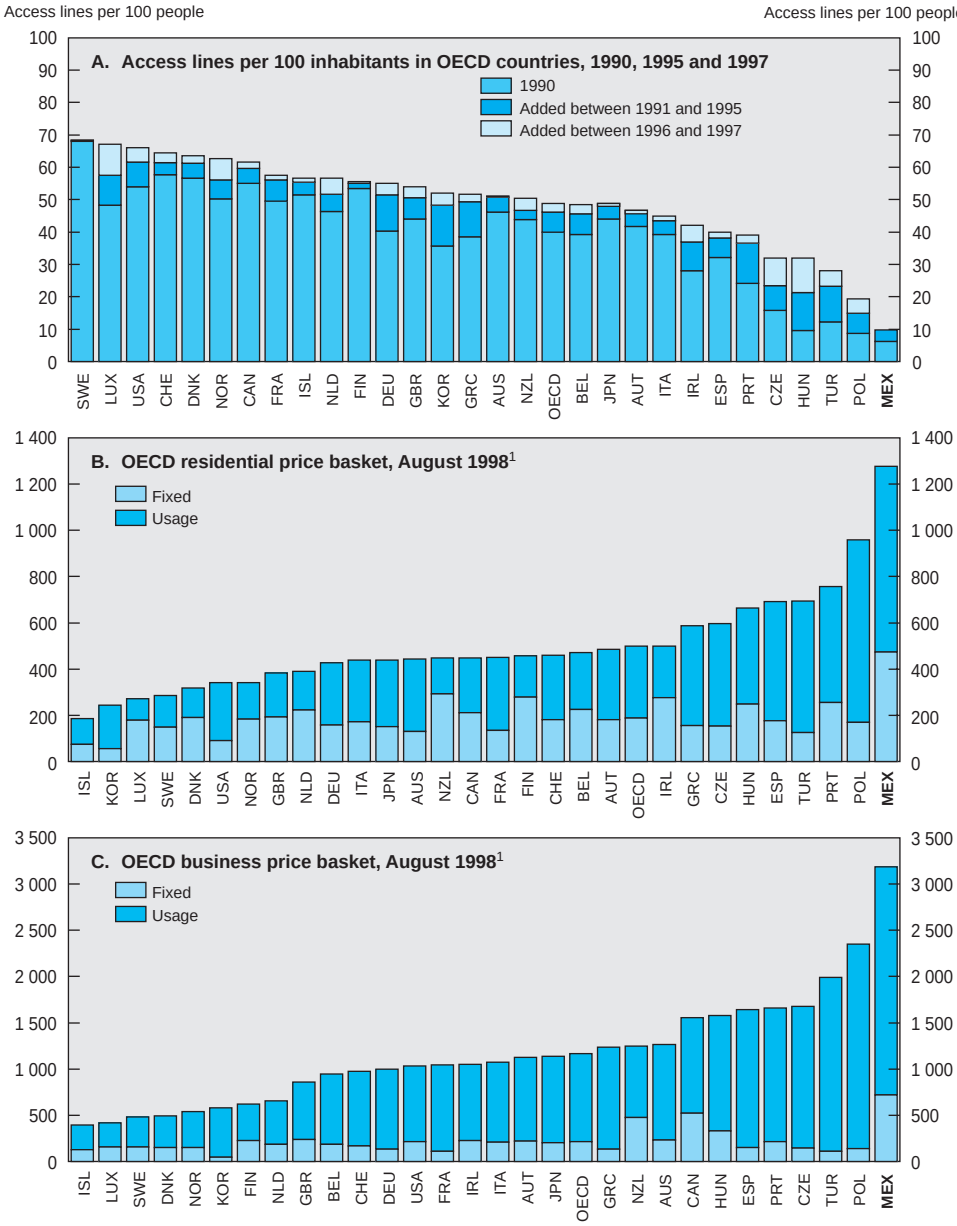
The local service market is now opening up to competition as well. In July 1998, Mexico completed the auction of 77 licenses (covering nine regions across the country) for mobile and fixed wireless services. The opening of the fixed wireless local services is expected to bring competition to rural communities, still under-equipped in traditional wire networks, more than to urban areas, where these are well developed. Whereas long-distance operators targeted multinational companies as customers, fixed wireless telecommunication is aimed at lower income and semi-rural consumers. By the end of 1998, seven new companies are expected to have entered the local service. The new carriers will be competing with Telmex on a growing market: Mexico has an average of 10 phone lines per 100 inhabitants, compared with 15 in Argentina, Brazil and Chile, and 40 or more in the majority of OECD countries. The Federal Telecommunications Commission published in August 1998 the calendar for the auctioning of concessions to use frequency bands in the radio-wave spectrum for television services and radio-communications: eight auctions have been programmed to take place before March 1999.

The performance of the telecommunications sector as a whole has been lagging behind that in most OECD countries, including low-income ones.[125] Network investment and penetration were still the lowest in the OECD in 1997; and, from a competitive standpoint, it should be noted that price levels were still the highest in the area, pointing to the need for further progress in implementing a pro-competitive regulatory regime (Figure 36).

In the natural gas distribution sector, where the legal framework and regulations were established in 1997, sales started in 1998 with a concession for the Monterrey region, two for the Mexico City Valley and one for the Queretaro area. Each purchaser is required to significantly increase coverage and meet precise objectives for investment over the next five years.[126]

With the demand for power expected to increase by more than 6 per cent annually over the years to come, Mexico faces a major challenge to add electricity-generating capacity. Until early 1999, deregulation of the power industry was

Figure 36. **Performance indicators in telecommunications**



1. Average annual spending, including tax, expressed in US$ PPP.
Source: OECD, *Communications Outlook.*

slow. The opening of the sector to private capital was limited to the generation business, while the state-owned companies (the Federal Electricity Commission, CFE and the smaller "Luz y Fuerza del Centro") retained their monopoly on transmission and distribution. The tendering of independent power projects to be built and operated by private firms started in 1998. But such government-backed projects, even though they involve private initiative, constitute only a first step in the opening of the sector. Hence, in February 1999, the government sent to Congress an ambitious bill that aims at preparing the ground for the sector's reform. The proposal is organised in three stages. First, the two state-owned companies are to be split into several generating and distribution companies and one national transmission company. In the transition period, while this restructuring goes on, a new regulatory and legal framework is to be established and an independent entity will be created with responsibility for system and market operations. In a second stage, power generation and sales will be fully opened to private investment, and generating companies, both state-owned and private, will be competing in the power market, while distribution will start to be concessioned. Finally, after the year 2000, the privatisation of state-owned companies will be allowed. This last step, which is intended to be implemented gradually is seen as crucial to complete the process. Although it is too early to assess the proposal and its likely outcome, it was certainly appropriate to move forward with the reform. Incorporating the public companies and separating their accounts from those of the government can make a significant advance towards better management. Other countries' experiences show that there is no single model to follow to deregulate the electricity sector, but successful examples have generally included a number of common features – many of which are part of the reform strategy in Mexico – such as, typically: segmenting the power sector; breaking up the national monopoly in transmission into local units; eliminating restrictions on producers; privatising parts of the industry; and, finally, implementing a comprehensive regulatory framework that includes the appropriate incentives, with a view to introducing competition whenever possible.

The oil sector, belonging to state-owned Petroleos Mexicanos (PEMEX) is another area where restructuring has been slow. Within PEMEX, the opening of the petrochemical division (one of the company's four subsidiaries) was announced in 1995, but plans had to be scaled down because of political resistance. Eight basic petrochemical products, considered as "strategic sectors", like oil extraction and production, continue to be reserved for state-ownership. In secondary petrochemicals, the private sector may only hold 49 per cent of the equity of existing plants, where PEMEX keeps a majority share. Full private (and foreign) ownership is allowed in new plants only. The restriction has delayed the whole privatisation process in the sector. The process started nonetheless in 1998, with the public offering of 49 per cent of the capital of one of the units

regrouping several secondary petrochemical plants; by the end of 1998, the bidding had started, although slowly.

Regulatory reform[127]

An important element affecting the competitive environment in Mexico, as in many other OECD countries, is the array of government and branch-imposed regulations which set standards for behaviour of established enterprises and condition entry in certain trades or professions. The Mexican deregulation effort started as an explicit national policy about ten years ago.[128] The first phase of the reform sought to adapt the regulatory framework to an increasingly open economy in order to create a level-playing field for private (domestic and foreign) investment. It concentrated on newly privatised sectors which were considered as priority sectors, such as telecommunications, transport, gas and electricity. This was followed by a second phase of deregulation, aimed at providing legal certainty and enhancing competition, through the adoption of a modern framework, with the enactment of the Competition Law in 1993 and the creation of the Federal Competition Commission to enforce it.

The most recent phase of regulatory reform started in 1995 under the new administration. Because extensive and time-consuming procedures to obtain licenses and permits contribute significantly to the cost of doing business, one of the explicit objective of the new strategy is to streamline all requirements that burden the operation (or start up) of firms in Mexico and to reduce the administration's response time. The 1995 Agreement for Deregulation in Entrepreneurial Activity (ADAE) set explicit standards for good regulation, which are comprehensive and well conceived. It has established a process of permanent scrutiny of existing formalities and new regulatory proposals. All government agencies are required to evaluate compliance of their regulations with the principles set out in the agreement. The review of formalities is proceeding across the whole administration. In the two years to July 1998, nine federal ministries had completed the scrutiny process; by then, they had reached agreements to remove or improve almost 50 per cent of the total number of mandatory formalities for businesses; and half of the changes agreed had been implemented. An innovative step was taken to facilitate communication with the creation in June 1998 of an Internet site posting all information about formalities (mandatory or voluntary) to start up or operate a business classified by sector of activity.[129]

Major reforms, some dating back to the early 1990s, have been undertaken to deregulate a wide-range of areas, among which: trucking and bus transportation at the federal level (1989-90); electricity for co-generation and self-supply by independent producers (1992-93); maritime transportation regarding concessions of port services (1991-93); land tenure (1992); court procedures in the Federal District and the State of Nuevo Leon (1996-97); mining (1996);

environment (1996); health care, allowing the creation of a private market for generic drugs in particular (1997). Limits on foreign investment have also been eased over the years in various sectors (1993 to 1996). In 1998, in the wake of global financial instability, the Ministry of Commerce (SECOFI) announced new rules to facilitate foreign direct investment.[130] The goal is to offer more legal security and transparency, as well as to eliminate red-tape and establish fixed delays for permit applications. By eliminating grey areas and introducing more certainty in rules, the government is also aiming at reducing opportunities for bribery.

As recognised by Mexican officials, compliance costs still put a heavy burden on small and medium-sized enterprises (SMEs) in particular, thereby impeding job creation and, more specifically, contributing to keep many activities in the informal sector. Initiatives such as the simplification of bureaucratic red-tape for SMEs or on-line information tools over the past three years were meant to address these issues. In conjunction with actions taken in other areas, such as lowering payroll taxes and improving the social security system, they can help lower some of the barriers to formal activity.

In addition to elaborating a sound policy and establishing the relevant institutions, developing a good regulatory environment also requires strong administrative capacities. In this area, although much progress has been made, there is still room for improving transparency, predictability and accountability of the administrations involved. The changes needed include a wide variety of practices, such as the standardisation of processes for making new regulations, issuing clear policy guidelines, using consultation with interested parties, drafting rules that are easy to understand and implement, and ensuring that appeal mechanisms are clear and consistent. Further improvement would also come from widening the focus of the reform strategy: so far the reform has been essentially business oriented, while regulations affecting consumers and citizens have not been given sufficient consideration, unlike policies in several other OECD countries. In short, to reap the full benefits from the ongoing reform process, it is necessary that the authorities move ahead with existing plans, broaden the scope of reform and carefully oversee the implementation stage.

Notes

1. In the absence of appropriation accounts for households and the corporate sector, developments in these sectors must be based on indirect evidence. For the formal sector, indicators of employment and wages can be found in the social security institute for private sector employees, IMSS.
2. There has been a major change in the structure of Mexican exports in the past two decades. In the early 1980s, oil-related products accounted for about half of total exports, whereas in the late 1990s, they accounted for only around 10 per cent.
3. According to the central bank survey of manufacturing, exporting firms have predominantly used credit from foreign suppliers, with major firms obtaining foreign bank credit. On the domestic side, medium-sized enterprises have mostly relied on reinvested profits, while smaller enterprises have been largely financed by domestic suppliers and Mexican banks.
4. Imports of consumer and investment goods, after a boom in the first quarter, decelerated through the year, suggesting a halt in the building-up of stocks in response to stronger import prices and weaker confidence.
5. The budget cuts are reviewed in detail in the fiscal policy section of Chapter II.
6. 1997 saw the introduction of projects with a deferred impact on the expenditure account (PIDIREGAS). These projects are considered as private investment for national account purposes. Once completed, they will be transferred back to the public sector for operation. Between January 1997 and July 1998, a total of 9.4 billion pesos, equivalent to 0.25 per cent of 1998 GDP, were invested under this programme.
7. Private consumption was particularly buoyant in the first half of the year. Sales of durable goods (up 18 per cent in the first half of 1998, from a year earlier) represent only 10 per cent of total household consumption and are highly concentrated in higher income categories. Sales of non-durables, by far the largest component of private consumption, include necessity items such as food. The moderate growth of non-durables and services sales (up 8.5 and 6.4 per cent, respectively, in the first half of 1998 over a year earlier) is an indication that the boom was concentrated on a limited segment of the population.
8. In the manufacturing sector, the real wage rate increased by a little more than 3 per cent in January-September 1998 over a year earlier, (after a decline of 1.2 per cent in 1997).
9. The remarkable growth of workers in the service sector, as recorded in social security statistics, reflects to some extent the attractiveness of the new capitalisation pension system and new health insurance conditions, which may have enticed some

employees to "formalise" their activity (mainly in the retail sector, which saw the number of affiliates jump sharply after the reform was implemented in July 1997).

10. Labour force data are derived from the Survey of Urban Areas whose coverage has been significantly expanded over time. On the basis of national surveys, the labour force aged 12 years and over has been growing by a little over 3 per cent per year on average since the beginning of the 1990s.

11. These data should be interpreted with caution. The low level of the official unemployment rate does not provide an accurate indication of the structural under-utilisation of labour, as surveys do not include rural areas. Moreover, the lack of unemployment insurance also limits the extent of open unemployment shown in the survey, as the majority of the working-age population have to undertake some type of work. As a result, these rates should be seen as the lower bound for unemployment.

12. Non-tradable goods' prices have a weight of approximately 50 per cent in the CPI index and are mainly determined by labour costs.

13. The consensus forecast for inflation in 1998 moved up from 13.2 per cent to 15.6 per cent between the first and the third quarter of 1998. During the same period, the consensus forecast for 1999 moved up from 11.1 per cent to 13 per cent.

14. In 1994, before the peso crisis, *Maquiladora* wages averaged about half of non-*Maquiladora* wages. By the end of 1998, the ratio had risen to about two-thirds.

15. Mexican exports are largely priced in dollars: this is especially the case for oil and manufactured goods for which Mexico is a price taker, in particular goods produced by foreign companies for export markets, such as cars.

16. The average price of the Mexican oil export basket fell from US$17 per barrel in October 1997 to less than US$8 in December 1998.

17. Mexico seems to have maintained its market share in its main export market, the United States. The share of imports of Mexican goods in total US imports continued to increase, reaching 10.3 per cent in January-July 1998, compared with 9.8 per cent in 1997 (and 7.4 per cent in 1994, following entry into force of NAFTA).

18. In 1998, the share of manufacturing exports was around 90 per cent of total merchandise exports and the share of oil-related exports was down to about 6 per cent in 1998, from 10 per cent a year earlier. Agricultural and mining products accounted for the rest.

19. Intermediate goods make up 77 per cent of total imports – with export-related intermediate goods accounting for the largest share (one-half of total imports) – while capital goods are 14 per cent of the total and consumption goods are only 9 per cent. More than half of total Mexican imports are accounted for by export-related intermediate goods.

20. The next OECD Secretariat projections will be made available in *Economic Outlook 65*, to be published in June 1999 (a preliminary version will be released in May).

21. Price stability was defined by the monetary authorities in 1995 as corresponding to consumer price inflation in the 0-3 per cent range.

22. Net international assets are defined as gross reserves, plus credit agreements with central banks with maturities beyond six months, minus the liabilities with the IMF and central banks originating from credit agreements with less than six month maturities. Moreover, changes in the US dollar value of currency purchases and sales are taken into account. Estimates for the time path of the monetary base have been

provided on a daily basis since 1997, reflecting the seasonal pattern of the demand for base money. The limit set on the expansion of the Bank's net domestic credit (defined as base money less net international assets) is a ceiling rather than a target; it is defined on a quarterly basis. For a detailed discussion of the monetary policy framework, its instruments and targets, see Chapter II of the 1998 *OECD Economic Survey*.

23. Since the Bank sets each year a minimum target for increases in net international assets and the policy framework includes mechanisms aimed at limiting exchange rate volatility, "floating" essentially means that the Bank of Mexico avoids predetermining the exchange rate or providing signals regarding desired levels of the exchange rate.

24. Conditions stated in the programme for 1998 as necessitating action from the monetary authorities were similar to those evoked in the Bank's report of September 1997. For instance, in the case of excessive speculation against the peso, the banking system would be put in "short" position – *i.e.* liquidity would be tightened by means of a reduction of the daily cumulative balance target for the accounts held by commercial banks at the central bank – to induce an increase in interest rates. The banking system would be put in "long" position in the opposite case.

25. The first two features together imply that the intermediate target for base money growth is conditional on a satisfactory balance-of-payments outcome: an unsatisfactory outcome, for instance, a lower-than-targeted accumulation of net international assets, would lead to lower monetary growth and set in motion an adjustment process.

26. The stated aim of the authorities in intervening in the exchange market was to reduce excessive volatility in the peso/dollar rate. As stated by the Bank of Mexico (Bank of Mexico, *The Mexican Economy*, June 1997), under conditions of uncertainty, there is a risk that the market becomes insufficiently liquid, so that even small scale trading can cause disproportionate changes in market rates. Both the Ministry of Finance and the Bank of Mexico are represented in the Foreign Exchange Commission. The resolutions of the Commission currently require a favourable vote of at least one representative of the Ministry of Finance. The portion of the financial reform which was not approved in December 1998 includes a proposal for giving the central bank full autonomy in exchange rate policy (Chapter IV, Box 14).

27. Net international assets (of the Bank of Mexico) have followed a very similar path.

28. The statistical interval is based on the standard deviation of the monetary base demand forecasting model used by the Bank of Mexico. The upper limit of the interval was crossed only a few times during the year, notably in August and September, when national holidays led to a temporary increases in the demand for bills and coins.

29. Statistics discussed here refer to credit by commercial banks that are not intervened nor in a special situation, including current, past-due and rediscounted portfolio, accrued, current, and past-due interest. Values resulting from portfolios ceded to FOBAPROA and trusts are excluded, as well as special CETES financing (corresponding to UDI restructuring) which result from transfer of portfolios to UDI trust funds.

30. Development banks are state-owned institutions specialised in providing credit for specific sectors: small and medium-sized enterprises (through NAFIN), the exporting sector (BANCOMEXT) and local infrastructure (BANOBRAS). These and three smaller development banks are second-tier institutions that supply funds to commercial

banks and credit unions rather than deal directly with retail customers from priority sectors. The financial operations of these institutions have not been recorded in public sector accounts since 1993, reflecting the fact that almost all of these operations are made at market conditions.

31. The public sector accounts discussed in this section follow the same definitions as those used in the Ministry of Finance reports to Congress. The public sector, also referred to as the broad public sector, comprises the federal government and public enterprises under budgetary control, but excludes states and local governments.

32. Starting in 1995, the administration introduced several programmes to shore up the financial sector which had been hard hit by the peso crisis. These included loan repurchases, capitalisation measures, bank take-overs and incentives for debt restructuring. For details on these programmes see Chapter IV below as well as previous *Economic Surveys of Mexico*.

33. See the banking section in Chapter IV. Estimates put the yearly debt service cost of the programmes at around 0.8 per cent of GDP for a number of years (assuming an implicit real interest rate similar to that on the outstanding public debt).

34. For details on the reform of social security institute for private sector workers (IMSS) see Chapter IV below, and *1998 Economic Survey*, Annex II.

35. There were also higher federal government transfers to the National, Railroads of Mexico (Ferronales) to support the privatisation programme. The 1997 social security reform applies only to the IMSS, the institution covering private sector workers. ISSSTE, the social security institution for government employees, saw its revenues increase.

36. "Programmable" expenditures are those over which the federal government has some control. Other items, such as transfers to state and local government and interest payments on public debt, over which the government has little control, are referred to as "non-programmable expenditures" in the Ministry of Finance terminology; they account for almost 40 per cent of the total federal budget.

37. The initial budget assumptions included an average price for the Mexican oil export mix of US$15.50 a barrel for 1998 (compared to US$16.46 in 1997). In the event, oil prices were much weaker than expected, falling throughout the year to average US$10.16 a barrel for 1998. Export volumes also fell, as Mexico joined other producers in limiting supply to support prices. The impact from lower oil prices comes through both as lower revenue for the federal government (from reduced contributions from PEMEX, the state-owned oil producing company) and for PEMEX.

38. It should be noted that infrastructure developed under the PIDIREGAS scheme since 1997, whereby projects are financed by the private sector under the public sector's impulse, have not been affected by budgetary cuts in 1998. The corresponding expenditures are not reflected in public sector budget accounts until the projects are completed and delivered to the public sector. At that time, payments will be scheduled by instalments from budget resources, while the flow of revenue from the project operation will lessen the budgetary cost.

39. Net interest paid on the public external debt is exposed to changes in international interest rates for the part of the debt contracted at variable rates (one-third of the government's foreign debt). Domestic interest rates also have an impact on interest payments abroad, to the extent that part of the domestic debt is held by non-residents. The interest paid on public sector debt refers to the Ministry of Finance definition which does not include official trust funds.

40. The budget was approved on 30 December, after difficult negotiations in Congress. It is broadly similar to the government's initial proposal, but the oil price assumption is lower, spending cuts are more substantial and tax measures differ slightly (more details on the tax measures are included in Box 10 of Chapter III, below).

41. The most meaningful indicator of the fiscal policy stance would be the change in the cyclically-adjusted primary surplus. However, since potential output for Mexico is not available, this indicator cannot be estimated. With output growth expected to decelerate substantially in 1999, the 1999 budget proposal implies a larger degree of discretionary fiscal tightening than suggested by the unadjusted balances.

42. Valuation effects related to the debt/GDP ratio are linked to the real appreciation of the peso during the period which decreased the peso value of the debt denominated in foreign currency. About two-thirds of Mexico's public external debt is US dollar denominated. Data on public debt refer to the Bank of Mexico methodology and include the federal government, public enterprises and official financial intermediaries (development banks and official trust funds). According to the data reported by the Ministry of Finance, which do not include official trust funds, the net public debt amounted to 27.2 per cent of GDP in December 1998 compared with 25.4 per cent in December 1997.

43. These are denominated in UDIs which are units of account indexed for inflation.

44. Detailed fiscal policy measures were only proposed for 1999.

45. A detailed discussion of the PRONAFIDE, 1997-2000 was presented in the 1998 *Economic Survey of Mexico*, Chapter II.

46. In rough terms, every US$1 fall in the price of the Mexican oil mix entails a fall in revenue equivalent to 0.25 per cent of GDP; an increase in real interest rates of 1 percentage points has an impact of 0.07 per cent of GDP on the payment of domestic debt service.

47. See Chapter III, section on budgetary volatility.

48. It should be emphasised that this chapter is mainly concerned with the tax system excluding revenues from PEMEX, and that issues relating to the extraction of rent from the state monopoly are not analysed. Moreover, the chapter generally treats social security contributions as taxes on labour even though these are not included in the tax laws and to some extent could be regarded as insurance. The *Economic Survey of Mexico*, 1998, gives a detailed exposition of the health insurance and pension components of the social security system.

49. Some of these issues have been discussed in special chapters of the three previous *Economic Surveys of Mexico* (1995, 1997, 1998).

50. Moreover, as the major privatisation operations have been completed earlier in the 1990s, there is no longer much potential to raise revenues from this source.

51. Since 1980 the standard deviation from PEMEX royalties has been 1 per cent of GDP, *i.e.* if normality is assumed then the movement from one year to another will not exceed 2 per cent of GDP with 95 per cent confidence. A rough rule of thumb could then be, that tax revenues, excluding PEMEX contributions, should be large enough to accommodate a 2 per cent of GDP drop in PEMEX contributions without causing the budget deficit to exceed a targeted maximum and without having to resort to discretionary expenditure cuts.

52. Figures based on the 1996 Household Income and Expenditure Survey (INEGI).

53. In the OECD 1997 *Economic Survey of Mexico*, references are given to studies that estimate informal employment to be 20-44 per cent of the urban labour force – a share which according to most of the indicators used have been increasing since the 1980s.

54. Other studies have estimated the size of the informal sector through electricity consumption or money demand, and have generally found that the informal sector could be as much as one-third of GDP. These studies, however, do not only capture informal activity, but also tax evasion by the formal sector, which may be a different issue for policy.

55. Background information for this section is presented in the annexes: I. Summary of tax reform measures, 1987-98; II. Main issues in the tax mix discussion in OECD countries in the 1990s; III. The statutory tax regime, 1998.

56. Annex II sets the Mexican reforms in a broader context by summarising some overall principles for taxation as well as some of the main themes of tax discussions in OECD countries since the 1980s. For a discussion of developments in Latin American countries, see Shome, 1992.

57. Taxes include federal, state and local taxes, social security contributions and royalty payments from PEMEX (the latter being classified as a tax in OECD's revenue statistics). In the public sector accounts of the Ministry of Finance (*Hacienda*) the contributions from PEMEX to the federal government budget are treated as non-tax revenues.

58. Countries generally apply widely different practices when compensating individuals, households or corporations through the tax and transfer system, which in turn implies different levels of expenditure and taxes even where the economic impacts are similar. Some rely primarily on expenditure, while others use tax allowances and/or credits more extensively. Moreover, some countries tax gross income, including social transfers, whereas others only tax income net of transfers. Finally, countries may enact policies, for instance regulation, that have similar goals and create similar economic incentives and disincentives as taxes and spending, but have different implication for the size of the government sector as measured by taxes or spending (Leibfritz *et al.*, 1997).

59. The structure and size of revenues from PEMEX are described in more detail in Box 4.

60. Revenues from customs and import duties declined from a peak of 1.2 per cent of GDP in 1992 to 0.6 per cent in 1998 as tariffs were reduced.

61. Unfortunately, the statistics do not allow a full correction of the Mexican data for these idiosyncratic features. Revenues from withholding taxes amounts to some 0.7 per cent of GDP (1997) or around one-quarter of total corporate tax revenue, but information on tax revenues from self-employment is not available.

62. However, as a large proportion of consumption by the lower income deciles are zero-rated goods and services, the VAT increase would not have had a full impact on the living standards for these groups.

63. The subsidy is creditable against tax and varies inversely with the proportion that fringe benefits represent in total earnings. The subsidy decreases with rising income thus adding to progressivity of the overall tax system. According to official estimates, fringe benefits amount to one-third of total remuneration for employees. Fringe benefits are only deductible for the firm if they are extended to all employees, and

for wage-earners with incomes above seven minimum wages they are only exempt up to the amount of one minimum wage.

64. The minimum wage is around 28 pesos per day (December 1998) or less than US$3.

65. Among the entire group of OECD countries, Japan and Korea also (in general) apply lower marginal wedges than Mexico.

66. Empirical work (Leibfritz *et al.*, 1997) shows that labour-demand elasticities are generally much higher than overall supply elasticities (where substitution and income effects move in opposite directions), so that labour taxes tend to be much more distortionary in countries where there are rigid real wages, and most of the tax effect falls on the demand rather than the supply of labour. Unfortunately, there exists no empirical work investigating this issue for Mexico.

67. The marginal effective tax rates are reflected in the required pre-tax rate of return an investment has to earn in order to provide a personal investor with the same post-tax return as a bank deposit earning a pre-tax 5 per cent real rate of interest. The estimates are based on the King-Fullerton methodology. The methodology has a number of shortcomings, including the absence of uncertainty and the rudimentary treatment of financial structures and of the intermediation process. For a thorough exposition see Gordon and Tchilinguirian, 1998.

68. In particular, it is noted how the full imputation system implies equal marginal effective rates on retained earnings and new equity.

69. Producers may be able to shift taxes on to consumers (via higher prices), depending on product market conditions, or onto labour (via lower wage cost, including through reducing employment), depending on labour-market conditions. Labour, on the other side, may also be able to shift taxes in wages and salaries to producers if labour market conditions allow after-tax wages to be held unchanged.

70. By some estimates only around one half of total consumption is taxed.

71. In 1997, the cash refunds amounted to 1.1 per cent of GDP, up from 0.7 per cent of GDP in 1994 and 0.3 per cent of GDP in 1993.

72. The VAT productivity is a standard measurement of the revenues collected per percentage point tax, *i.e.* VAT revenues in per cent of GDP divided by the standard VAT rate. Although a relatively rough indicator it provides some guidance for international comparison.

73. VAT revenue divided by total consumption exclusive of government wages.

74. The immediate deduction scheme for corporations outside the three largest metropolitan areas and for small- and medium-sized enterprises, although comprising some economic rationale in terms of supporting investment and regional development, also created unintended loopholes and distortions, *cf.* Box 10. With respect to consolidation rules, these were considerably more lax than in most other OECD countries (*e.g.* implying that transfer of negative income from a subsidiary to a parent company could take place even though the ownership of voting shares in the subsidiary was less than 50 per cent). According to the new rules, ownership now has to exceed 50 per cent and foreign resident subsidiaries can no longer be included in a Mexican tax consolidation group. Furthermore, the maximum amount of earnings or losses of a subsidiary that may be consolidated is now 60 per cent against previously 100 per cent. Consolidation rules are now more in line with rules in the rest of the OECD but there may still be room for some further tightening.

75. Moreover, enterprises in agriculture, fisheries and publishing only pay half the standard rate, *i.e.* 17 per cent of taxable income and 0.9 per cent in assets tax.

76. The most widely used fringe benefits in OECD countries are food, discounts, transportation, leisure facilities, accommodation, medical and dental insurance, financial insurance and education expenses. For Mexico the non-taxed fringe benefits also includes child care, while excluding food and accommodation. Nearly all OECD countries tax employer-provided motor vehicles and accommodation, and nearly half of the countries tax insurance arrangements and leisure facilities.

77. This assumes that: *i)* the potential tax-contributions from agriculture and transportation would be proportional to their contributions to GDP; *ii)* consolidation rules are tightened further; *iii)* taxable individual income would increase by one-third as fringe benefits are taxed; *iv)* the fiscal subsidy would be eliminated without compensatory reductions in rates, and *v)* finally – in accordance with the authorities' estimate – the effects from eliminating VAT exemptions and the zero-rate, while keeping the standard rate at 15 per cent, would increase VAT revenues by 1.6 per cent of GDP.

78. The question then arises why these workers do not take advantage of the negative tax credit. The main answer is that most workers in the informal sector are not there by deliberate choice but because of shortage of jobs in the formal economy (and self-employed in the formal sector are not entitled to the tax credit). Moreover, for some individuals there could also be an incentive to work as wage earners in the informal sector to the extent the "tax rebate" for the informal enterprise is shared with workers. Finally, the social benefits obtained by entering the formal sector (including health care, pensions and the potential tax credit refund) may not always be perceived as sufficiently attractive. The other side of the coin is that, in contrast with a number of other OECD countries, no benefits are lost when entering work in the formal sector – which provides incentives for integration.

79. According to the views expressed by several private sector representatives, the non-taxation of the informal sector is one of the most serious tax concerns of enterprises in the formal sector.

80. Around one-third of total employment are self-employed, which is comparable to Turkey, Greece and other Latin American countries, but is a much higher share than in other OECD countries. In the United States and Canada, for instance, self-employment is only around 10 per cent of total employment.

81. For example, only 13 per cent of employees of micro enterprises are declared to the social security system (IMSS). Source: *Encuesta Nacional de Micronegocios,* 1994.

82. SAT (Tax Administration Service) covers all the four main areas of tax administration: legal interpretation; auditing; collection; and customs. This is done via 65 local offices, each of which perform all four areas locally. SAT has some 30 000 employees. Internal control procedures are in place to prevent and detect fraud and corruption within SAT staff.

83. This strategy follows broadly the guidelines recommended in much of the literature on tax administration, including Mansfield (1988) and Tanzi and Pellechio (1995).

84. The problems related to income distribution and poverty in Mexico are treated in more depth in the 1995 *Economic Survey* on Mexico.

85. See Box 9 below, which elaborates on this discussion.

86. Whereas a high progressivity can be a desirable feature of the tax system from distributional objectives, it also tends to imply adverse economic consequences in

terms of distorted incentives (not least in terms of under-investment in human capital) and increased evasion and avoidance. Moreover, a high overall progression may in some cases be incompatible with the international mobility of production factors.

87. Unfortunately, the available data for Mexico do not allow for comparisons of income distributions before and after tax, as an alternative measure of progressivity. See for instance Norregaard (1990).

88. To the extent that changes in minimum wages have differed from general wage developments this conclusion could be modified. However, over recent years, minimum wages have generally moved with average wages, so that the comparison over time of progressiveness remains valid.

89. This has also provided a stronger incentive to work in the informal sector, at least for individuals with incomes above three times the minimum wage (*i.e.* where tax payments become positive). In practice, however, this effect is likely to be of minor importance given the low levels of income in the informal sector.

90. The 1999 tax package increases the top individual statutory rate from 35 per cent to 40 per cent (*cf.* Box 10).

91. Total consumption of food by private households times the standard rate of 15 per cent (*i.e.* account is not taken for the border rate of 10 per cent, but this is not likely to have significant effects). The consumption of food is approximated from total household expenditure according to the 1996 household survey and the relative weight on food and non-food per decile from the 1994 survey. The estimate of the subsidy does not consider substitution effects between consumption and savings or within the consumption basket when prices changes.

92. The "all else being equal" assumption implies, among other things, that potential positive revenue effects from higher compliance (in the absence of multiple regimes) are not taken into account.

93. This calculation is an approximation that does not consider compensation in terms of maintaining the same bundle of goods or the same utility level. At the practical level this is of minor importance.

94. Furthermore, to the extent such problems exist they could be viewed as a problem of PROGRESA targeting rather than one of registration.

95. This subject is treated at more length in the 1995 *Economic Survey of Mexico* (Annex II), including a description of activities and responsibilities of sub-national levels of government in Mexico and other OECD countries.

96. These numbers do not necessarily reflect the taxing autonomy of sub-national government, in particular because they do not reflect revenue sharing arrangements.

97. This exemption implies in certain circumstances that the federal government has an incentive not to increase the regulated fuel price by too much: if the (sharable) excises are high due to a large difference between the regulated price and the world market price, the (non-sharable) oil fees will have to be correspondingly lower in order not to exceed PEMEX's 60.8 per cent total contribution.

98. Recent changes in the tax responsibilities of local government are described in more detail in Annex I.

99. The episodes of 1983/84 and 1995/96 illustrate how the growth rate of the real economy (and thus the non-oil tax base) sometimes can be strongly negatively

correlated with the change in PEMEX contributions, whereas the episodes of 1987 and 1990 show the opposite pattern.

100. See also the Inter-American Development Bank (1997).

101. A third potential solution could be to link part of government expenditure to oil prices by issuing government debt indexed to the oil price. Such commodity linked securities are not an uncommon phenomenon. However, markets for such bonds are normally less liquid than markets for more ordinary instruments, implying a premium. Moreover, investors will have to be compensated for the oil-price risk, so that the overall additional costs of such a programme might outweigh the advantages.

102. Chile's copper stabilisation fund was established in 1987 in order to mitigate the fiscal impact of fluctuations in copper prices (copper is some 40 per cent of Chile's total exports and the state copper company CODELCO normally contributes 5-10 per cent of total tax revenues). The fund receives transfers from CODELCO when export prices exceed a certain threshold and transfer resources to the treasury when prices fall below the threshold. The impact of this stabilisation mechanism has tended to diminish in the 1990s as the share of copper produced by private companies has increased strongly. The Norwegian Petroleum Fund, created in 1991, serves a broader purpose: by investing the fiscal surplus in financial assets abroad, the government fund also aims to offset the net currency inflows associated with oil, in order to stabilise the exchange rate.

103. Foreign-currency denominated liabilities which accounted for 19 per cent of total liabilities in September 1994, rose to 29 per cent in March 1995 with the effect of devaluation.

104. Currency depreciation alone led to a reduction in bank capitalisation ratios, as outstanding loans denominated in foreign currency saw their peso value rise.

105. The banking system's equity in December 1994 was broadly equivalent to 3 per cent of GDP.

106. This includes the cost of rescuing the road operators, estimated at 0.5 per cent of GDP.

107. The following discussion refers to "normal" banks, *i.e.* it excludes banks under the control of authorities and excludes assets which have been transferred to FOBAPROA.

108. Banks have tightened credit standards. Furthermore, the supervisory commission for banks and securities (CNBV) has established new guidelines for the provisioning of new loans: banks that grant new loans to debtors with a bad credit record must establish reserves for 100 per cent of the loan value.

109. New rules brought Mexican accounting standards into line with internationally accepted principles. From January 1997, the whole amount of a loan that is delinquent is recorded as non-performing. In the past, only the portion of the loan that was due and unpaid was recorded as such.

110. Some foreign banks have entered the Mexican banking sector as part of bank rescue operations as the government offered incentives to attract new shareholders. Takeovers and mergers have allowed some foreign banks to gain quickly an established position in the market, while others have preferred to set up new fully-owned subsidiaries.

111. Until 1998, FOBAPROA liabilities were for the most part long-term obligations maturing between 2005 and 2007. FOBAPROA had to obtain market loans at interest rates

which were generally between 100 and 150 basis points above "recognised" public debt.

112. An important change in this regard is the new regulation from the Bank of Mexico, which forces banks to keep a liquidity coefficient for their dollar liabilities: banks must hold highly-liquid dollar denominated assets (such as US Treasury Bills) to back their short-term liabilities.

113. Past-due loans are likely to rise over the coming quarters as payments are delayed. Mortgages are also expected to suffer, as borrowers with variable rate loans see their monthly payments rise more rapidly than the value of their assets. In the case of mortgage loans denominated in UDIs (which contain fixed real interest rates and account for 60 to 70 per cent of mortgages held by major banks), higher inflation would add to the principal.

114. According to other countries' experience with the introduction of such pension systems, the most immediate impact is a shift from public saving to private saving, as tax revenue finances the transition cost from a pay-as-you-go to a capitalisation system.

115. The new financing scheme for IMSS health insurance, presented in the 1998 Survey, foresees a gradual decrease in the rate of the employer-employee proportional contribution. Another part of the financing is based on a fixed fee per worker (from the government and from employers).

116. Contributions to the pension system include compulsory contributions (by employers, employees and the government) and voluntary deposits that workers can make on their accounts (either directly at the banks which channel the funds to the AFOREs or through their employers). Contributions to the housing fund, INFONAVIT (which are compulsory) are deposited on different accounts and managed by INFONAVIT itself. Funds transferred from the previous SAR accounts, created in 1992 for retirement savings are split between the two accounts. Only the resources that are deposited on the retirement account (from compulsory or voluntary contributions plus those funds transferred from SAR accounts) are invested through the specialised mutual funds. The 1998 *Economic Survey of Mexico* (Box 2 in particular) provides details of the new pension system.

117. The contributions of workers who have not yet chosen a fund administrator are kept on a "concentrating" account at the central bank under the name of the IMSS. In addition to contributions of workers who have not yet selected a private fund manager, the concentrating account of IMSS at the central bank includes those for which the identification process is not completed.

118. In December 1998, 61.8 per cent of the AFOREs portfolio was invested in one– and two-year fixed-rate BONDES, 22 per cent in short-term fixed rate CETES and 11.9 per cent in three and five-year indexed UDIBONOS. Only 3.1 per cent of the total was invested in private sector securities (including banks), mostly short-term, although regulations allow the AFOREs to invest up to 35 per cent of their portfolio in private debt.

119. For example, traditional investment funds have been charging commissions amounting to 2.4 per cent of the outstanding balance on average, against 1.3 per cent in the case of AFOREs, according to estimates by the regulating agency CONSAR (National Commission for Retirement Savings). The CONSAR makes adjustments to the data on commissions to allow comparison across fund managers: because commissions are charged under different structures by different AFOREs (fees are front-end, back-

end, or a combination of both), they are not easy to compare before adjustment. The calculations made by the CONSAR improve transparency, which should help strengthen competition among AFOREs, in turn contributing to improve conditions for savers in terms of services and commissions.

120. For a detailed review of the Mexican health care system and the reform strategy, see the 1998 *Economic Survey of Mexico*, Chapter IV.

121. Detailed comparative statistics on education are available in OECD, *Education at a Glance*, 1998, based on information up to 1995.

122. In mid-1998, the number of children attending high-school was 2.3 per cent up on a year earlier. Altogether, coverage of basic education had risen by 0.6 per cent to a total of 16.5 million children; but school enrolment is still lagging considerably relative to other OECD countries.

123. Privatisation proceeds amount to a cumulated 1 per cent of GDP over 1995-98, compared with 7.9 per cent of GDP during the period of the previous administration, 1988-1994.

124. In the past there was a complex range of long-distance rates according to the distance; nowadays all carriers offer a single rate for national long-distance calls, regardless of the distance. This enables users to take informed decisions in the choice of carrier. In addition, the average public price paid for long-distance has dropped significantly – by June 1997, six months after the opening to competition, prices had dropped by 30 per cent on average.

125. A detailed review of the telecommunications sector is included in the OECD report on regulatory reform in Mexico (forthcoming).

126. For the Monterrey area the objective is to provide service to 500 000 users within five years. For the Mexico Valley, the concessions are for a 30-year period, with exclusive of distribution rights during five years: by then the purchasers are required to have increased coverage from 135 000 to 820 000 users; under the concession for the Queretaro area, a five-year exclusivity is expected to allow new investments so that the existing pipeline network is increased twofold to reach 50 000 users.

127. The following paragraphs draw on a preliminary report prepared by the OECD in the context of the multi-disciplinary review of regulatory reform in OECD countries (forthcoming).

128. Milestones in building the regulatory reform system have been: the creation of the Unit for Economic Deregulation in the Ministry of Trade and Industry (1989); the Federal Metrology and Standards Law (1992); the Federal Competition Law (1992), which created the independent Federal Competition Commission; the Federal Administrative Law (1994); and the Agreement for Deregulation in Entrepreneurial Activity, ADAE (1995), which strengthens the role of the Unit for Deregulation. Since 1996, a regulatory impact assessment (RIA) is required for all new regulations.

129. Information on the government agency in charge and on applicable fees and charges, as well as a time limit for the administration's response, are also included on the Internet site.

130. The new regulations published in September 1998 update the Foreign Investment Law of 1993 and the related Law Decree of 1996. The regulations now legally establish the long-term practice of creating trusts with Mexicans as a way to circumvent some restrictions: one example was the prohibition of foreign property ownership within 100 km of the Mexican borders and 50 km of the coastline; this mechanism will now be legal as long as it is for business purpose.

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Annex I

Summary of tax reform measures, 1987-98

Taxation of personal income

1988 Reduction of the number of brackets from 15 to 12. Reduction of maximum tax rate from 55 to 50 per cent. Indexation of brackets to minimum wage.[1] Exemption of vacation bonuses and profit sharing from taxable income (up to 15 times the minimum wage).[2]

1989 Introduction of tax credit to replace flat deduction. Reduction of maximum tax rate from 50 to 40 per cent.

1990 Reduction of maximum tax rate from 40 to 35 per cent. Introduction of inflation adjustment of tax brackets.

1991 Introduction of a tax discount (subsidies) in inverse proportion to the share of fringe benefits in total payroll.[3] Reduction of the rate charged on interest income of individuals.

1995 Tax credit increased for lower income brackets. Optional simplified rate (2.5 per cent) for street vendors created.

1997 Increased exemptions and tax credits to low income earners.

1998 Creation of a new regime for small taxpayers in order to combat fraud, to replace previous regimes (except that applying to primary sectors). For taxpayers with entrepreneurial activities earning less than 2.2 million pesos, the marginal tax rate on gross income varies from 0.25 to 2.5 per cent, after a deduction of three times the minimum wage.

Taxation of corporate income

1989 Tax base is fully adjusted for inflation, following a transition period. Corporate tax rate is lowered from 42 to 35 per cent.

1990 Introduction of the simplified tax regime for small taxpayers and specific sectors (regardless of size) replacing special tax bases.

1991 The special tax rate on income derived from the agriculture, livestock, fisheries and forestry sectors – both corporate and personal – is reduced to 50 per cent of

normal tax rate (instead of 40 per cent previously).[4] Fiscal losses can be carried forward over ten years (instead of five previously).

1994 Reduction of marginal tax rate from 35 to 34 per cent.

1995 Simplified tax regime is expanded by raising maximum income of firms eligible for the programme. Deductions for investment and R&D are increased; and depreciation rates are accelerated.

1997 Tax payment restructuring for firms facing liquidity problems. Tax arrears are partly forgiven and partly re-negotiated at more favourable terms.[5]

1998 Measures to promote long-term savings are introduced, including through employees' individual retirement accounts.

Asset tax

1989 Introduction of the asset tax at a 2 per cent rate (not applicable for the first three years after company starts operating). Asset tax is refunded against income tax – excess is carried forward for three years.

1990 The asset tax rate is reduced to 1 per cent for taxpayers who benefit from the reduced income tax rate regime (applying to the primary sector). Consolidation of asset tax introduced.

1991 The number of years over which asset tax refund is possible increases from three to five years.

1992 Exemption for street vendors who pay income tax under specific regimes (10 per cent of their purchases). Special, more favourable, regime created for the hotel sector.

1994 The number of years over which asset tax refund is allowed increases from five to ten years.

1995 Tax rate lowered to 1.8 per cent. New enterprises exempt for three years. Established firms with qualifying investment programmes can defer payments for four years.

1996 Income below 7 million pesos exempt. Investment and employment creation incentives are introduced for one year.[6]

1997 Widening of the tax to include most financial institutions (except for assets used in intermediation operations). Income below 8.9 million pesos exempt. Investment and employment creation incentives extended for one year.

1998 Investment and employment creation incentives were not extended.

Value added tax

1988-92 A number of changes in VAT rates and structure are introduced.

1992 General VAT rate lowered from 15 to 10 per cent and the special border rate raised from 6 to 10 per cent.

1995 General tax rate is raised from 10 to 15 per cent, except in border areas, where it remains at 10 per cent.

1996 Widening of the zero rate to include potable water, processed food items and medicines, at all stages of production. VAT applies only to real interest (instead of nominal previously) on consumer credit and credit card loans.

Car taxes

1987 Five tax rates are established on the sale of new cars, depending on the price of vehicle.

1989 Reduction of rates; limits are introduced on deductibility for businesses.

1990 Full deductibility is re-established for cars purchased by businesses.

1991 Tax rates on ownership and use of cars established at 2.0, 5.0 and 8.5 per cent.

1992 Tax rates on ownership and use of cars raised to 2.5, 6.5 and 10.0 per cent.

1997 Tax on new cars tax reduced (temporary measure applicable for one year only).

Excise taxes

1989 Rates reduced for tobacco and wine.

1993 Rates reduced for beer.

Revenue-sharing measures

1996 Measures to introduce a new federalism

Revenue-sharing rules

The general revenue-sharing pool is increased to 20 per cent of "sharable" federal tax receipts – 7 percentage points more than before.

The federal government will distribute 20 per cent of excise tax revenues on alcoholic beverages and beer, and 8 per cent of the tax on tobacco, directly to state and local governments.

Rescheduling of revenue-sharing transfers through the year to improve the cash flow of the lower echelons: 90 per cent of the funds are to be apportioned monthly and only 10 per cent set aside until the end of the fiscal year.

The federal government increases the proportion of bridge toll revenues to be distributed to the states and municipalities to 25 per cent of operating revenue (from 10 per cent previously). The resources may be earmarked for infrastructure and capital expenditure.

Devolution of responsibilities

Lower echelons are authorised to impose licensing fees for bars.

They are authorised to levy a tax on hospital services and on motor vehicle ownership or use. Beginning in 1997, the proceeds from the federal new car tax will be passed on to the lower echelons in their entirety.

The federal tax on the acquisition of real estate is rescinded, and the lower echelons will be authorised to impose it.

1997 Lowering of federal tax rate on lotteries (from 21 to 15 per cent); states can levy a 6 per cent tax on the same basis. Under an administrative co-operation agreement, the states are authorised to administer a number of federal taxes. This is an incentive for improving their collection capability insofar as they may keep any additional taxes imposed as a result of the tax audits they conduct (this being applicable to income tax, assets tax, excise tax for small taxpayers and VAT).

Notes

1. Indexation to the minimum wage implies some drift since minimum wage adjustments are systematically lower than inflation.
2. The exclusion of fringe benefits from the taxable basis was followed by a gradual increase in this component of income, to about ⅓ of the employers payroll (Francisco Gil, 1995).
3. The coefficient of subsidy is calculated at the level of the employer, not by worker. This measure has lowered effective marginal tax rates by significant amounts – by some estimates by as much as 50 per cent (Francisco Gil Diaz, 1995).
4. The tax rates are reduced by 75 per cent if taxpayers from these sectors put their production through industrial processing.
5. This measure should be seen in the context of a general effort to alleviate debtors' problems in the aftermath of the peso crisis, to prevent the development of a widespread non-payment culture. Support programmes for other debtors and banks are reviewed in Chapter IV.
6. This measure was meant to support the economic recovery; it was extended in 1997 to boost the domestic-oriented sector.

Annex II

Main issues in the tax mix discussion in OECD countries in the 1990s

The role of the tax system is to allow the Government to fulfil its inter-temporal budget constraint by generating the necessary revenues as efficiently, equitably and manageably as possible.[1] This implies that taxation should minimise disincentives to work, invest and save; that horizontal equity should be observed and vertical equity taken duly into account;[2] and that tax laws and regulations should not be unduly complex and difficult to enforce by the authorities. While these broad guidelines constitute a set of "first principles" for taxation, there is more ambiguity when it comes to practice. In theory the least distortionary tax is a lump-sum tax where each individual pays the same amount in tax. This raises problems with respect to vertical inequality as well as generating sufficient revenue. Most general equilibrium models imply that consumption taxation is less distortionary than income taxation (because consumption taxation can be viewed as a proportional tax on lifetime income or intergenerational income, that does not distort inter-temporal decisions, *i.e.* savings and investment). However, this again generates concerns of vertical inequality and possibly also compliance because the revenue-neutral rate is likely to be very high for most OECD-countries. Another issue is whether income from different sources should be taxed at rates according to the elasticity of the bases, or a single rate schedule should apply to aggregate income from different sources (the so-called Shantz-Haig-Simons principle). In practice, it is hardly possible to identify and implement such a thing as an "optimal tax system" and the exact design of the tax system will be influenced by a host of idiosyncratic, and often interrelated, features of the individual countries – such as the income level and distribution, regulation and flexibility of labour and product markets, relative magnitude of distortions from different taxes and their impact on economic activity and/or social welfare, industrial structure, considerations on tax competition and political preferences for government expenditure levels. Moreover, the overall political context is of course decisive for the design of the tax system as are the administrative capacities in place.

The issue of overall tax mix is perhaps the most fundamental problem in the design of tax systems. There has been considerable convergence in tax mixes in the OECD, but large differences among countries remain. Generally, the broad thrust of tax reforms have involved an increase in direct labour taxation in the form of social security taxation, a decrease in both statutory and average effective rates of corporate tax, and a shift in consumption taxes away from excise taxes and toward value added taxation. There has also been a slight increase in the use of environmental taxes, especially in Europe. These changes seem to have been driven by two considerations. First, there was a desire to reap efficiency gains from the reform of tax systems. Clearly, the priority assigned to finding less distorting ways of levying taxes had to increase as tax burdens themselves increased in the OECD. A second consideration was undoubtedly globalisation, as these reforms tended to lower marginal rates on more mobile tax bases (and, hence, on those with more elastic

responses to taxation) and to raise them on less mobile tax bases. In this sense, then, the reforms corresponded to a standard principle for the design of tax systems – that tax distortions are minimised if the burden is focused on tax bases that respond relatively little to taxation. At the same time, though, this shift in tax mix gives rise to fairly serious concerns for distribution and tax equity. Some of the main themes of the tax structure discussion among OECD countries are summarised below.

First, there has been some advocacy for a move from income taxation to general consumption taxes (VAT). This would reduce tax disincentives on inter-temporal decisions (*i.e.* savings and investment). Moreover, consumption taxes are generally less easy to avoid for the underground/informal economy. However, consumption taxes tend to be less progressive than income taxes, they may distort consumption patterns if rates are not uniform, or some goods and services are exempted, and taxpayer resistance is often significantly biased against consumption taxes (Messere, 1993). Moreover, such a shift would also have generational distributional consequences in favour of the younger generations during the transition since people with assets accumulated under the income tax regime would essentially "pay twice" as they consumed these assets under the consumption tax regime.

Second, a measure which has attracted significant attention in Europe is a switch from payroll taxes to VAT (see for instance OECD, 1995). This is particularly attractive to countries where payroll taxes have been high and increasing and where labour market arrangements convert part of the incidence of this tax into unemployment.[3] The effect of such a measure would be to shift some of the tax burden to those outside the labour force (*i.e.* receivers of capital and transfer incomes), thus reducing the disincentive for employers to take on labour – especially low-skilled and part-time employed to the extent social security contributions are levied from the first unit of income earned and reduced above certain ceilings. A variant of this proposal has focused on a shift from labour to energy taxes, in which case some of the tax burden could (but not necessarily) be shifted to energy consumers and producers. A number of countries have already experimented with such taxes (Belgium, Denmark, Finland, Netherlands, Norway, Sweden, and the United Kingdom). The few empirical studies of such measures suggest that the employment impact of such measures is extremely small .

One difficulty encountered in implementing policies that shift the tax burden onto transfer recipients is in reducing the standard of living of those receiving transfer payments. If they are fully compensated for the tax-induced increase in prices, then the scope for tax cuts on labour is much reduced. Moreover, such a shift might only have marginal effects on labour supply decisions to the extent these depend on the total tax burden of a household (*i.e.* including consumption taxes), *cf.* Layard *et al.* (1996). It has also been argued that such a shift might – at least in the short term – improve wage-competitiveness since VAT but not payroll taxes are rebated at frontiers. Although true, especially for countries with fixed exchange-rate regimes and slow wage adjustment, such temporary arguments should not, by themselves, validate adjustment to tax regimes.

Third, the links between benefits and tax systems have been widely analysed, in particular the existence of so-called poverty or unemployment traps in which low-income households face very high marginal effective tax rates when moving up the income ladder, or taking on work. In fact, marginal effective tax rates, while having generally been lowered for high incomes, have barely moved, or in several countries even increased for low-income taxpayers during the past two decades. However, as noted by the *Implementing the OECD Jobs Strategy (1997)*, marginal effective tax rates for low incomes have been reduced during the 1990s in several countries including France, the United Kingdom, Canada, Denmark, Iceland, Ireland, the Netherlands, New Zealand and Sweden. Measures to

reduce poverty or unemployment traps through the use of employment conditional benefits were introduced or expanded in the United States, the United Kingdom, Canada, Australia, Ireland and New Zealand. The same has been the case for Mexico with the introduction in 1994 – later expanded in 1995 – of a progressive salary tax credit.

Fourth, in a few countries, a move from employees' social security contributions towards an income tax has been advocated even though the proceeds may still be earmarked for social welfare benefits. The argument here is between those who regard social security contributions as insurance, and those who regard them as a regressive form of taxation, with flat rates, low ceilings and no allowances for family circumstances. In most OECD countries, the link between contributions and benefits is at best tenuous, and social security is *de facto* no more than a tax on labour. France and Sweden have recently reduced payroll taxes within an unchanged overall tax wedge of labour income. In Mexico, the social security reform also draws in this direction (1998, *Economic Survey* of Mexico).

Fifth, a general switch from mobile to immobile tax bases is being widely considered, as trade- and capital-market liberalisation increases the potential for tax competition between countries. The trade-off will be to what extent higher marginal tax rates on relatively immobile factors can replace taxes on mobile factors, thereby safeguarding domestic revenues, without causing excessive distortions in the supply and demand of the immobile factor. Up until now, there has been little evidence, that increasing globalisation and tax competition have had much influence on domestic tax profiles (leaving aside the European Union, where a substantial convergence has taken place in indirect taxation due to the gradual implementation of the single market).

Notes

1. See for instance Goode, 1984.
2. Horizontal equity refers to a situation in which people with the same level of income are taxed at the same level. Vertical equity implies that tax payments are increasing in income.
3. Although the incidence of social security contributions depends on the degree of labour-market flexibility, there is some empirical evidence that these taxes have a larger effect on employment than other taxes, *cf.* the *OECD Jobs Study (1994)*.

Annex III

The statutory tax regime, 1998[1]

Income from individual non-business activities is taxed at rates ranging from 3-35 per cent (Table A1). The statutory rates, both at the bottom and at the top, are lower than found in most other OECD country.[2] Income brackets are indexed to inflation on a semi-annual basis. In contrast with many other OECD countries, Mexico does not apply a zero-rated band for low incomes (Table A2). However, a basic allowance is given for holiday bonus and end-of-year bonus. Moreover, a salary tax credit implies that low-income earners pay negative taxes. Non-standard deductions include medical expenses, school transportation, funeral charges, charitable donations and voluntary contributions to pension schemes (up to a certain limit). Imputed rental income from owner occupied housing is not taxed, while interest expenditure, including on housing, are not deductible for individuals. Positive real interest income is taxed at a rate of 17 per cent, which applies only up to the first 10 percentage points of interest and only for financial instruments with less than one year's maturity. Exempt from tax is interest income from government securities and several other instruments. States and municipalities are not allowed to tax income, but states can levy payroll taxes.

The social security system has been fundamentally reformed in recent years. The pension scheme is being transformed from a pay-as-you-go system into a fully funded system and the financing of health insurance is being changed so as to reduce labour taxation, increase the financial viability of the system and make affiliation to social security more attractive for the uninsured population in the informal sector.[3] The new pension system strengthens the link between workers' contributions and benefits, which reduce incentives for workers to evade or under-declare income. In addition to mandatory contributions, workers can make voluntary deposits on their accounts, and the government pays a fixed contribution. The financing reform of the health insurance increases contributions from general taxes, while lowering the share of social security contributions (except for low incomes) and their marginal rates (Table A1). This could alleviate problems of under-reporting of income[4] and might also facilitate the integration of workers from the informal to the formal sector as payroll taxes fall most heavily on low-paid and part-time workers (OECD, 1998).

Corporate income is taxed at a uniform rate of 34 per cent – roughly in line with most other OECD countries (Table A3) and lower than in the United States and Canada. As in the individual income scheme, corporate income is fully indexed for the effects of inflation. Besides income tax, Mexican enterprises have to pay into a so-called profit sharing scheme by which 10 per cent of after-tax profits from the previous year is distributed to workers.[5] Depending on the incidence, this could raise the effective corporate income tax rate by up to 10 percentage points (*i.e.* from 34 to 44 per cent).[6] Such schemes are not used in other OECD countries. Mexico has recently implemented the OECD guidelines for transfer pric-

ing, and have also applied a set of specific rules for operations with residents in tax havens. Dividends are taxed at the corporate level, but individuals may choose to have dividends taxed in the individual income scheme (grossed up by a factor of 1.515) and credit the tax paid at the corporate level. No double taxation of dividends thus takes place, thereby ensuring an equal treatment of corporate financing through new equity or retained earnings. Compared to almost all other OECD countries, the wedge on dividends – *i.e.* the difference between gross earnings at the corporate level and net "take-home" dividend for the individual share-holder – is correspondingly low in Mexico (Table A3). In accordance with the integration of individual and corporate taxation, self-employed are free to chose between the individual and corporate income scheme.

In order to improve compliance of corporate taxation, a net worth tax ("asset tax") was introduced in 1989, whereby net assets are taxed by a flat rate of 1.8 per cent at present. The asset tax is creditable against income taxes paid, and the purpose of the tax is primarily to limit income-tax avoidance – for instance, through transfer pricing or overstatement of costs – by imposing a minimum tax on corporate activity above a certain threshold. The rationale for using an asset tax, as opposed to, for instance, a lump-sum tax, is that assets in principle reflect the earnings capacity of a company. From an administrative point of view, companies have less incentive to under-report investment, as compared to income, as this would have negative consequences for their depreciation allowances. The current tax rate of 1.8 per cent assumes that the annual real return to private capital should exceed 5.3 per cent on average. If this is the case, the assets tax does not distort investment incentives (because the income tax generated would exceed the asset tax) – and a lower rate of return would, in any case, indicate an inefficient use of the assets. The assets tax thus encourages an efficient capital allocation. The assets tax has been highly criticised by the business sector, not least because it applies in years when firms are making losses,[7] but also because it discourages foreign investors by not allowing start-up concessions for these investors. Although it is normally not recommended to tax investment directly because of the possible adverse consequences for capital accumulation – and, indeed, net worth taxes are only levied in a few other OECD countries (Canada, Switzerland, Finland and Iceland) – it has a clear justification in the case of Mexico, given the administrative problems of controlling and enforcing corporate income-tax payments. One study thus concluded that every peso of assets tax collected brings in an additional 3.5 pesos of corporate income taxes, thus corroborating the important control function of the assets tax.[8] Assets taxes or net wealth taxes are also levied in a number of other Latin American countries.[9]

Besides the general corporate tax regime, three partly overlapping preferential regimes are in place: the simplified regime for the agriculture,[10] fishery and land transportation sectors; the special rate regime for agriculture and publishing; and the small taxpayers regime. The simplified regime applies the standard corporate tax rate of 34 per cent, but calculates income on a cash-flow basis. All expenditure is thus immediately deductible (including investment in physical and financial assets), and only resources taken out of the company are subject to taxes. In practice, this leaves companies under this regime, which are typically family-run businesses, with ample room to avoid any taxation. The special-rate regime applies only half of the standard corporate tax rate, *i.e.* 17 per cent on income and 0.9 per cent on assets, on the enterprises included.[11] Finally, the small taxpayers regime includes companies with an annual gross income below 2.2 million pesos. This scheme was especially set up to encourage small firms and the self-employed to register in the formal sector and to reduce the government's administrative costs linked to these entities. Besides avoiding the complexities of the general regime, notably inflation accounting, this regime also applies lower reporting requirements. Gross income is taxed

at rates between 0.25-2.5 per cent. Even though preferential regimes like the ones in Mexico are not uncommon in other OECD and Latin American countries, the economic justification for preferential treatment of certain sectors or types of enterprises are most often relatively weak. This is to a large extent the case for Mexico, where the concessions granted – especially in terms of the simplified and low-rate regimes – seem to reflect a favouring of political pressure groups more than profound macroeconomic and structural considerations.

Taxation of consumption has typically accounted for 40-50 per cent of total tax revenue (excluding royalties from PEMEX) with the sources being split almost equally between VAT revenues and other consumption taxes, mainly excises, customs and import duties. Revenues from VAT have been significantly influenced by numerous changes to the system over the years – during the 1990s most notably the decision in 1990 to switch the VAT administration from states to the federal government, the standard rate reduction in 1992 and its reversal in 1995 (as noted in Annex I). In broad terms, the Mexican VAT system follows a set-up not very different from that of most other countries. It applies a standard VAT rate (at present 15 per cent) while having some goods and services (food, medicine, newspapers, etc.) being exempted or taxed at a lower rate, in this case zero. Furthermore, the standard VAT rate is reduced to 10 per cent in areas close to the US border. However, Mexico deviates from many other OECD countries by the amount of exemptions and zero-rated goods.[12] In most other OECD countries the lower rates are applied to a much smaller variety of goods and services (Table A4). Mexico also allows wider concessions to small traders than seen in other OECD countries except for Japan. One departure in the other direction is the inclusion of some financial services, including consumer credits and personal loans, in the tax base. Overall, however, the revenues from VAT are relatively low in an international comparison – of OECD countries applying VAT only Japan and Canada have lower revenue/GDP ratios.

The problem of exempting and zero-rating a very wide range of goods and services is not only the revenue foregone *per se* by narrowing the base, but also that it opens up multiple possibilities for non-compliance and avoidance – for instance by falsely attributing sales to zero-rated or exempted transactions – thus making it difficult for the tax administration to enforce the VAT-regime appropriately. This is exacerbated by the possibility for companies to get cash refunds for VAT credits. In most countries a zero-rate only applies in the export sector, and the only businesses that can be expected to systematically show tax credit in excess of tax debits are exporting enterprises. In such countries authorities often choose to make the option of VAT refunds available to exporting companies only, so as not to burden the administration with the processing and control tasks involved in refund claims. In Mexico, the economic rationale for using consumption taxes (*i.e.* they do not distort savings and investment decisions) is also diluted to the extent that different rates distort relative prices. Finally, by allowing a large spectrum of goods and services preferential treatment, pressures from rent seeking may mount as taxpayers in the ordinary regime are encouraged to also try to become exempt or zero-rated.

Excise taxes are based on the same system of credits, debits and reimbursements as the VAT, whereas in many other countries excises are levied only at the import – or production – point. Although technically elegant, the Mexican system is difficult to administer, which could give rise to problems of non-compliance in this system as well. Like most other countries, Mexico levy excises on fuels, alcoholic beverages, tobacco and motor vehicles. No excises are levied on environmentally damaging activities. Revenues from excises to GDP have fluctuated sharply over the 1980-96 period, with a coefficient of variation being double that of VAT-revenues to GDP. The main reason is the excise on fuel, which fluctuates inversely proportionally with the oil price. The fuel price for the final

consumer at the pump is regulated by the authorities on a monthly basis (*i.e.* the price is linked to expected inflation) with the schedule being determined in the annual budget.[13] The excise is then determined residually as the difference between the regulated consumer price and the import price on oil adjusted for a mark-up and commercialisation costs at PEMEX. When market prices of oil fall, the regulated consumer price remains unchanged, and excise revenues will increase accordingly. However, the overall revenue intake will be dominated by a drop in other oil-related revenues. The revenue raised by PEMEX are described in more detail in Box 4 of the main text.

Notes

1. A detailed overview of the Mexican tax system in 1998 is presented in Table A1.
2. Switzerland has lower rates at the central government level, but not when taking local government income taxes into account. New Zealand has a lower top rate, *i.e.* 33 per cent.
3. More details are given in the 1998, *Economic Survey* of Mexico and Chapter IV of this survey.
4. IMSS registers show that in 1993 one out of five workers were contributing to IMSS on the basis of earnings of one minimum wage or less, while according to the national Employment Survey only one out of 40 workers were earning such low wages.
5. Formally, this is not a tax since it is instituted in the Labour Law rather than the Tax Laws. However, it has the same *de facto* effects as an income tax.
6. Most likely, however the incidence will be on workers, and the effective increase in the rate is likely to be significantly less than 10 percentage points. It should also be added, that the profit sharing is deductible from corporate income to the extent it exceeds fringe benefits. As these typically vastly exceeds 10 per cent of profits the effective deductibility of profit sharing is probably small.
7. However, assets taxes paid can be refunded by the amount by which corporate income tax exceeds the assets tax in any of the following ten years. Likewise the amount by which income taxes have exceeded assets taxes during the previous three years are creditable against the assets tax in a given year. New enterprises are exempt for 3 years, and enterprises with incomes below 8.9 mill. pesos (1998) are also exempt.
8. Gil Diaz (1995).
9. Shome (1992).
10. Including livestock and forestry.
11. Workers in agriculture and fishery are also totally tax exempt for incomes below 20 times the minimum wage.
12. Exemptions differ from zero-rating in that businesses that sell exempt products cannot claim credit for the VAT included in their purchases (*i.e.* the VAT paid is then shifted on to consumers, and all transactions up to the exempted sale will be taxed). Moreover, if a business' sales consist exclusively of exempt commodities, it can be considered outside of the VAT chain and treated as a final consumer. Exemptions should thus be considered a superior option to zero-rating from the administrative as well as from the revenue viewpoints but not from a distributive viewpoint as replacing zero-rating with exemptions would increase prices for a range of basic necessities.
13. The Ministry of Finance has the option of raising the regulated price at its discretion without Congressional approval.

Table A1. **Main features of the tax system, 1998**

Tax	Nature of tax	Deductions and exemptions	Rates
Federal taxes **1. *Taxes on Income:*** 1.1. *Individual:* 1.1.1. Individual income tax IIT)	The IIT is levied on an individuals income from all sources, including wages and salaries, income from business activities, service fees, real property and interest. Changes in the real value of assets and liabilities due to inflation are taken into account in determining the tax base. Income also includes gains from the effect of exchange rate changes on the value of debts denominated in foreign currency. Tax-files must be returned before end of April of the year following the year in which the income arose. Individuals who are exempt from filing include employees whose salary income arose exclusively from one single employer and those whose final taxes have already been paid by way of withholding at source. Income tax provisions include a separate set of rules governing computation of income derived from business activity that are substantially the same as the computation of income under the corporate income tax scheme (CIT). The self-employed with business activities may choose to include business profits multiplied by 1.515 on their final tax return as part of their global taxable income and take a credit for advance at the corporate tax rate, or pay taxes on that income separately under the scheme for corporate business profits. If gross income the preceding year did not exceed 2.2. million pesos (1997) they may opt for the special regime for small taxpayers, where gross income is taxed at rates from 0 to 2.50% (7 brackets). The simplified tax system permitted under the CIT is also applicable to individuals with income from business activities in the same sectors determining eligibility for corporations (see below).	Certain items of income are exempt from income tax when paid to individuals; these items include: income from agriculture, livestock raising, forestry, or fishery to the extent that income does not exceed 20 times annual minimum salaries; interest paid by credit institutions on savings deposits which do not exceed two times general minimum salaries for one year and whose interest does not exceed the rate fixed every year by the Congress; interest paid on investments in registered securities of the Federal Government; interest on bonds, securities, and other credit documents with terms exceeding one year and publicly issued under rules issued by the SHCP; interest paid on bonds and debts issued by international credit institutions of which the Mexican Government is a member; inheritances; fringe benefits calculated on the minimum wage provided they do not exceed the limits established under the Fiscal Law; certain social security benefits and other benefits received by workers. Deductions for personal expenses: expenses incurred for transportation to school of taxpayer's direct descendants in zones where transportation is necessary; medical, dental, and hospital expenses of the taxpayer and dependants; funeral expenses for the dependants up to the annual minimum wage; and donations made to qualified institutions. In addition, residents may deduct deposits made in special personal savings accounts, pension plan insurance premiums paid to Mexican institutions and the cost of shares in specified investment companies (up to a certain limit)	

Taxable income[1]	Percentage on excess
0	3
3 639	10
30 884	17
54 275	25
63 093	32
75 539	33
152 352	34
240 128	35

Fiscal subsidy taxable income[1]	Per cent subsidy on marginal tax
0	50
3 639	50
30 884	50
54 275	50
63 093	50
75 539	40
152 352	30
240 128	20
304 704	10
365 644	0

Income brackets are adjusted semi-annually for inflation.

1. Each line refers to an income tax bracket whose lower limit is given on the line and upper limit on the next line.

Source: 1998 *Tax Law*.

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
	Income from the following sources receives special treatment: personal services, leasing of real property, sales of personal (non real estate) assets, up to net earnings three times the minimum wage. Special treatment encompasses the definition of deductible expenses, including interest and depreciation, and estimated tax payments and withholding. Taxpayers with income from these sources are eligible for the subsidy and credit (see Deductions and exemptions) granted under the general provisions of the IIT. Dividends and profits distributed are exempt from IIT when paid out of the special account for distributable profits (see CIT below) or subject to a final tax at the CIT rate when distributed out of other resources. This final tax is applied to the dividend or distribution multiplied by 1.515. Individuals my choose to include dividends and profits in taxable income in which case dividends or profits received are multiplied by 1.515 and a credit against income tax liability is taken at the CIT rate. For interest that is not exempt under the individual income tax, the withholding tax is 17 per cent on the first 10 percentage points of interest. Interest paid to individuals on special savings accounts is subject to a withholding tax of 35 per cent that is creditable against the final liability. A 15 per cent withholding tax is applied to indexed saving accounts (UDIs). Interest and gains from derivative financial transactions are subject to a 15 per cent provisional withholding tax, creditable against the annual tax liability.	Tax on income from agriculture, cattle breeding, forestry, fishery and book editing is reduced by 50 per cent. Fiscal subsidy. Individuals receive a credit against their income tax liability which is calculated as shown under rates. General credit. Individuals are entitled to a general tax credit of 1 303.86 pesos on an annual basis (1998). Salary credit. Instead of the flat general credit, individuals earning salary income is entitled to a tax credit determined according to the schedule shown under rates. Foreign tax credit. Taxpayers are granted a credit for tax paid abroad on income from a foreign source. This credit is limited to the amount of foreign-source income times either the CIT rate for taxpayers who compute their tax according to the provisions for income from business activities or the IIT rate for taxpayers who compute their tax under the general IIT provisions.	

Salary credit (1998)

Taxable income[1]	Annual salary credit
0.01	2 985
12 972	2 983
19 100	2 983
19 457	2 982
25 467	2 880
25 943	2 805
27 760	2 805
32 604	2 598
34 591	2 382
39 125	2 160
45 646	1 859
52 166	1 596
54 135	1 304

1. Each line refers to an income bracket whose lower limit is given on the line and upper limit on the next line.

Source: 1998 *Tax Law.*

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
1.1.2. Withholding tax	Employment income is subject to monthly withholding of income tax at source at the rates shown under rates. The withholding tax is creditable against final liability. Annual tax. In addition to the monthly withholding tax, the withholding agent must calculate every year the annual tax on employment income due by each employee at the annual IIT rates, considering the ordinary fiscal subsidy and a credit table that combines the credit tables used during the year for employment income proper or the general credit for other income. The obligation of employers to file an annual tax return on behalf of employees does not apply to employees who promise to file an annual return.		
1.1.3. Capital gains	Capital gains realised by individuals from the transfer or sale of immovable property, securities, and personal property are taxed as income from the transfer or disposition of goods. The sale price of assets is reduced by their acquisition cost adjusted for inflation to compute capital gains. A deduction for depreciation of buildings is also made. Capital gains arising from the acquisition of property are taxed as ordinary income.	Capital gains are not taxed if they raise in the following situations: gain from the sale of a personal residence if the individual lived in the residence for at least two years prior to the sale; transfer of ownership by reason of death or donations to immediate family members; gains from the sale of securities when the transaction is carried out through the Mexican stock exchange and certain exchanges of treaty countries; the gains from sale of personal property up to an annual amount not exceeding three times the minimum wage.	
1.2. *Tax on foreign remittances*	Withholding tax on wages, interest and royalties paid to non-residents.	Exempt interest includes interest on loans granted to the Federal Government.	

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
1.3. *Business* 1.3.1. Corporate income tax (CIT)	A broad-based tax on income of legal entities. Income includes changes in the value of assets and liabilities due to inflation, for example, gain accruing to taxpayer as a result of the reduction in real value of debts. Taxable income equals gross income less expenses and net losses carried forward from preceding periods. Corporate income tax returns are due on March 31. Receipts are deemed to accrue for the supply of goods and services on the day the relevant contract is concluded or upon any of the following events: issuance of a voucher for the price or delivery of the goods or some advance or partial payment for the goods, whichever occurs first. Interest is included in the taxable income of recipient companies. Royalties are normally included in the recipient's gross income. Capital gains obtained by business companies are normally included in gross receipts. In determining taxable capital gain on sales of land, buildings, and machinery and equipment, the historical cost of the undepreciated balance is adjusted for inflation. Exchange rate gains and losses are treated as interest. Companies must maintain a special account for distributable profits defined as profits net of CIT; non-deductible expenses other than reserves; and dividends received from other resident companies. Dividends and other distributions are deducted from the special account, as are any capital deductions treated as dividends.	Dividend and profit distributions are exempt when paid out of the distributable profits account (*i.e.* after corporate income tax). Non taxable income includes: *i)* capital increases; *ii)* payments made by shareholders to cover losses; *iii)* premiums received upon issuance of shares; *iv)* appreciation of shares in other companies arising from the use of the equity accounting method of valuation; *v)* revaluation of assets and capital; *vi)* dividends and profits received from other resident legal entities. These must be included in the income subject to profit sharing. Depreciation. Fixed assets can be depreciated at fixed yearly percentages (straight-line method) which for the following broad asset categories are: Buildings 5[1] Machinery and equipment 5-25[1] Computers and peripherals 30 Vehicles 25 Basis for annual depreciation is adjusted for inflation. 1. A 10% rate applies to constructions catalogued as archaeological, artistic, historical or patrimonial monuments. 2. A few machineries have rates of 50-100%.	34 per cent.

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
		Immediate deduction: Taxpayer may choose, instead of taking annual depreciation allowances under ordinary rules, an immediate depreciation deduction for certain assets. This deduction is a percentage of original cost, for example: Buildings 74 Machinery and equipment 74-95.7 Computers and peripherals 94 The immediate deduction equals the present value of the annual depreciation allowances using a real discount rate of 3 per cent. Immediate deduction is not applicable to furniture, office equipment, vehicles and aeroplanes. Immediate deduction is available to taxpayers outside the three largest metropolitan districts – Mexico City, Monterey, and Guadalajara – and taxpayers regardless of their location with gross income and assets not exceeding 7 and 14 million pesos respectively.	
	Simplified regime. Includes agriculture, livestock, forestry, fishery and land transportation activities. Taxable income is calculated on a cash-flow basis, where only resources taken out of the entity are subject to tax.		34 per cent.
	Special rate regime. Includes agriculture, livestock, forestry, fishery, silviculture and publishing.		17 per cent (or 25.5 per cent if these taxpayers, except those dedicated to publishing, commercialise or industrialise their products).
	Special regime for small taxpayers. Includes individuals carrying out business activities with a yearly gross income below 2.2 million pesos.		0-2.5 per cent of gross income.

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
		Inventory. Not recognised for tax purposes. All purchases and costs of production are deductible when incurred. Interest and inflationary losses are deductible provided they are computed under rules that require the taxpayer to compute the inflationary profit derived from monetary liabilities. Loss carry-forward. Net losses may be deducted from taxable profits of the ten years following the year in which they were sustained. No carry-back is allowed. Net losses are adjusted for inflation. Losses realised on sales of shares, debt and other transferable securities may be deducted only from gains realised on the sale of such assets during the year of the loss or subsequent five years. Foreign tax credit. A credit is granted for tax paid abroad on income from a foreign source. This credit is limited to the amount of foreign-source income times the CIT rate.	
1.3.2. Tax on business assets	A tax payable by corporations, unincorporated businesses and individuals (whether resident or non-resident) carrying out business activities or leasing to businesses subject to the tax, including maquiladoras. The assets tax is reduced by the amount of CIT paid. Further, the amount by which the CIT paid exceeds the assets tax in a given year can be used to obtain a refund of assets tax paid in any of the previous 10 years. The tax base is the average value of assets held during the taxable year or the prior four years at the taxpayer's option, less the average value of the liabilities to other business enterprises resident in Mexico, except those contracted with the financial sector. Tax is filed together with the income tax return. Procedures for adjusting the value of assets for inflation are specified in detailed rules.	Companies with income below 8.9 million pesos are exempt. For financial institutions the tax is levied on assets and liabilities which are not subject to financial intermediation. Companies are not liable during: *i)* the pre-operational period; *ii)* first year of operation; *iii)* two fiscal years following the first year of operation; and *iv)* last fiscal year for companies in liquidation. When the CIT exceeds the tax on assets, the excess may be carried back for 10 years and credited against the minimum tax paid in any of those years, and a refund may be requested. The excess of the CIT over the tax on assets during the previous three years may be credited against the tax on assets in the current year.	1.8 per cent. (0.9 per cent for entities in the special rate regime, *cf.* above.)

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
2. *Social security contributions*	The social security system for private sector employees (IMSS) underwent a major reform in 1997 concerning both pensions and health insurance. The financing of the pension system, now based on individual accounts, is shared between employers, employees and the government according to the following scheme:		

Contributions to pension scheme and disability insurance after the IMSS reform

	Premium (per cent of wage)	Cap (times the minimum wage)	Financing
Disability and life insurance	2.5	15-25[1]	Tripartite[2]
Retirees' medical care	1.5	15-25[1]	Tripartite[2]
Retirement and old-age severance	4.5	15-25[1]	Tripartite[2]
Government's contribution	365 pesos[3]	–	Government
Retirement saving: individual account	2.0	15-25[1]	Employer
INFONAVIT (housing)	5.0	10	Employer
Voluntary contributions	–	–	Employee/employer

1. Rising from 15 minimum wages in 1997 to 25 minimum wages in 2007.
2. Employer 70 per cent, worker 25 per cent and government 5 per cent of total.
3. At 1997 constant prices, amount equivalent to 5.5 per cent of the minimum wage in the Federal district, indexed to this minimum wage.

The financing of the health insurance system is being changed gradually over a transition period of ten years (1997 to 2007).

IMSS health fund financing

	1997	2007
Government		
Fixed	13.9 × min. w.	13.9 × min. w.
Maternity and sickness	0.05% of wage	0.05% of wage
Employers		
Fixed	13.9 × min. w.	20.4 × min. w.
Proportional	6% of wage > 3 × min. w.	1.1% of wage > 3 × min. w.
Maternity and sickness	0.70% of wage	0.70% of wage
Employees		
Proportional	2% of wage > 3 × min. w.	0.4% of wage > 3 × min. w.
Maternity and sickness	0.25% of wage	0.25% of wage

Source: National Authorities; OECD *Economic Survey Mexico,* 1998.

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
3. *Domestic taxes on goods and services* 3.1. *Value-added tax*	VAT is levied on the following transactions: transfer of goods; rendering of independent services, including professional services and interest not specifically exempt; granting of temporary use or right to use goods (leasing); imports of goods and services. Tax point. A taxable transfer of goods is deemed to be carried out and the tax is due when: *i)* the property is physically delivered; *ii)* the price is wholly or partially paid; *iii)* the relevant document covering the transfer is issued. Taxable amount. VAT is calculated on the price, inclusive of any taxes, fees, interest, or penalties. If there is no price, the base is the market value. If there is no market value, then an estimated value is used.	Tax credit. Taxpayers may deduct from their VAT liability the VAT charged on inputs or paid by the taxpayer on imports. Tax refund. If credits in a month exceed the liability to VAT, the taxpayer may carry the balance forward to the next month or apply for a tax refund of the full balance. Exempt persons. Individuals engaging in business activities who supply goods or services exclusively to final consumers are exempt provided their receipts in the previous year do not exceed one million pesos (a number adjusted annually for inflation). Individuals engaging in agriculture, forestry, or fishery are exempt within the same limit. Exempt transactions: inheritances; gifts; land; construction attached to land provided that it is used for dwelling homes; books, periodicals, magazines, and copyrights; used personal property; tickets for lotteries and games and prizes paid; Mexican and foreign currency; company shares, stock shares, and bonds. Exempt services: employee services rendered for remuneration; education at recognised schools and institutions; public transportation of persons by land (transportation by railroad is taxable); insurance of risks of agriculture and cattle raising; life insurance covering pensions and annuities; interest on most business loans and for first mortgage loans to individuals and other kind of interest specified in the law; derivative financial transactions; medical services.	15 per cent. 10 per cent for border regions. Zero-rated goods and services: non processed animal and vegetable products except rubber; food except processed food and smoked salmon and caviar; patented medicines; milk; ice and water; specified services, machinery and equipment for agriculture and fishery; wholesale of gold, gold bullions and jewellery with a content of at least 80 per cent of gold. Exports are zero rated.

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
3.2. *Excises*	An excise tax is levied on specified goods and services. The tax is levied under a value-added system up to the wholesale level. Cigarettes, gasoline, and diesel are taxed only at the producer or importer level. Taxable amount. With respect to imports, the excise tax is normally calculated on the import value plus customs duties and other duties paid on importation, except VAT.		Fuel: Based on the difference between the import parity and the regulated final consumer price. Beer.: 25-60 per cent. Wine: 25-30 per cent. Alcoholic beverages: 60 per cent. Mineral oils: 39.3-78.9 per cent. Tobacco: 20.9-85 per cent. New automobiles: 2 to 17 per cent plus a fixed amount, both based on the purchase price. *Source:* OECD, *Consumption Tax Trends*, 1998 edition.
4. *Taxes on international trade and transactions* 4.1. *Import duties*	Customs duties are levied on the customs value of most imported goods. Compensatory duties may be imposed on merchandise that the Ministry of Commerce has demonstrated to have been imported under conditions involving unfair practice in international trade. In addition to import duties, imports may be subject to VAT and excise taxes.	Goods may be temporarily imported duty free. The customs authorities may permit that merchandise be processed, transformed, or repaired within bonded premises to be subsequently exported or returned abroad.	The tariff provides for rates of up to 20 per cent. The average tariff rate has been reduced to about 11 per cent (1995).
4.2. *Export duties*	Export duties are levied on a few exported goods, for example, live rattlesnakes and turtles, although the export of most manufactured goods is not taxed.		
State taxes **1. *State payroll tax***	The Federal District and some states levy a small tax on payrolls to be withheld by employers.		
2. *Tax on the transfer of property*	The sale of immovable property is subject to a federal tax and a similar tax levied by each state under an agreement to avoid double taxation. The tax base is the sale price.	No inheritance tax is levied at either the federal or local level.	The rate of the federal tax is 2 per cent, and local taxes may not be higher than the federal tax.

Table A1. **Main features of the tax system, 1998** *(cont.)*

Tax	Nature of tax	Deductions and exemptions	Rates
3. *Tax on motor vehicles*	The states operate a tax on motor vehicles older than 10 years, based on the residual value of the vehicle. More importantly, they administer and appropriate the proceeds from the federal tax on motor vehicles that applies to vehicles less than 10 years old. This tax is based on the purchase price adjusted for inflation and depreciation.		
Municipal taxes			
Local tax on property	The tax is levied on the cadastral value of immovable property.		

Table A2. **Rates and structure of income tax on individuals, 1998**

	Taxes raised by Central government				Rates of sub-national taxes, per cent	Top marginal tax rate on wage income, per cent	Deductibility of interest payments[2]		Imputed rental income
	Highest and lowest marginal rate per cent[1]	Number of tax schedules[1]	Zero rated band	Tax credit			Housing	Consumption	
Mexico	**3-35**	**8**	**Yes**	**No**	**–**	**35.0**	**N**	**N**	**E**
United States	15-39.6	5	Yes	No	0-11.6	48.1	FA(C)	N	E
Canada	17.5-31.3	4	No	Yes	0-22.8	54.1	N	N	E
Germany	25.9-53.0	3	Yes	No	–	55.9	N	N	E
Greece[3]	5-40	4	Yes	No	–	40.0	FA	N	T
New Zealand	15-33	3	No	Yes	–	33.0	N	N	E
Portugal	15-40	4	No	Yes	–	46.6	FA(C)	N	E
Spain	17-47.6	8	Yes	No	3-8.4	56.0	PC	N	T
Turkey	25-55	7	Yes	No	–	61.3	FA	N	T

1. Excluding zero band or allowance.
2. Key to abbreviations: FA = Full deduction of tax allowance; PC = Partial (proportionate) deduction of tax credit; (C) = Deduction subject to a ceiling; N = No deduction; E = Exempt; T = Taxed.
3. 1997 instead of 1998.

Source: National Authorities; The OECD Tax Data Base, 1998; European Tax Handbook, 1998.

Table A3. **Statutory rates and rules for taxation of corporations, 1998**

	Central government basic rate per cent	Top marginal rate[1] per cent	Net worth tax	Dividends wedge[2]	Special rates (incl. small profits rate) per cent
Mexico	**34**	**34**	**Net worth tax of 1.8 per cent.**	**35**	**17/25.5**
United States	35	39.5	No tax on corporate net worth	67.7	15
Canada	29.1	46.1	No tax on corporate net worth	73.6	13.12/22.12
Germany[3]	47.5/30	58.2	Business tax on capital abolished as of January 1998	49.3	44.31/23.74
Greece[4]	35/40	40/35	No tax on corporate net worth	53.8	–
New Zealand	33	33	. .	33	–
Portugal	34	37.4	No tax on corporate net worth	53.1	–
Spain	35	35.8	No tax on corporate net worth	60.4	Several, depending on activity
Turkey	25	44	No tax on corporate net worth	66.4	20
Memorandum items:					
Argentina		33			
Brazil		25			
Chile		15			

1. Including local taxation and surcharges.
2. Differences between the pre-tax profit earned by the distributing company and the net dividend received by the shareholder.
3. These rates apply to retained profits only.
4. 1997 instead of 1998.

Source: The OECD Tax Data Base, 1998; National sources and the European Tax Handbook, 1998; KPMG Corporate tax rate survey, March 1998.

Table A4. **Main features of VAT systems in selected countries**

	Standard VAT rate	Concessions to small traders (in US dollars)	Departures from standard exemptions[1]		Coverage of lower rates	
			Exemption other than "standard exemptions"	Taxation of "standard exemptions"	Zero rate[2]	Lower rates
Mexico	15	112 194	The sale of books, newspapers, magazines, gold and silver coins and share, the exchange of foreign currency, retailing of gold bullion, authors' rights, public transport of passengers by land, agriculture, forestry and fishing activities.	Postal and insurance services except life and agricultural insurance. The letting of commercial buildings and financial services for consumer credits and personal loans.	The sale of non processed animals and vegetables except rubber; patent medicines, milk, water, ice, food except processed food and smoked salmon and caviar; agricultural equipment and machinery and fishing boats; the wholesale of gold, gold bullion and jewellery, some agricultural and fishing services, the letting of some agricultural machinery and equipment, the export of goods and services.	The sale of goods and services in the border regions. Rate = 10 per cent.
United States	–	–	–	–	–	–

Table A4. **Main features of VAT systems in selected countries** *(cont.)*

	Standard VAT rate	Concessions to small traders (in US dollars)	Departures from standard exemptions[1]		Coverage of lower rates	
			Exemption other than "standard exemptions"	Taxation of "standard exemptions"	Zero rate[2]	Lower rates
Canada[3]	7/15	22 760	Child care, legal aid, ferry, road and bridge tolls, standard municipal services.	Lotteries and gambling, supply and leasing of commercial land and buildings, domestic postal services.	Medicine, basic groceries, exports, certain financial services (usually to non-residents), certain agricultural and fishing products, medical devices, international travel and transportation services, international organisations and officials, agriculture, precious metals (sales of 25 cents or less made through mechanical coin-operated devices).	–
Germany[4]	16	21 352	–	–	–	Books, food, newspapers, certain cultural events, charitable work if not exempt, transport (only applies to passenger transport by ship and to local public passenger transport). Rate = 7 per cent.
Greece[5]	18	7 444	–	Culture (lower rate).	–	Books, culture, food, medicine, newspapers. Rates = 4 and 8 per cent.

Table A4. **Main features of VAT systems in selected countries** *(cont.)*

	Standard VAT rate	Concessions to small traders (in US dollars)	Departures from standard exemptions[1]		Coverage of lower rates	
			Exemption other than "standard exemptions"	Taxation of "standard exemptions"	Zero rate[2]	Lower rates
New Zealand[7]	12.5	21 126	Supply of fine metal (gold, silver and platinum).	Postal services, medical care, dental care, charitable work (other than unconditional gifts), education, non-commercial activities of non-profit making organisations (other than unconditional gifts), cultural services, insurance and reinsurance (other than life insurance and reinsurance), letting of immovable property (other than residential accommodation), lotteries and gambling, supply of land and buildings (other than those which have been used for the provision of residential accommodation for five years or more).	The supply of a taxable activity (business) as a going concern; the supply of fine metal (gold, silver or platinum) from a refiner in fine metal; the supply by a local authority of the local authorities petroleum tax (the distribution of the local authorities petroleum tax between local authorities); the exportation of goods and services (being goods and services to be consumed outside NZ's borders).	For long term stay in a commercial dwelling. Certain services. if provided as part of the right to occupancy are subject to tax at the standard rate on 60% of the value of the supply (an effective lower rate on such services of 7.5 per cent).
Portugal[6]	17	12 790	Agriculture	–	–	Food, essentials, medicines, books, newspapers, electricity, water distribution, accommodation services, passenger transport, admission to sporting events. Rates = 5 and 12 per cent.

Table A4. **Main features of VAT systems in selected countries** *(cont.)*

	Standard VAT rate	Concessions to small traders (in US dollars)	Departures from standard exemptions[1]		Coverage of lower rates	
			Exemption other than "standard exemptions"	Taxation of "standard exemptions"	Zero rate[2]	Lower rates
Spain	16	–	Copyright to literature and works of art.	Some cultural services provided consumers pay for them.	–	Books, social lodgement, catering, certain cultural and entertainment services, food, hotels, restaurants, supplies to the handicapped, medicines, transport, newspapers, public amenities, burial services, agricultural and forestry. Rates = 4 and 7 per cent.
Turkey[8]	15	–	Deliveries and repairs of maritime, aircraft or railway transport vehicles used for business activities, social services, water used in agriculture.	Educational and cultural service, news papers, books, magazines (lower rate), postal services, sale of commercial buildings, letting, radio and television broad casting, hospitals, lotteries and gambling.	–	Agricultural products, leasing, second hand cars, news papers, books, magazines, basic food stuffs, natural gas, cinema, theatre, opera, ballet and education. Rates = 1 and 8 per cent.

1. Standard exemption are the following: postal services, transport of sick/injured persons; hospital and medical care; human blood, tissues and organs; dental care; charitable work; education; non-commercial activities of non-profit making organisation insurance and reinsurance; letting of immovable property; financial services; betting, lotteries and gambling; supply of land and buildings; certain fund-raising events.
2. All countries apply zero rates to exports.
3. 15% Harmonised Sales Tax (HST) applies in those provinces that have harmonised their provincial retail sales tax with the federal GST (the 15% HST is composed of a provincial component of 8% and a federal component of 7%).
4. This rate is applied as of 1 April 1998.
5. Tax rates are reduced by 30% in some remote areas.
6. 17% in Madeira and Azores autonomous regions.
7. For long term stay in a commercial dwelling GST at standard rate is levied on 60% of the value of the supply.
8. Turkey applies a 23 to 40 % rate on certain luxurious goods.

Source: *Consumption Tax Trends,* OECD 1998.

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